

EART11200 - The Natural Scientists Toolkit

Mathematics and Physics

Semester 2 Handbook - 2024-25 Academic Year

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Created for the Department of Earth and Environmental Science, University of Manchester

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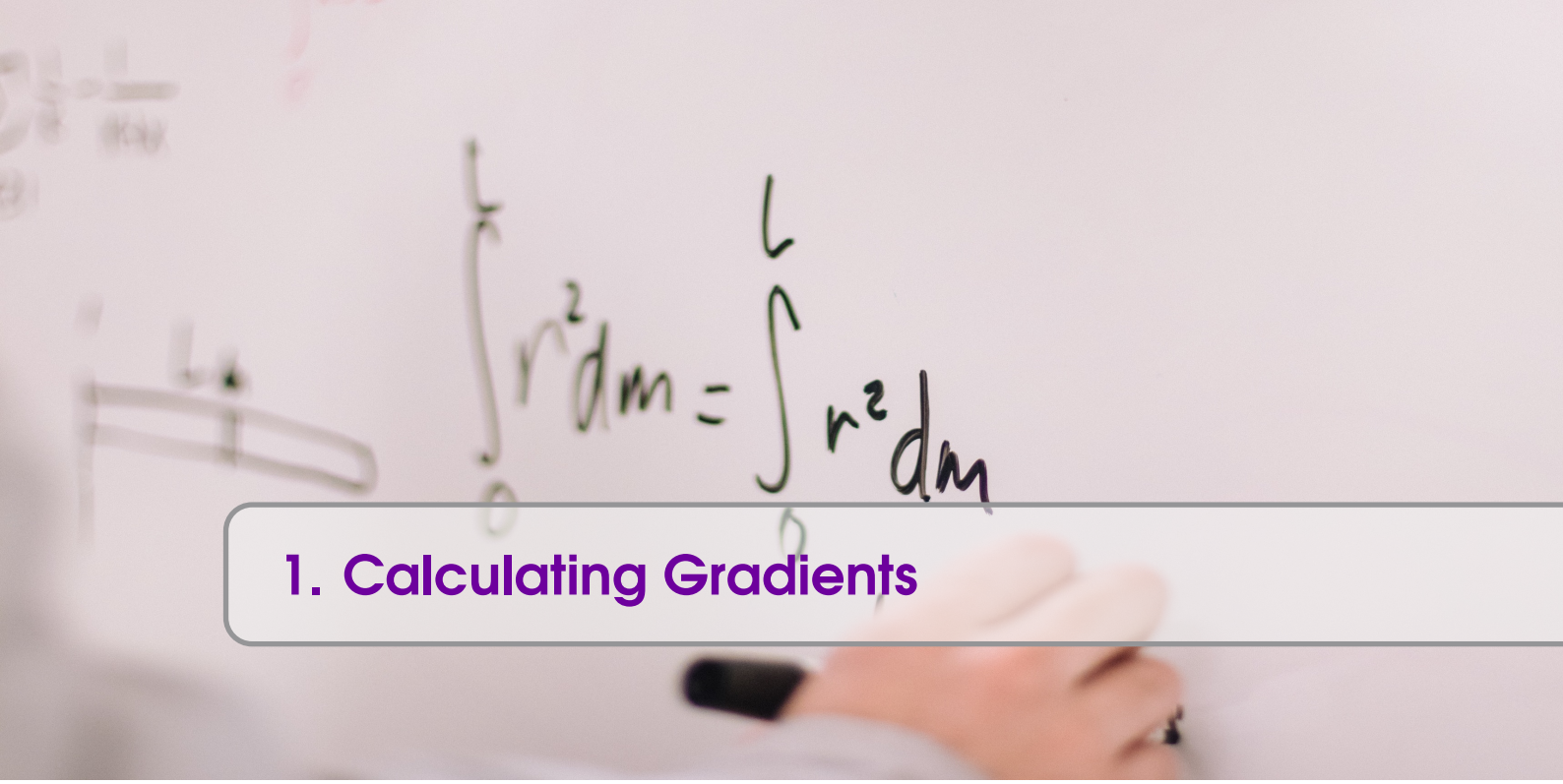
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1. Calculating Gradients

1.1 Maths, Calculus and Physics

In Earth and Environmental Science we are typically interested in understanding our past, and predicting our future. We live in a dynamic world where processes change in a matter of milliseconds to millennia. But what is Physics? If we check Encyclopedia Britannica we see that:

> *Physics can, at base, be defined as the science of matter, motion, and energy. Its laws are typically expressed with economy and precision in the language of mathematics.*

Essentially, Physics presents us with equations that describe the movement of energy through motion. Imagine relevant questions within our scientific domain:Where will warm air move from the tropics? How long will it take for the glacier to melt? But how do we solve these equations? We use the laws of Calculus. Again, if we check Encyclopedia Britannica we find one definition:

> *Calculus makes it possible to solve problems as diverse as tracking the position of a space shuttle or predicting the pressure building up behind a dam as the water rises.*

Calculus gives us the tools to calculate how quickly something is changing for example, or using that information to calculate quantities and states in the past or future. Is that relevant to understanding our changing climate? Physics is intricately built around laws of mathematics, and calculus provides us with the tools to understand the past, present

and future of our planet and our role on it. If we treat mathematics as the language of the universe then, through this course, we are going to expand our vocabulary. Understanding the basic laws of calculus and being able to implement fundamental laws of physics has been central to progressing all physical sciences.

What equations do we start with to practice implementing common rules of Calculus? This depends on the problem we wish to solve and each week we will focus on common problems in physics that are relevant to earth and environment sciences. As a start, we need to become familiar with a number of descriptions and concepts. For example, the following list of equations were taken from Ian Stewart in his book *17 Equations That Changed The World*. Let us quickly review some of these. You may have come across all of them or a subset, in your previous studies:

1. The first is the Pythagorean theorem. It was Pythagoras of Samos (c.570 – c.495 BC), who was an ancient Greek philosopher that discovered this:

   $$a^2 + b^2 = c^2$$

   This formula describes the relationship between the sides of a right triangle on a flat plane.

2. The second ones are a handful of laws of logarithms:

   $$\begin{aligned} log(xy) &= log(x) + log(y) \\ log(x/y) &= log(x) - log(y) \\ log(x^y) &= log(x) + log(y) \end{aligned}$$

   You will find in the crib sheet provided in this course booklet. This simple rule turns multiplication into addition, division into subtraction and powers into multiples! Logarithms were introduced by John Napier in 1614 to simplify calculations. Indeed, this was the most common way to quickly multiply together large numbers, greatly speeding up calculations in physics, astronomy, and engineering.

3. The third we look at is is Newton's law of gravitation:

   $$F = G\frac{m_1 m_2}{r^2}$$

   It describes the force of gravity between two objects, F, in terms of a universal constant, G and the masses of the two objects, m_1 and m_2, and the distance between the objects, r. Newton's law is remarkable in that it explains, almost perfectly, why

the planets move in the way they do. Notice the power; how would you rearrange this to calculate r?.

4. The fourth was derived by Albert Einstein as part of the theory of special relativity, and has adorned many a t-shirt:

$$E = mc^2$$

This famous formula states matter and energy are equivalent to each other.

5. The fifth describes the **derivative**:

$$\frac{df}{dt} = \lim_{h \mapsto 0} \frac{f(t+h) - f(t)}{h} \tag{1.1}$$

Derivatives are central to Calculus, and tell us how much one variable is changing with respect to one or many others. The expression $\frac{df}{dt}$ in the English equivalent reads: the rate of change of f with respect to variable t, which is often time. We use Calculus to arrive at an expression for $\frac{df}{dt}$, rather than calculating this by hand. In this equation, we are using an approximation which forms the basis of the final exercise in today's class.

6. The final is actually called a Fourier Transform:

$$f(\omega) = \int_{\infty}^{\infty} f(x) e^{-2\pi i x \omega} dx$$

These take a messy wave function like a recording of a person talking, and allow us to break the messy function into a combination of a number of simple waves. This transform is at the heart of modern signal processing and analysis, and data compression. However the important component of this equation is the symbol $\int$, which represents the act of **integration**.

The final two equations are particularly relevant to this module. Not because of the definition of the variable f ($f(x)$ tells us that f is a function of variable x e.g. temperature as a function of latitude), but because of the procedures they represent. Whilst a derivative function tells us how much one variable is changing as a function of time, for example, the integral reverses this operation. Specifically, we can convert a derivative at a given point in time (or space) to an absolute or net quantity. For example, if water is flowing out of a reservoir at a variable rate, we could work out the total water lost or gained over a set period of time through the act of integration. Conversely, if we measure the actual value of water in the reservoir at certain points in time, we could then derive a derivative expression that tells us the rate of change at any point in time.

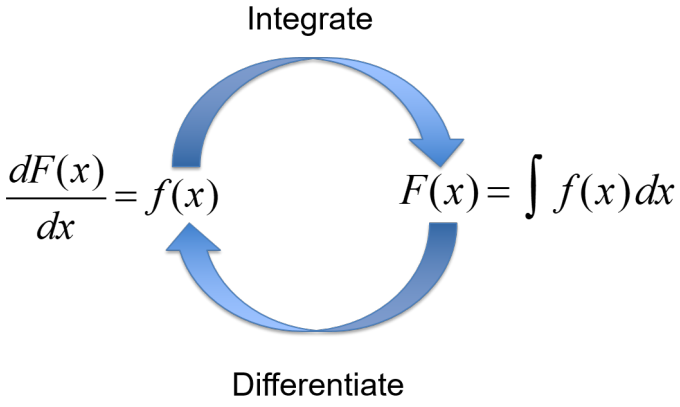


Figure 1.1: Interplay between differentiation and integration throughout this course. We start with differentiation and move in to integration later on

We have some key definitions we use throughout the course:

1. **Differentiation** allows us to quantify the rate at which much one variable is changing with respect to one or many others.
   - Algebraically, this is expressed using a **differential equation**. For example, $\frac{df}{dt}$ represented by equation 1.1 is a **differential equation**. The expression $\frac{df}{dt}$ in the English equivalent reads: the rate of change of f with respect to variable t.
   - We can derive the **differential equation** using the laws of calculus. Imagine we measured variable f as a function of variable t. We then fit a line through so we have an expression for $f(t)$. We might be able to use calculus to derive an equation for $\frac{df}{dt}$ directly so we can calculate the gradient at any point. We say we have derived an **analytical** expression.
   - In many cases we cannot derive an expression for $\frac{df}{dt}$ directly. It may simply be too complicated to use the laws of calculus. However, we can approximate a derivative using the difference between two values of both f and t. We say we have used a **numerical** approximation. Equation 1.1 represents this operation and we will be using this from today.

2. **Integration** reverses the act of differentiation.
   - If we are given a differential equation, integration allows us to predict values in the past or future based on this gradient expression. We will see integration effectively sums derivatives to arrive at absolute quantities of values we are interested in: heat, mass, velocity etc.
   - In some cases we can derive the **integral** using the laws of calculus. We say we have derived an **analytical** expression.
   - In many cases we cannot derive an expression for $\int \frac{df}{dt}$ directly. It may simply

be too complicated to use the laws of calculus. However, we can approximate an integral using using a technique very similar to approximations for derivatives. We say we have used a **numerical** approximation.

Figure 1.1 summarises how these types of operations are linked through our course. In weeks 1-5 we will be driven by different ways to arrive at differential equations. We use both numerical and analytical approaches and start with equations that express one quantity as a function of another, so we can extract the gradient. From week 6 onward we apply the process of integration. Here we start with a differential equation, and then integrate this to calculate quantities in the past or future. In both sections we will find we need to solve problems based on a subset of laws in Physics as applied to problems in earth sciences. These will be introduced each week, when relevant, to frame the problems we wish to solve. You can also take a look at more detail in the supporting material section, specifically Chapter 11. Here I provide a little more information on each theory should it help with the course content.

1.2 Approximating gradients

Figure 1.2 plots the change in variable y as a function of variable x according to the following equation, between the limits x=0 and x=20:

$$y = x^{0.5} + \left(\frac{x-10}{3}\right)^2$$

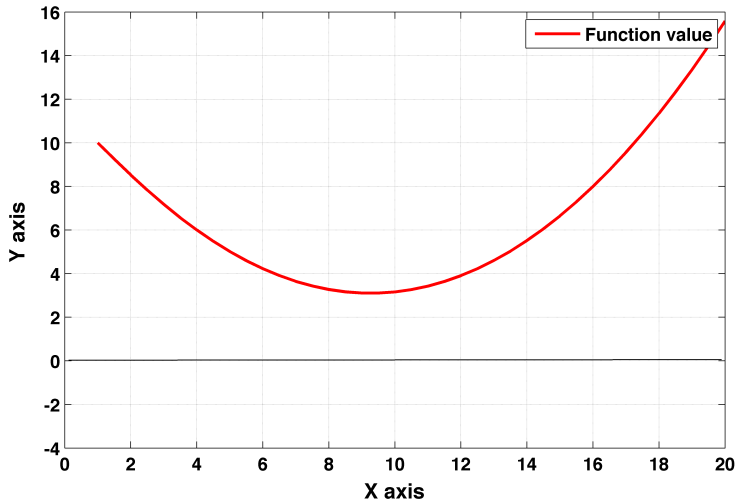


Figure 1.2: Generic function in which variable y changes with x. The line does not have a constant gradient.

How might we obtain a numerical approximation for the gradient at $x = 2$, $x = 4$, $x = 6$ and so on up to $x = 20$? A **numerical approximation** represents a manual derivative where

you take two values of both x and y, to obtain a derivative. Many times in research, we do not have a function from which we can obtain an **analytical derivative** and have to use numerical methods.

You may have seen previously that to calculate a gradient, you divided the change in y by the change in x. We are going to continue to use this idea to build up the principle of differentiation. Thus we say:

$$\frac{dy}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x}$$

There are many methods to do calculate a gradient. The first approach we'll consider is to take the current y value, subtracted from the previous y value and divided by the increment in x:

$$\frac{dy}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x} \equiv \frac{y_{current} - y_{previous}}{x_{current} - x_{previous}}$$

In the current problem, this procedure restricts us to start at x=4. Think about why this might be quite restrictive. We will call this the **Euler backward** method.

What happens if we change the direction of our increment to obtain this numerical approximation? To test this, rather than taking the value from a previous point, we could take the value at a future point. Thus, we start at $x = 2$ and take the value at $x = 4$. The approximate formula then becomes:

$$\frac{dy}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x} \equiv \frac{y_{future} - y_{current}}{x_{future} - x_{current}}$$

Here you can start at $x = 4$ but only obtain the gradient up to $x = 18$. Again, why is this restrictive? We will call this the **Euler forward** method.

Let us re-write the Euler's forward and backward methods using slightly different notation:

$$\frac{dy}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x} \equiv \frac{y_{x+\Delta x} - y_x}{\Delta x}$$
$$\frac{dy}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x} \equiv \frac{y_x - y_{x-\Delta x}}{\Delta x}$$

Here, Δx refers to the step-size or increment used between values in the approximation. Both of these approximations are attempting to calculate the same gradient but using different parts of the function curve, or function values.

The **central difference** method is another way to obtain a numerical approximation for the gradient and is more accurate for smooth functions, though this accuracy can vary depending on the shape of our curves. In this approach we take function values either side

of our current value and use the difference in a similar way. If we have two equally spaced function values, we can write the central approximation approach as:

$$\frac{dy}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x} \equiv \frac{y_{x+\Delta x} - y_{x-\Delta x}}{2\Delta x}$$

Can you see why we now use $2\Delta x$ in the denominator?

You may come across other (slightly) different notation when writing differential equations. Sometimes we write $\frac{dy}{dx}$ as y'. All of the following expressions are equally valid:

$$y' = \frac{dy}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x} \equiv \frac{y_{x+\Delta x} - y_{x-\Delta x}}{2\Delta x}$$
$$f' = \frac{df}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta f}{\Delta x} \equiv \frac{f_x - f_{x-\Delta x}}{\Delta x}$$
$$p' = \frac{dp}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta p}{\Delta t} \equiv \frac{p_{future} - p_{current}}{t_{future} - t_{current}}$$

Now we are done with numerical approximations. The analytical derivative of Equation 1.2 above happens to be given by the following:

$$\frac{dy}{dx} = 0.5x^{-0.5} + \frac{2}{3}\left(\frac{x-10}{3}\right)$$

In the first practical session you will be asked to compare the accuracy of all approaches.

1.3 Working in the practical classes

For the exercises given out each week, you will find the solutions at the end of each Chapter. Rather than print these out in class, I would encourage you to try these first and then take a look at your working. Do try to resist looking at these solutions until you have given each problem a shot. Once you read through the solutions to check your answer, if anything is not clear then please do ask for help during each class. I will walk through example solutions, often at the beginning of the following weeks lecture.

You are not expected to retain a carbon copy of each solution in your head as we move through the course content. Do feel free to revisit material and operations as appropriate. The point of this course is to become familiar with numerical operations used to solve common problems in physical sciences. The more time you spend on these problems, the easier they will become.

Exercises

Introductory Exercises

We first need to practice manipulating components of equations. In any area of science, as we pull apart various theorems (*a statement proven to be true and expressions through mathematical operations and arguments*), we are often left with a core set of operations. The analogy is pulling apart a physical instrument or device that is then left as a collection of nuts, bolts, wires, connections etc. Recognising how to manipulate powers and logarithms, for example, allows us to understand relationships between variables we are interested in whilst also building our own theoretical basis. When you move on to the programming course in the toolkit, we will then understand the process that allows us to implement these formulas on a computer through a programming language. In this module we start with the underlying mathematics.

In this section we practice manipulating powers, re-arranging equations and understanding relevant trigonometric relationships. All of these are relevant to the problems we will come across in the course. Some of the following examples in the first two sections are taken in part from Foundation Maths Third Edition by Anthony Croft & Robert Davison [ISBN 0 130 45426-5], which we provide as a PDF on the course blackboard site. You might find these examples easy, or not, but I would recommend running through them nonetheless.

Powers and indices

The function

$$y = x^3$$

states that variable y is a function of x to the power of 3.

Question 1.1: Without using a calculator, work out the value of:

(a) 4^2
(b) 5^3
(c) $\frac{2}{5}^3$

Question 1.2: Write the following expressions more concisely by using an index:

(a) $a * a * a * a$
(b) $(yz) * (yz) * (yz)$
(c) $\left(\frac{a}{f}\right) * \left(\frac{a}{f}\right)$

To manipulate expressions involving indices we use common laws given below:

$$a^m * a^n = a^{m+n} \qquad \frac{a^m}{a^n} = a^{m-n} \qquad (a^m)^n = a^{m*n}$$

Question 1.3: In each of the following, chose an appropriate law to simplify the expression:

(a) $5^{13} * 5^3$

(b) $8^{13}/8^5$

(c) $\left(a^3\right)^4$

(d) $\frac{y^7}{y^3}$

Logarithms and exponentials

Logarithms provide an alternative way of writing expressions involving powers. If:

$$a = b^c.$$

then:

$$log_b a = c$$

For example, using the above we could solve the expression:

$$e^x = 14$$

to find x as:

$$x = log_e 14 = 2.639$$

You will often use what we call natural logarithms and logarithms with a base 10. A natural logarithm is also written as:

$$log_e x = ln(x)$$

Question 1.4: Write down the expressions below in their simplest form:

(a) $\frac{\exp(-8)}{\exp(-10)}$

(b) $\frac{\exp(3)}{\exp(-15)}$

(c) $\frac{\exp(-1)\exp(-1)}{\exp(-5)}$

(d) $\frac{\exp(-3)^2}{\exp(-6)}$

Question 1.5: Solve the following for b:

(a) $\log_{10}\left(4b^2\right) = 3.$

(b) $\exp\left(\frac{2b-1}{3}\right) = 5$

Question 1.6: Solve the following for t:

(a) $\exp(2t) = 5$

(b) $\exp(5t) \times \exp(-t) = 5$

(c) $\exp(10t) \div \exp(-t) = 5$

(d) Two equations: $\exp(4t) = 0.1N$ and $\exp(t) = N$ [hint: divide one equation by the other to cancel N]

Question 1.7: Evaluate the following (to 3 decimal places):

(a) $x = \log_3(3)$. Or what power do I need to raise 3 by to equal 3?

(b) $x = \log_5(10)$. Or what power do I need to raise 5 by to equal 10?

(c) $x = \log_2(8)$. Or what power do I need to raise 2 by to equal 8?

(d) $x = \log_3(59049)$. Or what power do I need to raise 3 by to equal 59049?

Rearranging expressions

Question 1.8: Rearrange the following equations to make x the subject:

(a) $y = 2x + 4x + 5x$

(b) $y = \frac{1}{2x+4x+5x}$

(c) $y = \frac{1}{2x^2+4x^2+5x^2}$

(d) $y = \frac{x}{2x^2+4x^2+5x^2}$

(e) $y = \sqrt{x+1}$

(f) $M = \frac{\pi\rho x^3}{6}$

Question 1.9: Expand the following:

(a) $3\left(x^2+3y\right)$

(b) $7x\left(-\frac{1}{x}+y^3\right)$

(c) $3xy\left(x+\frac{2}{xy}\right)$

Question 1.10: The equation describing radioactive decay is:

$$\frac{dN}{dt} = -\lambda N$$

where N is the number of unstable nuclei, t is time and λ is called the decay constant. This is a differential equation. The solution to this differential equation is given by the following:

$$N(t) = N_0 e^{-\lambda t}$$

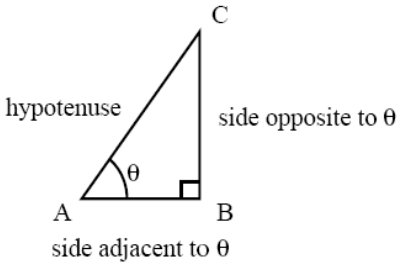
The half-life is the time it takes for the number of unstable nuclei to decrease by one half. Your task is to prove that the decay constant is given by the following expression:

$$\lambda = \frac{\ln 2}{t_{1/2}}$$

where $t_{1/2}$ is the half-life of the sample.

Trigonometry

Question 1.11: The questions below are about triangles labelled below:



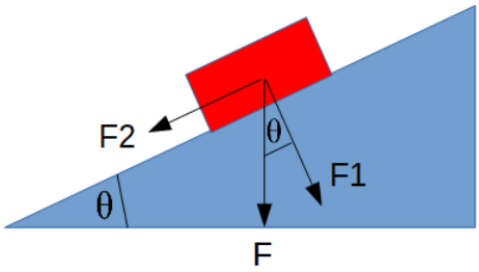
(a) A right angled triangle, ABC, has AB= 10 and BC= 25, what is the angle θ?

(b) A right angled triangle, ABC, has $\theta = 40^\circ$ and AB= 5, what is the length of the hypotenuse?

(c) A right angled triangle, ABC, has $\theta = \frac{\pi}{8}$ radians and BC= 10, what is the length of the hypotenuse?

(d) A right angled triangle, ABC, has a hypotenuse of length 50 and BC= 10, what is the angle θ in radians?

Question 1.12: Below is a schematic of a block placed on an incline: You will come



across this schematic a number of times when we solve problems using Newton's law of motion. In questions involving trigonometry thus far, you have been calculating lengths of opposite and adjacent sides of a triangle. We can use the same procedure to calculate the magnitude of different forces acting on a body. In this example, we have a block sat on an incline that is subject to the Earths gravitational pull. The gravitational force acts vertically, from the centre of the block to the centre of the earth. Assuming the triangle is a hard object, a component of this force will drag the block down the slope, labeled as $F2$. Another component will act perpendicular to the slop, which we have labeled as $F1$ and is known as the normal force. If the magnitude of the gravitational pull is given by the formula $m.g$, where m is the mass of the block and g is a constant, what is the magnitude of (a) $F1$ and (b) $F2$?

Calculating a gradient

In this section we practice calculating gradients and approximating differentiation using Euler's backward method, Euler's forward method and the central differences method. Recall that the formulae for Euler's backward method, Euler's forward method and the central differences method are as follows:

$$\text{Euler's backward method: } \frac{dy}{dx} = \frac{y_x - y_{x-\Delta x}}{\Delta x}$$
$$\text{Euler's forward method: } \frac{dy}{dx} = \frac{y_{x+\Delta x} - y_x}{\Delta x}$$
$$\text{Central differences method: } \frac{dy}{dx} = \frac{y_{x+\Delta x} - y_{x-\Delta x}}{2\Delta x}$$

Question 1.13: Figure plots the change in variable y as a function of variable x according to the following equation, between the limits x=0 and x=20:

$$y = x^{0.5} + \left(\frac{x-10}{3}\right)^2$$

(a) Your task is obtain a numerical approximation for the gradient at $x = 2$, $x = 4$, $x = 6$ and so on up to $x = 20$. Recall, this represents a manual derivative where you take two numbers for both x and y, to obtain your derivative. Many times in research we do not have a function from which we can obtain an analytical derivative and have to use numerical methods. Evaluate the values of y at each value of x and use the Euler's backward method, Euler's forward method and the central differences method to fill in the first four columns of the following table:

x	y	Backward	Forward	Central	Analytical
2		–	–	–	–
4					
6					
8					
10					
12					
14					
16					
18					
20		–	–	–	–

(b) Now we are done with numerical approximations, the analytical derivative is given by:

$$\frac{dy}{dx} = 0.5x^{-0.5} + \frac{2}{3}\left(\frac{x-10}{3}\right)$$

Fill in the final column with the values of the table with the analytical derivative.

(c) Compare the approximations with the analytical derivative. Is one method better than the other?

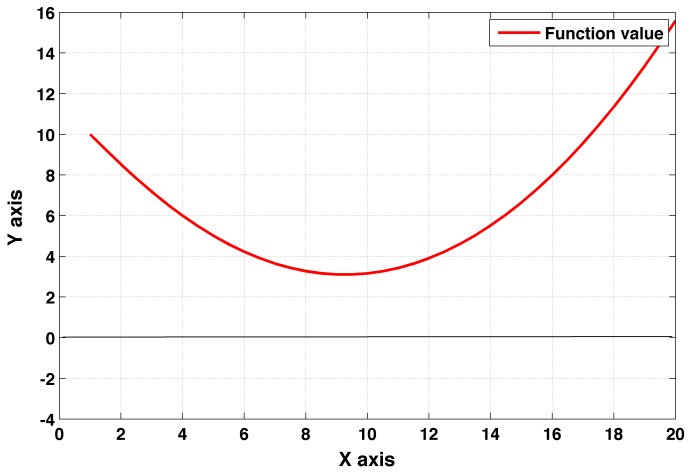


Figure 1.3: Generic function in which variable y changes with x. The line does not have a constant gradient.

Question 1.14: The equation describing radioactive decay is given by:

$$\frac{dN}{dt} = -\lambda N \tag{1.2}$$

where N is the number of unstable nuclei, t is time and λ is called the decay constant. This is a **differential equation**. The solution to this differential equation is given by the following:

$$N(t) = N_0 e^{-\lambda t} \tag{1.3}$$

which means the more radioactive atoms we have the higher the decay rate. In future classes we can use both **differentiation** and **integration** to move between the differential equation 1.2 and equation 1.3. For now we want to quantify the accuracy of a **numerical** approach to calculating the how N varies with time, thus approximate $\frac{dN}{dt}$ over and over again, when compared with the expression given by equation 1.3.

Radioactive caesium has a half-life of 30 years and so the decay constant, $\lambda = \frac{\ln 2}{30} \cong 0.0231$ yr^{-1}. Using the radioactive decay equation we know that the gradient in the number of atoms vs time is equal to $-\lambda N$ (where N is the number of radioactive atoms). The initial number of radioactive Cs atoms is 1000 at $t = 0$.

(a) Fill in the values in the table below using both the *Euler forward step* finite difference approximation and equation 1.3, using $\Delta t = 1$ year:

t	$N(t)$	**Numerical** (ΔN)	**Analytical** ($-\lambda N(t)$)
0	1000.00	-22.84	-23.10
1	977.16	-22.31	-22.57
2			
3			
4			
5			
6			
7			
8			
9			
10			

(b) How is the agreement between the analytic and approximate methods?

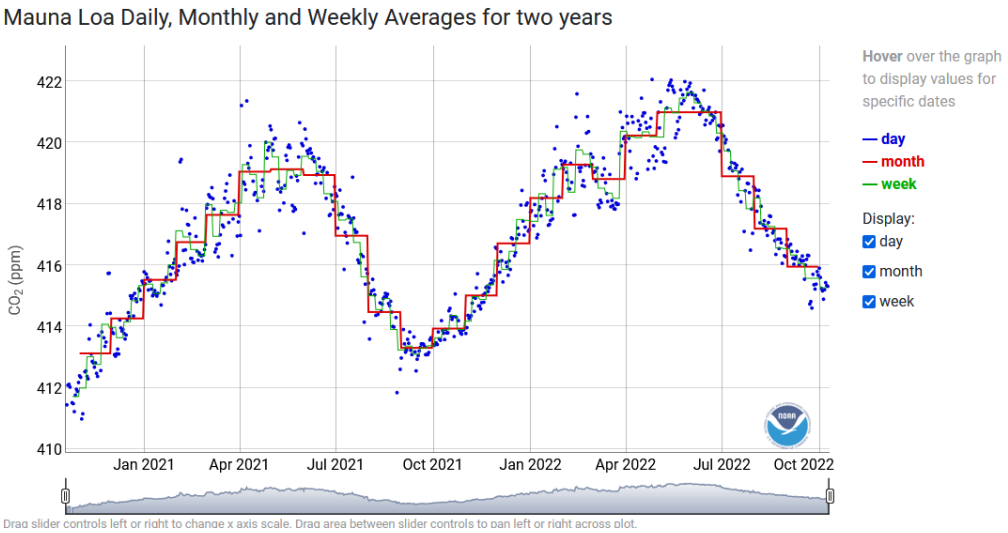


Figure 1.4: Mauna Loa Daily, Monthly and Weekly Averages for two years. See gml.noaa.gov/ccgg/trends/ for more details

Question 1.15: The table below provides monthly mean CO_2 (in parts per million, or ppm) constructed from daily mean values measured at the Mauna Loa observatory (See here for additional details). These are also plotted as lines amongst raw data in figure 1.4.

year	month	decimal date	monthly average (ppm)
2021	4	2021.2917	419.05
2021	5	2021.3750	419.13
2021	6	2021.4583	418.94
2021	7	2021.5417	416.96
2021	8	2021.6250	414.47
2021	9	2021.7083	413.30
2021	10	2021.7917	413.93
2021	11	2021.8750	415.01
2021	12	2021.9583	416.71
2022	1	2022.0417	418.19
2022	2	2022.1250	419.28
2022	3	2022.2083	418.81
2022	4	2022.2917	420.23
2022	5	2022.3750	420.99
2022	6	2022.4583	420.99
2022	7	2022.5417	418.90
2022	8	2022.6250	417.19

(a) Use a numerical approximation to derive rates of change in June for both 2021 and 2022. By using Δt of 1 month, compare the results from a central difference approximation to a forward difference approximation.

(b) Repeat the above operation but this time using a Δt of 2 months.

Question 1.16: Consider the following cubic equation:

$$f(x) = x^3 + 2x^2 - 5x + 16$$

(a) Calculate the numerical approximation of the derivative of $f(x)$ for $x = 1$ using Eulers Backward, Eulers Forward, and the Central Differences Methods using a stepsize of $\Delta x = 1$, $\Delta x = 0.1$, $\Delta x = 0.01$, and $\Delta x = 0.001$.

(b) The analytical derivative is given by:

$$\frac{df}{dx} = 3x^2 + 4x - 5$$

Calculate the differences (or the percentage difference) between the analytical derivative and the numerical approximation at $x = 1$ that calculated from part (a). What do you notice as we increase the step size?

Solutions

Question 1.1:

(a) $4^2 = 16$

(b) $5^3 = 125$

(c) $\frac{2}{5}^3 = \frac{8}{125}$

Question 1.2:

(a) $a * a * a * a = a^4$

(b) $(yz) * (yz) * (yz) = (yz)^3$

(c) $\left(\frac{a}{f}\right) * \left(\frac{a}{f}\right) = \left(\frac{a}{f}\right)^2$

Question 1.3:

(a) $5^{13} * 5^3 = 5^{16}$

(b) $8^{13}/8^5 = 8^8$

(c) $\left(a^3\right)^4 = a^{12}$

(d) $\frac{y^7}{y^3} = y^4$

Question 1.4:

(a) $\frac{\exp(-8)}{\exp(-10)} = \exp(2)$

(b) $\frac{\exp(3)}{\exp(-15)} =$

(c) $\frac{\exp(-1)\exp(-1)}{\exp(-5)} = \exp(3)$

(d) $\frac{\exp(-3)^2}{\exp(-6)} = 1$

Question 1.5:

(a) $\log_{10}\left(4b^2\right) = 3 \implies b = 15.811.$

(b) $\exp\left(\frac{2b-1}{3}\right) = 5 \implies b = 2.914$

Question 1.6:

(a) $\exp(2t) = 5 \implies t = 0.805$

(b) $\exp(5t) \times \exp(-t) = 5 \implies t = 0.402$

(c) $\exp(10t) \div \exp(-t) = 5 \implies t = 0.146$

(d) Two equations: $\exp(4t) = 0.1N$ and $\exp(t) = N \implies t = -0.768$

Question 1.7:

(a) $x = \log_3(3) \implies x = 1.000$

(b) $x = \log_5(10) \implies x = 1.431$

(c) $x = \log_2(8) \implies x = 3.000$

(d) $x = \log_3(59049) \implies x = 10.000$

Question 1.8:

(a) $y = 2x + 4x + 5x \implies x = \frac{y}{11}$

(b) $y = \frac{1}{2x+4x+5x} \implies x = \frac{1}{11y}$

(c) $y = \frac{1}{2x^2+4x^2+5x^2} \implies x = \pm\sqrt{\frac{1}{11y}}$

(d) $y = \frac{x}{2x^2+4x^2+5x^2} \implies x = \frac{1}{11y}$

(e) $y = \sqrt{x+1} \implies x = y^2 - 1$

(f) $M = \frac{\pi\rho x^3}{6} \implies x = \sqrt[3]{\left(\frac{6M}{\pi\rho}\right)}$

Question 1.9: Expand the following:

(a) $3\left(x^2 + 3y\right) \implies 3x^2 + 9y$

(b) $7x\left(-\frac{1}{x} + y^3\right) \implies -7 + 7xy^3$

(c) $3xy\left(x + \frac{2}{xy}\right) \implies 3x^2y + 6$

Question 1.10: The equation describing radioactive decay is:

$$\frac{dN}{dt} = -\lambda N$$

where N is the number of unstable nuclei, t is time and λ is called the decay constant. This is a differential equation. The solution to this differential equation is given by the following:

$$N(t) = N_0 e^{-\lambda t}$$

Here we can simply set the value of N to be half of N_o. Then, by using common laws regarding re-arranging logarithms, we can use the following procedure:

$$\begin{aligned} \frac{N_o}{2} &= N_o e^{-\lambda t_{1/2}} \\ ln\left(\frac{1}{2}\right) &= -\lambda t_{1/2} \\ ln\left(\left(\frac{1}{2}\right)^{-1}\right) &= \lambda t_{1/2} \\ \lambda &= \frac{ln2}{t_{1/2}} \end{aligned}$$

where $t_{1/2}$ is the half-life of the sample.

Question 1.11:

(a) 68.2°

(b) 6.53

(c) 26.13

(d) 0.20 radians

Question 1.12:

1. $F1 = mg\cos(\theta)$
2. $F2 = mg\sin(\theta)$

Question 1.13:

(a) - (b) Filled in table:

x	y	Backward	Forward	Central	Analytical
2	8.53	–	–	–	–
4	6.00	-1.26	-0.89	-1.07	-1.08
6	4.23	-0.89	-0.48	-0.68	-0.69
8	3.27	-0.48	-0.06	-0.27	-0.27
10	3.16	-0.06	0.37	0.16	0.16
12	3.91	0.37	0.81	0.59	0.59
14	5.52	0.81	1.24	1.02	1.02
16	8.00	1.24	1.68	1.46	1.46
18	11.35	1.68	2.12	1.90	1.90
20	15.58	–	–	–	–

(c) Looking at the raw differences:

x	Backward	Forward	Central
4	-0.18	0.20	0.01
6	-0.20	0.21	0.00
8	-0.21	0.21	0.00
10	-0.21	0.22	0.00
12	-0.22	0.22	0.00
14	-0.22	0.22	0.00
16	-0.22	0.22	0.00
18	-0.22	0.22	0.00

and also looking at the percentage difference:

x	Backward	Forward	Central
4	16.55	18.18	0.81
6	29.44	30.31	0.44
8	78.28	79.34	0.53
10	134.97	135.99	0.51
12	36.63	36.80	0.09
14	21.23	21.30	0.03
16	14.95	14.99	0.02
18	11.54	11.56	0.01

We can see that the central differences method is very accurate. Recall from lectures that the central differences method is more accurate for smooth functions, though this accuracy can vary depending on the shape of our curves.

Question 1.14:

(a) Filled in table:

t	$N(t)$	**Numerical** (ΔN)	**Analytical** ($-\lambda N(t)$)
0	1000.00	-22.84	-23.10
1	977.16	-22.31	-22.57
2	954.85	-21.80	-22.06
3	933.05	-21.31	-21.55
4	911.74	-20.82	-21.06
5	890.92	-20.34	-20.58
6	870.58	-19.88	-20.11
7	850.70	-19.43	-19.65
8	831.27	-18.98	-19.20
9	812.29	-18.55	-18.76
10	793.74	-18.13	-18.34

(b) The agreement is pretty good in this case. Note that when using the approximate method the step size chosen must be small enough. You will only figure this out by trial and error (e.g. try a step size and then try a smaller step size to see if the results are the same).

Question 1.15:

(a) - (b) The table below gives the solutions to both parts where the time difference moves from 1 to 2 months. Notice how just using these derived gradients could be misleading. We need to see longer term trends to infer any global changes.

year	month	average (ppm)	forward	central	forward	central
2021	4	419.05	–	–	–	–
	5	419.13	–	–	–	–
	6	418.94	-1.980	-1.085	-2.235	-1.145
	7	416.96	–	–	–	–
	8	414.47	–	–	–	–
2022	4	420.23	–	–	–	–
	5	420.99	–	–	–	–
	6	420.99	-2.090	-1.045	-1.900	-0.760
	7	418.90	–	–	–	–
	8	417.19	–	–	–	–

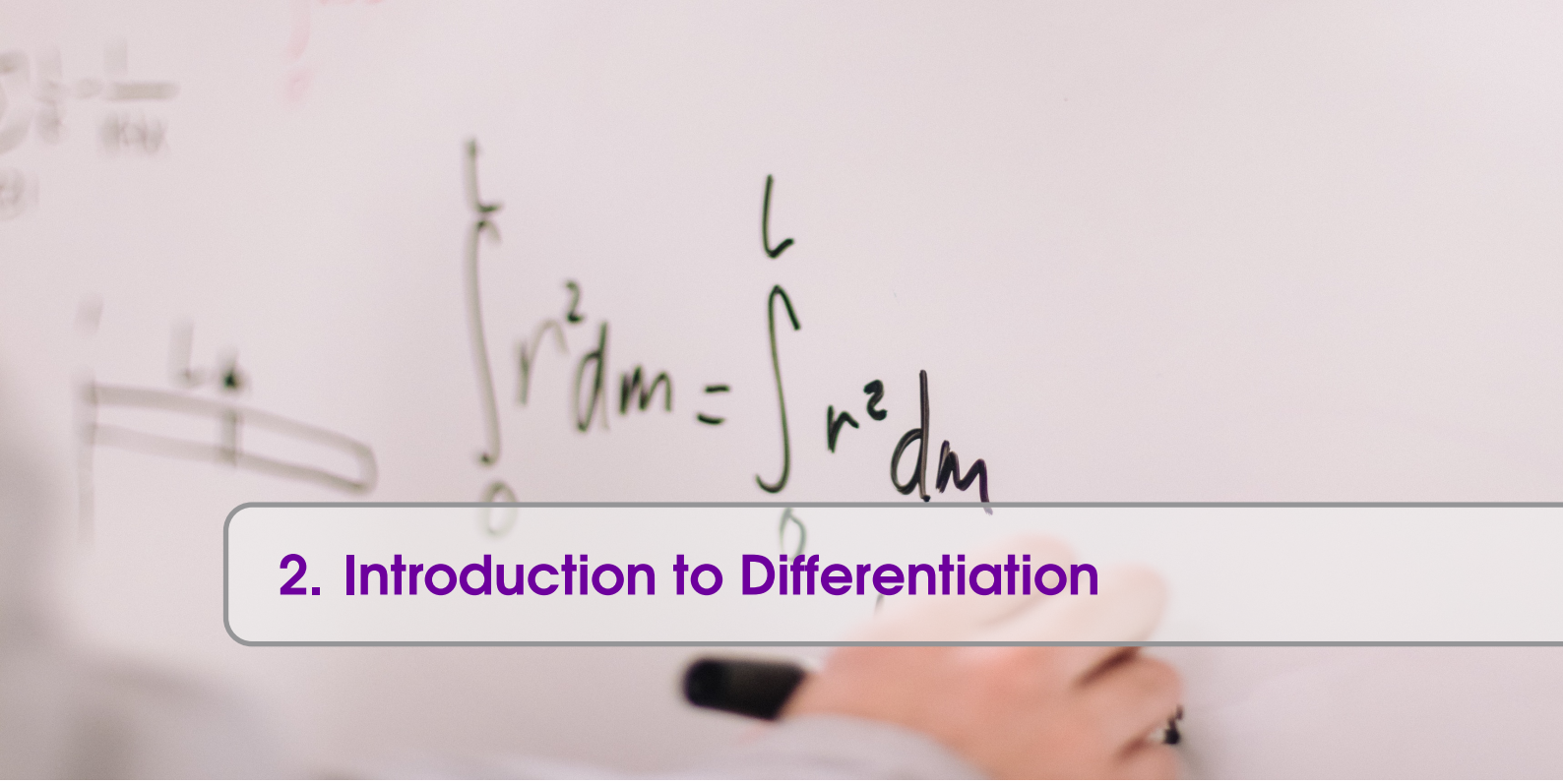
Question 1.16:

(a) The table below gives the solutions for the numerical approximation of the derivative of $f(x)$ for $x = 1$ using Euler's Backward, Euler's Forward, and the Central Differences Methods using a step size of $\Delta x = 1$, $\Delta x = 0.1$, $\Delta x = 0.01$, and $\Delta x = 0.001$. We observe that $f(x) = 14$ when $x = 1$.

Δx	$f(x-\Delta x)$	$f(x+\Delta x)$	**Backward**	**Forward**	**Central**
1	16.000	22.000	-2.000	8.000	3.000
0.1	13.849	14.251	1.510	2.510	2.010
0.01	13.980	14.021	1.950	2.050	2.000
0.001	13.998	14.002	1.995	2.005	2.000

(b) We observed that $f'(x) = 2$ when $x = 1$. The table below gives differences between the numerical and analytical derivatives. We can see that as we increase the step size that the accuracy of the numerical derivative increases also. This is one of the considerations when finding derivative numerically, increasing the step size increases the accuracy but also increase the computation, so there is a trade off in big applications.

Δx	**Backward**	**Forward**	**Central**
1	-4.000	6.000	1.000
0.1	-0.490	0.510	0.010
0.01	-0.050	0.050	0.000
0.001	-0.005	0.005	0.000



2. Introduction to Differentiation

2.1 Overview

Take a look at the graph of the function $f(x) = 2x^3 - 20x^2 + x + 4000$ in Figure 2.1. Clearly it can be seen that the gradient of this line is not constant with x. For example:

1. at $x = -5$ the gradient is quite steep and positive.
2. at $x = 0$ the gradient is almost zero.
3. at $x = 5$ the gradient is fairly shallow and negative.

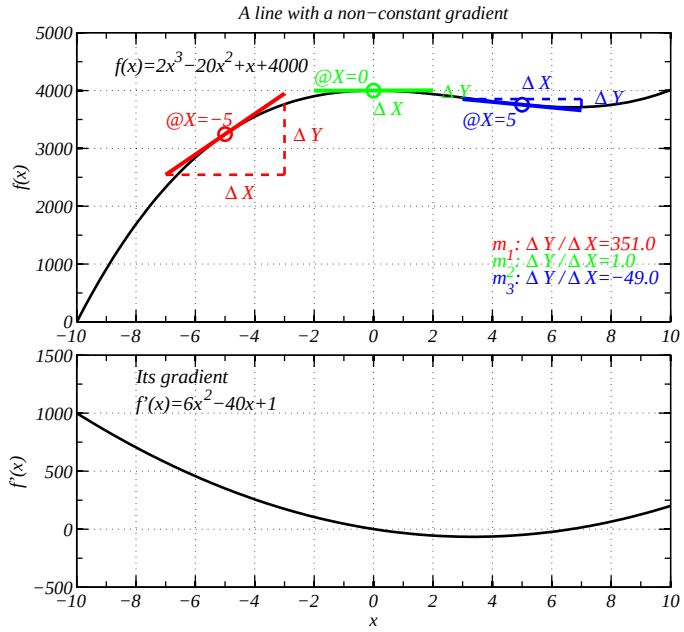


Figure 2.1: Top shows the function $f(x) = 2x^3 - 20x^2 + x * 4000$; bottom shows the gradient $\frac{dy}{dx}$, or $f'(x)$, which is positive until about $x = 0$, negative until about $x = 6$ and positive thereafter.

You could find the gradient at every point by drawing a tangent to the line at that point and determining how much y changes for a given change in x, like we did in Chapter 1. However, we can use **differentiation** to find a function that describes the gradient, which we will call $\frac{dy}{dx}$. In this case the function $6x^2 - 40x + 1$ describes the gradient of $f(x)$. This is shown in the bottom panel of Figure 2.1, labelled as $f'(x)$ which is often used in place of $\frac{dy}{dx}$. Once we have $\frac{dy}{dx}$, we can evaluate it for any value of x to find the gradient.

2.2 Defining differentiation

How can we arrive at the formula used to generate the line in the bottom figure? We use the rules of **differentiation**. These rules are central to the application of calculus. We do not cover the derivation of all rules here; rather we learn how and where to apply them. You should treat these rules as an engineer might treat a screwdriver or hammer. Please keep a copy of your crib sheet nearby throughout the course. You can use this to determine which rule to apply for a given problem that requires differentiation. For now, let us take a look at a straight line and quadratic.

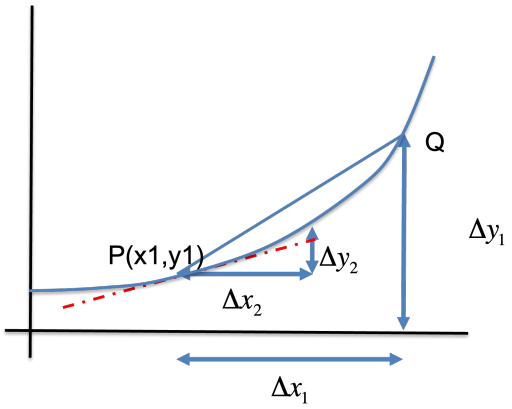


Figure 2.2: A hypothetical curve with a y value that increases non linearly as x increases. The gradient at points P and Q are different.

In Figure 2.2 we find a curved line and compare two positions on that line, namely P and Q. These position have coordinates, or values for x and y as (x, y) and $(x + \Delta x_1, y + \Delta y_1)$ respectively. In other words, position Q is found by movement along from P by an increment given by Δx_1 and Δy_1. We want to derive an expression for $\frac{dy}{dx}$, and we keep referring to approximating this using a finite difference approach (please refer to week 1). However, the choice of the increment used to arrive at this approximation is important. You can tell from our figure that we will arrive at a different gradient if we used a combination of increments Δx_1 and Δy_1 compared with Δx_2 and Δy_2. If this isn't clear, please ask in class. As our increment gets smaller and smaller, we get closer to the accurate value for $\frac{dy}{dx}$.

Indeed, we can write the following equation:

$$\frac{dy}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x}$$

which states that as our increment, Δx tends to 0, we arrive at our derivative value. This gives us a mechanism for deriving a formula for our gradient. For example, take the straight line displayed in Figure 2.3.

Derivative of a straight line

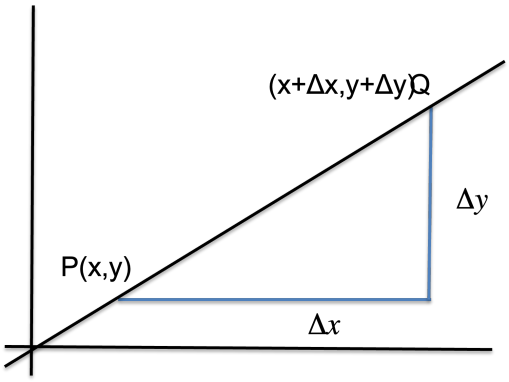


Figure 2.3: A hypothetical curve with a y value that increases linearly as x increases. The gradient at points P and Q are the same so we expect our gradient, thus derivative, to be constant.

The equation of a straight line can be expressed as the following:

$$y = ax$$

where the constant a is our gradient. We know our derivative $\frac{dy}{dx}$ should therefore be equal to a. But can we show this through a derivation? Using the equation of the straight line we can work through the following steps to arrive at an expression for $\frac{\Delta y}{\Delta x}$:

$$\begin{aligned} y + \Delta y &= a\,(x + \Delta x) \\ &= ax + a\Delta x \end{aligned}$$

If we then subtract $y = ax$ from both sides we get :

$$\Delta y = a\Delta x$$

Rearranging this gives us:

$$\frac{\Delta y}{\Delta x} = a$$

If we let $\Delta x \mapsto 0$ then we see that

$$\frac{\Delta y}{\Delta x} \to \frac{dy}{dx} = a$$

So perhaps if we can keep deriving expressions for $\frac{\Delta y}{\Delta x}$ we can see if that expression simplifies as Δx tends to 0 and identify a pattern which we could wrap up in a general rule.

Derivative of a quadratic line

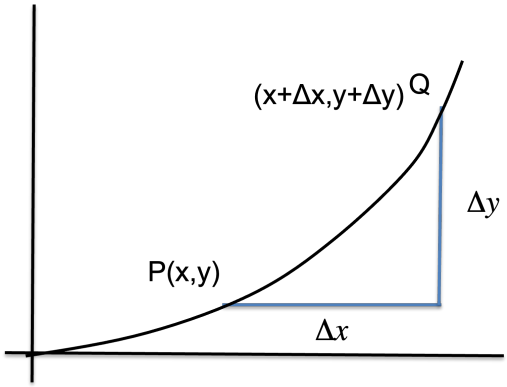


Figure 2.4: A hypothetical curve with a y value that increases quadratically as x increases. The gradient at points P and Q are different.

The equation of a quadratic can be expressed as the following:

$$y = ax^2$$

In Figure 2.4 we again position two points on our curve. Let us derive an expression for $\frac{\Delta y}{\Delta x}$ in moving between these points by substituting the x and y coordinates and isolating expressions for Δx and Δy:

$$\begin{aligned} y + \Delta y &= a(x + \Delta x)^2 \\ &= ax^2 + 2ax\Delta x + \Delta x^2 \end{aligned}$$

Lets subtract $y = ax^2$ from both sides of the equation:

$$\Delta y = 2ax\Delta x + \Delta x^2$$

Now let's divide the equation by Δx:

$$\frac{\Delta y}{\Delta x} = 2ax + \Delta x$$

If we let $\Delta x \mapsto 0$ then we see that

$$\frac{\Delta y}{\Delta x} \to \frac{dy}{dx} = 2ax$$

If we repeat this procedure for higher order polynomials, we calculate that the gradient function for some simple functions of x by **multiplying the original function of x by its power** and **reducing the power that x is raise to by 1**:

$$\begin{aligned} y = x^n &\implies \frac{dy}{dx} = nx^{n-1} \\ y = ax^n &\implies \frac{dy}{dx} = anx^{n-1} \end{aligned}$$

2.3 Common differentials

What if we didn't have a polynomial in x? There are many other types of functions, so we will also need expressions for other common types of functions. Table 2.1 we list all of the common differentials you will use in this course. Learning how to apply them is a key aspect of the problems we will encounter.

Table 2.1: List of common differentials

Function $y = f(x)$	**Derivative** $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

2.4 Differentiating sums of functions

So far we have seen how to differentiate simple functions (see Table 2.1. What happens when we have a sum of functions such as:

$$y = 2x^2 + 3x$$

It turns out that differentiation is invariant to sums of functions. This means that when you differentiate the sum of two or more functions:

$$y = f(x) + g(x)$$

the **derivative of the sum** is equal to the **sum of the derivatives** of the individual functions.

$$\frac{dy}{dx} = \frac{df}{dx} + \frac{dg}{dx}$$

Therefore, using the list of common derivatives, we see that the derivative of $y = 2x^2 + 3x$ is:

$$\frac{dy}{dx} = 2x + 3$$

We'll see what happens when we get more complicated combinations of functions (i.e. functions of functions, products, fractions) in the next Chapter!

Exercises

In the following exercises we can start to practice using them to derive derivative functions and thus calculate gradients. Please read the questions carefully before answering them. Throughout the course questions will be split and laid out as follows:

1. **Algebraic problems**: These questions are what I often call "pure" mathematical problems in the sense that they are not driven by a specific real word application. The functions might represent any given process in the earth system, however. The key point here is to continue practicing methods for differentiating.

2. **Real-world problems**: Here you are given a scenario that requires framing of a specific problem. You wont necessarily be asked to derive long derivatives here, but work with expressions of derivatives to get a better understanding of their role.

I would recommend you first try them without looking at crib sheets, but go back to that when you encounter any problems. For those who have not encountered the $\sum$ operator before, this represents a summation over the function given to the right hand side. For example, $\sum_{x=1}^{4} 3x$ is saying we need to sum the values obtained from from x=1 to x=4, stepping over increments of 1. In the problem provided below, try expanding the expression for a number of values before re-expressing your answer using the $\sum$ operator. Also note that an expression that uses $\sqrt{}$ as the square root operator can be re-expressed using the power $\frac{1}{2}$ (i.e. $\sqrt{x} = x^{\frac{1}{2}}$).

Algebraic problems

Question 2.1: Differentiate the following functions with respect to x:

(a) $f(x) = 5x^2$

(b) $g(x) = 10x^3$

(c) $h(x) = 11x^1$

Question 2.2: Differentiate the following polynomials with respect to x:

(a) $f(x) = 5x^3 + 2x^2 + 3x + 5$

(b) $g(x) = 10x^3 + x + 5$

(c) $h(x) = 11x^5 + 2x^4 + 2x^3 + x^2 + 3x + 2$

Question 2.3: Find the gradient of the following functions at the values of x specified:

(a) $f(x) = x^2 - 3x$ at $x = 10$

(b) $f(x) = 5x^5 - 2x$ at $x = 0$

(c) $f(x) = 10x - 3x^7$ at $x = -2$

Question 2.4: What function when differentiated gives the following [Hint: you will need to do the process of differentiation in reverse]:

(a) $\frac{dy}{dx} = 2x$
(b) $\frac{dy}{dx} = -3x^3$
(c) $\frac{dy}{dx} = 7$

Question 2.5: Obtain the derivative of the following functions:

(a) $y = 10x(5x + 3x)$
(b) $y = 50x^2 + 30x^2$
(c) $y = 3ce^{ix}$
(d) $y = \sqrt{x}$
(e) $y = \cos(x)$
(f) $y = e^{5x}$
(g) $y = 3x + 5y$
(h) $y = 2ln(x)$

Question 2.6: Obtain the derivative of the following (slightly harder) functions:

(a) $y = \sum_i 3ce^{ix}$
(b) $y = \sqrt{\frac{4x}{\pi}}$
(c) $y = \cos(x) + \sin(x)$
(d) $y = e^{5x+4}$
(e) $y = ln(2x)$
(f) $y = 4x^4 + e^x + \sin(x)$

Real-world problems

Question 2.7: The equation describing radioactive decay is:

$$\frac{dN}{dt} = -\lambda N \tag{2.1}$$

where N is the number of unstable nuclei, t is time and λ is called the decay constant. $\lambda = \frac{\ln 2}{t_{1/2}}$ where $t_{1/2}$ is the half-life of the sample.

The solution to equation 2.1 is

$$N(t) = N_0 e^{-\lambda t} \tag{2.2}$$

where N_0 is the starting number of unstable nuclei (a constant).

In radioactive decay the number of unstable nuclei reduces with time. After a period of time known as the half-life. $t_{1/2}$, we have half the initial number of nuclei, $\frac{N_0}{2}$ due to radioactive decay of the other nuclei. After another half-life we will have half those number of nuclei again (i.e. $\frac{N_0}{4}$).

This behaviour can also be described using:

$$N\left(t_{1/2}\right) = N_0 \times 2^{-n} \tag{2.3}$$

where n is the number of half-lifes that have elapsed.

(a) Differentiate Equation 2.2 and substitute to show that it satisfies Equation 2.1.

(b) A radioactive material has a half-life of $t_{1/2} = 30$ years and there are $10,000$ nuclei in the sample. Using Equation 2.3, calculate how many nuclei there are after 90 years (i.e. 3 half-lifes).

(c) Repeat the calculation using Equation 2.2.

Question 2.8: The hydrostatic equation describes how pressure changes with altitude in relation to other variables such as gravity and temperature. It is a fundamental principle in fluid mechanics and atmospheric science. We will use this a lot more later, but it has a similar formulation to the radioactive decay equation and is given (here) by

$$\frac{dP}{dz} = -\frac{P}{R_a T} g$$

where g, R_a and T are assumed to remain constant. Show that the following equation:

$$P(z) = P_0 \exp\left(-\frac{g}{R_a T} z\right)$$

is a solution to the differential equation by differentiating and substituting it back into the differential equation.

Question 2.9: Imagine we are studying the volume of water in an ecosystem reservoir over time.

(a) If we say that the volume of water in the system can be described as:

$$V(t) = V_0 - (V_0)\frac{t}{25}$$

what is the drainage rate? Where V_0 is the volume of water in our system when $t = 0$, in m^3. (take $V_0 = 1 \times 10^{-2}$)

(b) In reality the water would drain at a non-constant rate. If we say that the volume of water in the reservoir can be described as:

$$V(t) = V_0 - (V_0)\frac{t^2}{625000}$$

what is the drainage rate at time 600s? Use the process of differentiation to determine the gradient analytically, but also compare to a gradient calculated numerically using a finite difference method with a time step Δt of 20 seconds, 10 seconds either side of 600s.

(c) If we say that the volume of water in the sink can be described as:

$$V(t) = V_0 - (V_0)\frac{t^3}{312500000}$$

what is the drainage rate at time 600s? Use the process of differentiation to determine the gradient analytically, but also compare to a gradient calculated numerically using the finite difference method with a time step Δt of 20 seconds, 10 seconds either side of 600s.

(d) If we say that the volume of water in the sink can be described as:

$$V(t) = V_0 \exp\left(-\frac{t}{300}\right)$$

what is the drainage rate at time 600 s? Use the process of differentiation to determine the gradient analytically, but also compare to a gradient calculated numerically using the finite difference method with a time step Δt of 20 seconds, 10 seconds either side of 600s.

Solutions

Algebraic problems

Question 2.1: Differentiate the following functions with respect to x:

(a) $f(x) = 5x^2 \implies \frac{df}{dx} = 10x$
(b) $g(x) = 10x^3 \implies \frac{dg}{dx} = 30x^2$
(c) $h(x) = 11x^1 \implies \frac{dh}{dx} = 11$

Question 2.2: Differentiate the following polynomials with respect to x:

(a) $f(x) = 5x^3 + 2x^2 + 3x + 5 \implies \frac{df}{dx} = 15x^2 + 4x + 3$
(b) $g(x) = 10x^3 + x + 5 \implies \frac{dg}{dx} = 30x^2 + 1$
(c) $h(x) = 11x^5 + 2x^4 + 2x^3 + x^2 + 3x + 2 \implies \frac{dh}{dx} = 55x^4 + 8x^3 + 6x^2 + 2x + 3$

Question 2.3: Find the gradient of the following functions at the values of x specified:

(a) $f(x) = x^2 - 3x$ at $x = 10 \implies \frac{df}{dx} = 2x - 3, \ \frac{df}{dx}(10) = 17$
(b) $f(x) = 5x^5 - 2x$ at $x = 0 \implies \frac{df}{dx} = 15x^4 - 2, \ \frac{df}{dx}(0) = -2$
(c) $f(x) = 10x - 3x^7$ at $x = -2 \implies \frac{df}{dx} = 10 - 21x^6 \ \frac{df}{dx}(-2) = -1334$

Question 2.4: What function when differentiated gives the following [Hint: you will need to do the process of differentiation in reverse]:

(a) $\frac{dy}{dx} = 2x \implies y = x^2$
(b) $\frac{dy}{dx} = -3x^3 \implies y = -\frac{3}{4}x^4$
(c) $\frac{dy}{dx} = 7 \implies y = 7x + d$

[d is a constant. Why do you think we need this? Ask in class if this is unclear]

Question 2.5: Obtain the derivative of the following functions:

(a) $y = 10x(5x + 3x) = 80x^2 \implies \frac{dy}{dx} = 160x$
(b) $y = 50x^2 + 30x^2 = 80x^2 \implies \frac{dy}{dx} = 160x$
(c) $y = 3ce^{ix} \implies \frac{dy}{dx} = 3cie^{ix}$
(d) $y = \sqrt{x} = x^{\frac{1}{2}} \implies \frac{dy}{dx} = \frac{1}{2}x^{-\frac{1}{2}} = 1/2\sqrt{x}$
(e) $y = \cos(x) \implies \frac{dy}{dx} = -\sin(x)$
(f) $y = e^{5x} \implies \frac{dy}{dx} = 5e^{5x}$
(g) $y = 3x + 5y \implies y = -\frac{3x}{4} \implies \frac{dy}{dx} = -\frac{3}{4}$
(h) $y = 2ln(x) \implies \frac{dy}{dx} = 2/x$

Question 2.6: Obtain the derivative of the following (slightly harder) functions:

(a) $y = \sum_i 3ce^{ix} \implies \frac{dy}{dx} = 3cie^{ix}$ (Remember the differential of a sum is the sum of the

differential. See Section 2.4)

(b) $y = \sqrt{\frac{4x}{\pi}} = \frac{2}{\sqrt{\pi}}\sqrt{x} \implies \frac{dy}{dx} = \frac{2}{\sqrt{\pi}}\frac{1}{2\sqrt{x}} = \sqrt{\frac{1}{\pi x}}$

(c) $y = \cos(x) + \sin(x) \implies \frac{dy}{dx} = -\cos(x) + \sin(x)$

(d) $y = e^{5x+4} \implies \frac{dy}{dx} = 5e^{5x+4}$

(e) $y = ln(2x) = ln(2) + ln(x) \implies \frac{dy}{dx} = \frac{1}{x}$

(f) $y = 4x^4 + e^x + \sin(x) \implies \frac{dy}{dx} = 8x^4 + e^x + \cos(x)$

Question 2.7:

(a) Differentiating Equation 2.2 gives us:

$$N(t) = N_0 e^{-\lambda t} \quad \implies \quad \frac{dN}{dt} = -N_0 \lambda e^{-\lambda t}$$

substituting this in Equation 2.1 gives us

$$-N_0 \lambda e^{-\lambda t} = -\lambda N_0 e^{-\lambda t}$$

Alternatively, arranging the equation we see that:

$$\begin{aligned} \frac{dN}{dt} &= -\lambda \underbrace{N_0 e^{-\lambda t}}_{N} \\ &= -\lambda N \end{aligned}$$

Therefore Equation 2.2 is indeed the solution to the radioactive decay equation!

(b) A radioactive material has a half-life of $t_{1/2} = 30$ years and there are $10,000$ nuclei in the sample, so at 90 years, the number of nuclei is

$$\begin{aligned} N(90) &= N_0 \times 2^{-n} \\ &= 10000 \times 2^{-3} \\ &= \frac{10000}{8} \\ &= 1250 \end{aligned}$$

There are 1250 nuclei remaining after 90 years.

(c) A radioactive material has a half-life of $t_{1/2} = 30$ years we see that the decay parameter is:

$$\begin{aligned} \lambda &= \frac{\ln 2}{t_{1/2}} \\ &= \frac{\ln 2}{30} \end{aligned}$$

So after 90 years we see that

$$\begin{aligned} N(90) &= N_0 \times \exp(\lambda \times 90) \\ &= 10000 \times \exp\left(\frac{\ln 2}{30} \times 90\right) \\ &= 10000 \times \exp(3 \ln 2) \\ &= 1250 \end{aligned}$$

There are 1250 nuclei remaining after 90 years (i.e. the same).

Question 2.8: Differentiating $P(z)$ with respect to z gives us:

$$P(z) = P_0 \exp\left(-\frac{g}{R_a T} z\right) \quad \Longrightarrow \quad \frac{dP}{dz} = -P_0 \frac{g}{R_a T} \exp\left(-\frac{g}{R_a T} z\right)$$

substituting these in Equation 2.1 gives us

$$-P_0 \frac{g}{R_a T} \exp\left(-\frac{g}{R_a T} z\right) = -P_0 \frac{g}{R_a T} \exp\left(-\frac{g}{R_a T} z\right)$$

Alternatively, arranging the equation we see that:

$$\begin{aligned} \frac{dP}{dz} &= -\frac{g}{R_a T} \underbrace{-P_0 \exp\left(-\frac{g}{R_a T} z\right)}_{P} \\ &= -\frac{P}{R_a T} g \end{aligned}$$

Therefore $P(z)$ is indeed the solution to the hydrostatic equation!

Question 2.9:

(a) Differentiating with respect to t to get the (analytical) drainage rate:

$$V(t) = V_0 - (V_0)\frac{t}{25} \quad \Longrightarrow \quad \frac{dV}{dt} = -\frac{V_0}{25}$$

We can see that the drainage rate is constant here.

(b) Differentiating with respect to t to get the (analytical) drainage rate:

$$V(t) = V_0 - (V_0)\frac{t^2}{625000} \quad \Longrightarrow \quad \frac{dV}{dt} = - - (V_0)\frac{2t}{625000}$$

At time $t = 600$, the drainage rate is:

$$\frac{dV}{dt}(600) = -10^{-2} \times \frac{2 \times 600}{625000} = -1.92 \times 10^{-5}$$

Using the central differences method:

$$V(590) = 10^{-2} - 10^{-2}\frac{590^2}{625000} = 0.00443$$
$$V(610) = 10^{-2} - 10^{-2}\frac{590^2}{625000} = 0.00405$$
$$\frac{\Delta V}{\Delta t} = \frac{V(610) - V(590)}{610 - 590}$$
$$= \frac{0.00405 - 0.00443}{20} = -1.90 \times 10^{-5}$$

(c) Differentiating with respect to t to get the (analytical) drainage rate:

$$V(t) = V_0 - (V_0)\frac{t^3}{312500000} \quad \Longrightarrow \quad \frac{dV}{dt} = -(V_0)\frac{3t^2}{312500000}$$

At time $t = 600$, the drainage rate is:

$$\frac{dV}{dt}(600) = -10^{-2} \times \frac{3 \times 600^2}{312500000} = -3.46 \times 10^{-5}$$

Using the central differences method:

$$V(590) = 10^{-2} - 10^{-2}\frac{590^3}{312500000} = 0.003428$$
$$V(610) = 10^{-2} - 10^{-2}\frac{590^3}{312500000} = 0.002737$$
$$\frac{\Delta V}{\Delta t} = \frac{V(610) - V(590)}{610 - 590}$$
$$= \frac{0.002737 - 0.003428}{20} = -3.46 \times 10^{-5}$$

(d) Differentiating with respect to t to get the (analytical) drainage rate:

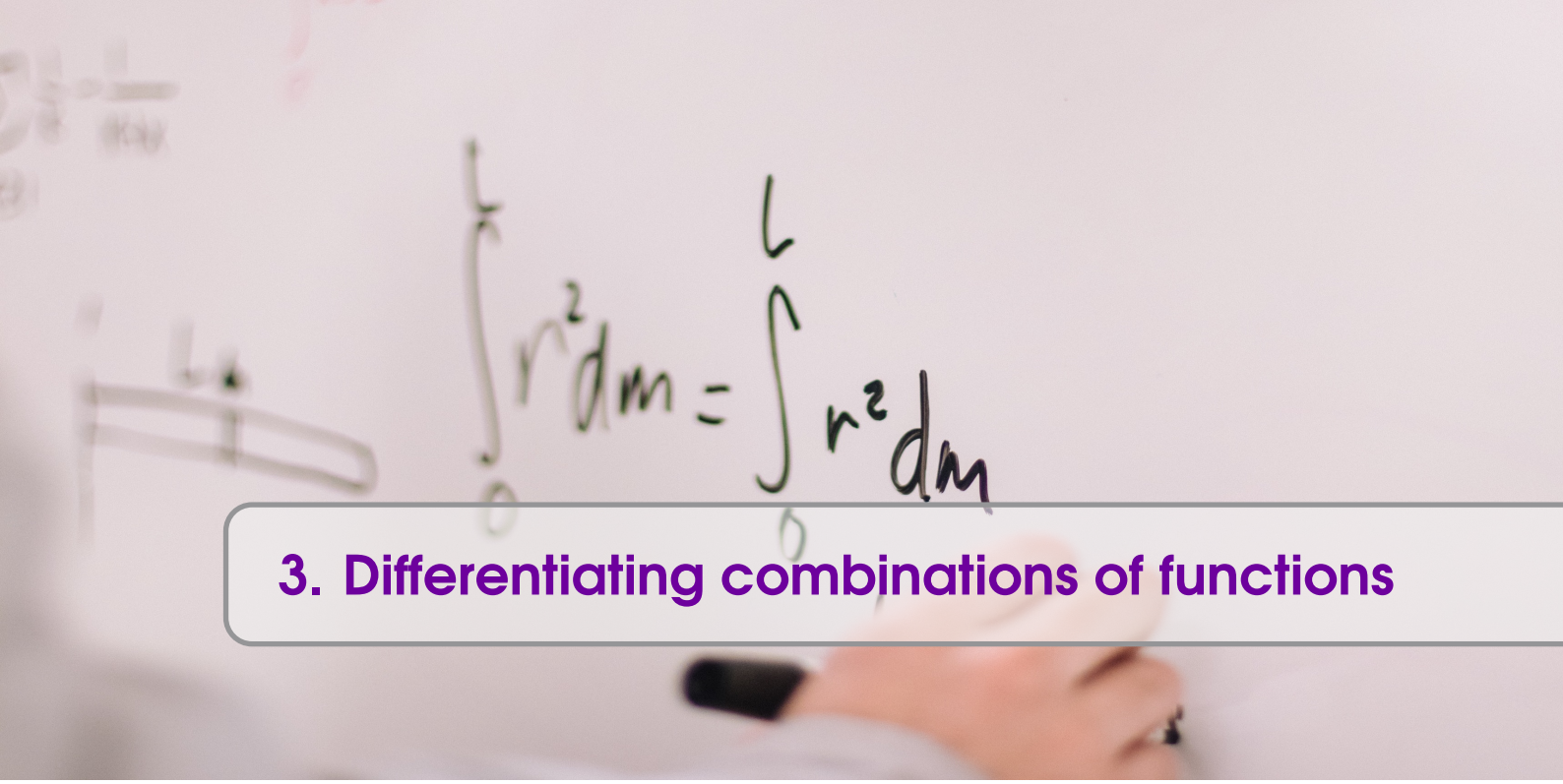
$$V(t) = V_0 \exp\left(-\frac{t}{300}\right) \quad \Longrightarrow \quad \frac{dV}{dt} = -\frac{V_0}{300}\exp\left(-\frac{t}{300}\right)$$

At time $t = 600$, the drainage rate is:

$$\frac{dV}{dt}(600) = -\frac{10^{-2}}{300}\exp\left(-\frac{600}{300}\right) = -4.5 \times 10^{-6}$$

Using the central differences method:

$$V(590) = 10^{-2}\exp\left(-\frac{600}{300}\right) = 0.001399$$
$$V(610) = 10^{-2}\exp\left(-\frac{600}{300}\right) = 0.001309$$
$$\frac{\Delta V}{\Delta t} = \frac{V(610) - V(590)}{610 - 590}$$
$$= \frac{0.001309 - 0.001399}{20} = -4.5 \times 10^{-6}$$



3. Differentiating combinations of functions

3.1 Overview

We saw Chapter 2 that for a sum of functions, the **derivative of the sum** is equal to the **sum of the derivatives** of the individual functions.

$$y = u(x) + v(x) \quad \Longrightarrow \quad \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

However, in many cases we are faced with more complicated combinations of functions such as **functions of functions**

$$y = u(v(x))$$

or **products of functions**

$$y = u(x)v(x)$$

or **quotients of functions**

$$y = \frac{u(x)}{v(x)}$$

and the basic laws of calculus can no longer apply to find the derivative of y. Another scenario arises where we wish to obtain a derivative of a function that does not fit under any of the criteria we use in our common rules of differentiation. In other words it does not look like we can obtain an analytical derivative and we may need to use a numerical approach (like we saw in Chapter 1). We will now look at expanding the laws of differentiation to deal with these new types of functions.

3.2 Functions of functions

Lets first consider the case when we have **functions of functions** such as:

$$y = u(v(x))$$

An example of this would be $y = \ln(2x)$ where we can split this into two functions, $u(x) = \ln(x)$ and $v(x) = 2x$. How do we obtain the derivative $\frac{dy}{dx}$? Luckily, there are properties of derivatives that we can use to slightly modify

$$\frac{dh}{dx}$$

and make it look like things we have seen before and differentiate as normal. This is called the chain rule and is best illustrated with an example. One of our common rules of differentiation shows us that the derivative function of the natural logarithm (i.e. $y = \ln(x)$ is given as follows:

$$\frac{dy}{dx} = \frac{d}{dx}(\ln(x)) = \frac{1}{x}$$

But what if our function looked slightly different? Let's return to $y = \ln(2x)$. What is the derivative?

$$\frac{dh}{dx} = \frac{d}{dx}(\ln(2x)) = ??$$

It is tempting to assume the answer would be $\frac{dy}{dx} = \frac{1}{2x}$ wouldn't it? We know that this is wrong since we can rewrite $h(x) = \ln(2x) = \ln(2) + \ln(x)$ and therefore we see that

$$y = \ln(2x) \quad \Longrightarrow \quad \frac{dy}{dx} = \frac{1}{x}$$

Fortunately, we have the **chain rule**. We can re-write the derivative of a function of a function such that it resembles a form we can derive an analytical derivative from. In this case, what if we assumed we could replace the expression $2x$ by a new variable z. Our expression now reads:

$$\frac{dy}{dx} = \frac{d}{dx}(\ln(z)) = \frac{1}{z}$$

We are not quite done, however. The chain rule allows us to treat derivatives in the same way we can multiply fractions. We **chain** together operations. Formally, it states that to get the derivative of a function, f, which can be written as a function of another function, $g(x)$ you differentiate the function, f, wrt the function g and multiply the result by the derivative of g with respect to x:

$$\frac{d}{dx}(f(g(x))) = \frac{df}{dg} \times \frac{dg}{dx} \tag{3.1}$$

This tells us that we first obtain the analytical derivative of our function f with respect to a new variable g and then multiply the derivative of our new variable g with respect to our original variable x. Can you see how the terms dg cancel out?

Example 3.1: Differentiate $y = (3x+1)^2$. We could expand the bracket and differentiate as we have seen before:

$$y = 9x^2 + 6x + 1 \quad \Longrightarrow \quad \frac{dy}{dx} = 18x + 6$$

or we could use *chain rule*. Let

$$u(v) = v^2 \quad \text{and} \quad v(x) = 3x + 1$$

and differentiate as usual

$$\frac{du}{dv} = 2v \quad \text{and} \quad \frac{dv}{dx} = 3$$

Therefore using the chain rule formula and substitution the solution is:

$$\begin{aligned} \frac{dy}{dx} &= 2\,(3x+1) \times 3 \\ &= 18x + 6 \end{aligned}$$

which is what we expected!

Example 3.2: Differentiate $y = \sqrt{13x^2 - 6x + 8}$. There is no obvious way to differentiate this using methods we have used before, we must use *chain rule*. Let

$$u(v) = \sqrt{v} = v^{0.5} \quad \text{and} \quad v(x) = 13x^2 - 6x + 8$$

and differentiate as usual

$$\frac{du}{dv} = 0.5v^{-0.5} = \frac{1}{2\sqrt{v}} \quad \text{and} \quad \frac{dv}{dx} = 26x - 6$$

Therefore using the chain rule formula and substitution the solution is:

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2} \times \frac{1}{\sqrt{13x^2 - 5x + 8}} \times (26x - 6) \\ &= \frac{13x - 3}{\sqrt{13x^2 - 6x + 8}} \end{aligned}$$

3.3 Products of functions

In many cases we are faced with the **product of two, or more, functions**. We might described a general case where we have two functions multiple as :

$$y = u(x) \times v(x)$$

where u is a function that varies only with respect to variable x and v is also a function that varies only with respect to x. How do we obtain the derivative $\frac{dy}{dx}$? Using the tools we have met so far, then we might first attempt to expand terms and separate variables. But what if we decided that the derivative could be calculated by simply multiplying the associated derivatives from each function. So, we might assume we can use the formula:

$$\frac{dy}{dx} = \frac{du}{dx} \times \frac{dv}{dx} \tag{3.2}$$

This would be useful if our functions were somewhat complex but could be treated in isolation i.e. we did not want to expand all of the terms. Let us look at a relatively simple example to see which is a valid approach, one where we can access the expression $u(x)v(x)$. If we assume $u(x) = x^3$ and $v(x) = x^6$ then we know that:

$$\begin{aligned} y &= u(x) \times v(x) \\ &= x^3 \times x^6 \\ &= x^9 \end{aligned}$$

From this we can obtain the derivative as follows:

$$\frac{dy}{dx} = \frac{d}{dx}(x^9) = 9x^8$$

But what if we had chosen to use the approach written in align 3.2. We would arrive at the following formula:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dx} \times \frac{dv}{dx} \\ &= \frac{d}{dx}(x^3) \times \frac{d}{dx}(x^6) \\ &= 3x^2 \times 6x^5 \\ &= 18x^7 \end{aligned}$$

which is wrong! We saw that it is wrong as our expansion leads to a derivative of $9x^8$. So here we can see that we have to be careful when working with products of functions.

Fortunately, in cases where we cant easily expand our expression, there is the **product rule** which states that to get the derivative of a function, y, which can be written as the *product* of two functions, u and v we calculate $\frac{du}{dx}$ and $\frac{dv}{dx}$ and calculate:

$$\frac{dy}{dx} = u \times \frac{dv}{dx} + v \times \frac{du}{dx} \tag{3.3}$$

(3.4)

Example 3.3: Differentiate $y = (3x+1)^2$. We could expand the bracket and differentiate as we have seen before:

$$y = 9x^2 + 6x + 1 \quad \Longrightarrow \quad \frac{dy}{dx} = 18x + 6$$

or we could use *product rule*. Let

$$u(x) = 3x + 1 \quad \text{and} \quad v(x) = 3x + 1$$

and differentiate as usual

$$\frac{du}{dx} = 3 \quad \text{and} \quad \frac{dv}{dx} = 3$$

Therefore using the product rule formula the solution is:

$$\begin{aligned} \frac{dy}{dx} &= (3x+1) \times 3 + (3x+1) \times 3 \\ &= 18x + 6 \end{aligned}$$

which is what we expected!

Example 3.4: Differentiate $y = x\exp(-3x)$. There is no obvious way to differentiate this using methods we have used before, we must use *product rule*. Let

$$u(x) = x \quad \text{and} \quad v(x) = \exp(-3x)$$

and differentiate as usual

$$\frac{du}{dx} = 1 \quad \text{and} \quad \frac{dv}{dx} = -3\exp(-3x)$$

Therefore using the product rule formula the solution is:

$$\begin{aligned} \frac{dy}{dx} &= x \times -3\exp(-3x) + \exp(-3x) \times 1 \\ &= (1 - 3x)\exp(-3x) \end{aligned}$$

3.4 Quotients of functions

What if we **divided one function by another**? We find similar problems. We might described a general case where we have two functions multiple as :

$$y = \frac{u(x)}{v(x)}$$

where u is a function that varies only with respect to variable x and v is also a function that varies only with respect to x. How do we obtain the derivative $\frac{dy}{dx}$? Using the tools

we have met so far, then we might first attempt to simplify into one expression. But what if we decided that the derivative could be calculated by simply dividing the associated derivatives from each function? We might assume we can use the formula:

$$\frac{dy}{dx} = \frac{du}{dx} / \frac{dv}{dx} \tag{3.5}$$

Again, this would be useful if our functions were somewhat complex but could be treated in isolation i.e. we did not want to expand and simplify all of the terms. Let us look at a relatively simple example to see which is a valid approach, one where we can access the expression $u(x)v(x)$. If we assume $u(x) = x^3$ and $v(x) = x^6$ then we know that:

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{x^3}{x^6}\right)$$

Once again we can rearrange this expression and arrive at the correct derivative through the following:

$$\frac{dy}{dx} = \frac{d}{dx}\left(x^{-3}\right) = -3x^{-4}$$

But what if we had chosen to use the approach written in align 3.5? We would arrive at the following formula:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{du}{dx} / \frac{dv}{dx} \\
&= \frac{d}{dx}(x^3) / \frac{d}{dx}(x^6) \\
&= \frac{3x^2}{6x^5} \\
&= \frac{1}{2x^3}
\end{aligned}$$

which is wrong! We saw that it is wrong as our expansion leads to a derivative of $-3x^{-4}$. So here we can see that we have to be careful when working with fractions of functions.

Fortunately, in cases where we cant easily expand and simplify our expression, there is the **quotient rule** which states that to get the derivative of a function, y, which can be written as the *quotient* of two functions, u and v we calculate $\frac{du}{dx}$ and $\frac{dv}{dx}$ and calculate:

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2} \tag{3.6}$$

Example 3.5: Differentiate $y = \frac{3x+1}{x^2}$. We could split this up into two fractions and differentiate,

$$y = \frac{3x+1}{x^2} = \frac{3}{x} + \frac{1}{x^2} \quad \Longrightarrow \quad \frac{dy}{dx} = -3x^{-2} - 2x^{-3} = \frac{-3x-2}{x^3}$$

or we could use *quotient rule*. Let

$$u(x) = 3x+1 \quad \text{and} \quad v(x) = x^2$$

and differentiate as usual

$$\frac{du}{dx} = 3 \quad \text{and} \quad \frac{dv}{dx} = 2x$$

Therefore using the quotient rule formula the solution is:

$$\begin{aligned}\frac{dy}{dx} &= \frac{x^2 \times 3 - (3x+1) \times 2x}{x^4} \\ &= \frac{-3x^3 - 2x}{x^4} \\ &= \frac{-3x-2}{x^3}\end{aligned}$$

which is what we expected!

Example 3.6: Differentiate $y = \frac{3x+1}{x^2+1}$. There is no obvious way to differentiate this using methods we have used before. We must use *Quotient Rule*. Let

$$u(x) = 3x+1 \quad \text{and} \quad v(x) = x^2+1$$

and differentiate as usual

$$\frac{du}{dx} = 3 \quad \text{and} \quad \frac{dv}{dx} = 2x$$

Therefore using the product rule formula the solution is:

$$\begin{aligned}\frac{dy}{dx} &= \frac{(x^2+1) \times 3 - (3x+1) \times 2x}{(x^2+1)^2} \\ &= -\frac{3x^2+2x-3}{(x^2+1)^2}\end{aligned}$$

Exercises

Algebraic problems

Question 3.1: Use the chain rule to differentiate the following functions:

(a) $y = \exp(x^2 + 5x + 2)$

(b) $y = \cos(3x + 5)$

(c) $y = \sin(x)^2$

Question 3.2: Use the product rule to differentiate the following functions:

(a) $y = x\exp(x)$

(b) $y = x^2\exp(4x)$

(c) $y = (x^2 + 9)\exp(-x)$

Question 3.3: Use the quotient rule to differentiate the following functions:

(a) $y = \frac{\exp(-x)}{x}$

(b) $y = \frac{x}{x^2+25}$

(c) $y = \frac{x^2}{x^3+25x}$

Question 3.4: Use the chain rule for (a) and the result of (a) in (b)-(e) find $\frac{dy}{dx}$ of the following equations (Hint: you may need to use a mixture of chain, product and quotient rules):

(a) $y = \sqrt{x^2 + 25}$

(b) $y = \sin\sqrt{x^2 + 25}$

(c) $y = x\sqrt{x^2 + 25}$

(d) $y = \frac{x}{\sqrt{x^2+25}}$

Question 3.5: Find where the following functions are decreasing and increasing. These problems require a number of steps. We know that where the derivative is 0 it is likely the function will be changing direction either side. We will come across the important property a zero derivative has soon. But for now, once you find where the derivatives are 0 you can then explore regions either side to determine if the gradient is positive or negative.

(a) $V(t) = (4 - t^2)(1 + 5t^2)$

(b) $f(x) = \frac{x-x^2}{1+8x^2}$

Question 3.6: Use the chain, product or quotient rules to find the derivative of the given function.

(a) $Z(s) = (\sin(x^2))^2$

(b) $S(x) = (1 + \sqrt{x^3})(x^{-3} - 2\sqrt[3]{x})$
(c) $R(w) = \frac{3w+w^4}{2w^2+1}$

Real-world problems

Question 3.7: Imagine we are studying the volume of water in an ecosystem reservoir over time (in hours).

(a) If we say that the volume of water in the sink can be described as:

$$V(t) = V_0 \exp\left(-\frac{t^2}{600}\right)$$

Where V_0 is the volume of water in our system when $t = 0$, in m^3. Use the chain rule to determine the gradient analytically. What is the drainage rate after 10 hours when $V_0 = 10$?

(b) If we say that the volume of water in the sink can be described as:

$$V(t) = \frac{V_0(t+10)}{10} \exp\left(-\frac{t}{5}\right)$$

Where V_0 is the volume of water in our system when $t = 0$, in m^3. Use the product rule to determine the gradient analytically. What is the drainage rate after 10 hours when $V_0 = 10$?

(c) If we say that the volume of water in the sink can be described as:

$$V(t) = \frac{V_0(x+10)}{(x^2+10)}$$

Where V_0 is the volume of water in our system when $t = 0$, in m^3. Use the quotient rule to determine the gradient analytically. What is the drainage rate after 10 hours when $V_0 = 10$?

Solutions

Question 3.1: Use the chain rule to differentiate the following functions:

(a) Split the function and differentiate:

$$\begin{aligned} u(v) = \exp(v), \quad &\Longrightarrow \quad \frac{du}{dv} = \exp(v), \\ v(x) = x^2 + 5x + 2 \quad &\Longrightarrow \quad \frac{dv}{dx} = 2x + 5 \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= (2x+5)\exp(x^2+5x+2) \end{aligned}$$

(b) Split the function and differentiate:

$$\begin{aligned} u(v) = \cos(v), \quad &\Longrightarrow \quad \frac{du}{dv} = -\sin(v), \\ v(x) = 3x + 5 \quad &\Longrightarrow \quad \frac{dv}{dx} = 3 \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= -3\sin(3x+5) \end{aligned}$$

(c) Split the function and differentiate:

$$\begin{aligned} u(v) = v^2, \quad &\Longrightarrow \quad \frac{du}{dv} = 2v, \\ v(x) = \sin(x) \quad &\Longrightarrow \quad \frac{dv}{dx} = \cos(x) \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= 2\sin(x)\cos(x) \end{aligned}$$

Question 3.2: Use the product rule to differentiate the following functions:

(a) Split the function and differentiate:

$$\begin{aligned} u(x) = x, \quad &\Longrightarrow \quad \frac{du}{dx} = 1, \\ v(x) = \exp(x) \quad &\Longrightarrow \quad \frac{dv}{dx} = \exp(x) \end{aligned}$$

Apply the product rule:

$$\begin{aligned}\frac{dy}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= \exp(x) + x\exp(x) \\ &= (1+x)\exp(x)\end{aligned}$$

(b) Split the function and differentiate:

$$\begin{aligned} u(x) = x^2, \quad &\Longrightarrow \quad \frac{du}{dx} = 2x, \\ v(x) = \exp(4x) \quad &\Longrightarrow \quad \frac{dv}{dx} = 4\exp(4x)\end{aligned}$$

Apply the product rule:

$$\begin{aligned}\frac{dy}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= 4x^2\exp(4x) + 2x\exp(4x) \\ &= (2x+4x^2)\exp(4x)\end{aligned}$$

(c) Split the function and differentiate:

$$\begin{aligned} u(x) = x^2+9, \quad &\Longrightarrow \quad \frac{du}{dx} = 2x, \\ v(x) = \exp(-x) \quad &\Longrightarrow \quad \frac{dv}{dx} = -\exp(-x)\end{aligned}$$

Apply the product rule:

$$\begin{aligned}\frac{dy}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= 2x\exp(-x) - (x^2+9)\exp(-x) \\ &= -(x^2-2x+9)\exp(-x)\end{aligned}$$

Question 3.3: Use the quotient rule to differentiate the following functions:

(a) Split the function and differentiate:

$$\begin{aligned} u(x) = \exp(-x), \quad &\Longrightarrow \quad \frac{du}{dx} = -\exp(-x), \\ v(x) = x \quad &\Longrightarrow \quad \frac{dv}{dx} = 1\end{aligned}$$

Apply the quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{v \times \frac{du}{dx} - u \times \frac{dv}{dx}}{v^2} \\ &= \frac{-x\exp(-x) - \exp(-x)}{x^2} \\ &= -\frac{(1+x)\exp(-x)}{x^2}\end{aligned}$$

(b) Split the function and differentiate:

$$\begin{aligned} u(x) = x, \quad &\Longrightarrow \quad \frac{du}{dx} = 1, \\ v(x) = x^2 + 25 \quad &\Longrightarrow \quad \frac{dv}{dx} = 2x \end{aligned}$$

Apply the quotient rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{v \times \frac{du}{dx} - u \times \frac{dv}{dx}}{v^2} \\ &= \frac{x^2 + 25 - 2x^2}{(x^2+25)^2} \\ &= \frac{25 - x^2}{(x^2+25)^2} \end{aligned}$$

(c) Split the function and differentiate:

$$\begin{aligned} u(x) = x^2, \quad &\Longrightarrow \quad \frac{du}{dx} = 2x, \\ v(x) = x^3 + 25x \quad &\Longrightarrow \quad \frac{dv}{dx} = 3x^2 + 25 \end{aligned}$$

Apply the quotient rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{v \times \frac{du}{dx} - u \times \frac{dv}{dx}}{v^2} \\ &= \frac{2x(x^3+25x) - x^2(3x^2+25)}{(x^3+25x)^2} \\ &= \frac{2x^4 + 50x^2 - 3x^4 - 25x^2}{(x^3+25x)^2} \\ &= \frac{25x^2 - x^4}{(x^3+25x)^2} \end{aligned}$$

Question 3.4: Use the chain rule for (a) and the result of (a) in (b)-(e) find $\frac{dy}{dx}$ of the following equations (Hint: you may need to use a mixture of chain, product and quotient rules):

(a) Split the function and differentiate:

$$\begin{aligned} u(v) = \sqrt{v}, \quad &\Longrightarrow \quad \frac{du}{dv} = \frac{1}{2\sqrt{v}}, \\ v(x) = x^2 + 25 \quad &\Longrightarrow \quad \frac{dv}{dx} = 2x \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= \frac{2x}{2\sqrt{x^2+25}} \\ &= \frac{x}{\sqrt{x^2+25}} \end{aligned}$$

(b) Split the function and differentiate:

$$\begin{aligned} u(v) = \sin(v), \quad &\Longrightarrow \quad \frac{du}{dx} = \cos(v), \\ v(x) = \sqrt{x^2+25} \quad &\Longrightarrow \quad \frac{dv}{dx} = \frac{x}{\sqrt{x^2+25}} \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= \frac{x}{\sqrt{x^2+25}} \cos(\sqrt{x^2+25}) \end{aligned}$$

(c) Split the function and differentiate:

$$\begin{aligned} u(x) = x, \quad &\Longrightarrow \quad \frac{du}{dx} = 1, \\ v(x) = \sqrt{x^2+25} \quad &\Longrightarrow \quad \frac{dv}{dx} = \frac{x}{\sqrt{x^2+25}} \end{aligned}$$

Apply the product rule:

$$\begin{aligned} \frac{dy}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= \sqrt{x^2+25} + \frac{x^2}{\sqrt{x^2+25}} \end{aligned}$$

(d) Split the function and differentiate:

$$\begin{aligned} u(x) = x, \quad &\Longrightarrow \quad \frac{du}{dx} = 1, \\ v(x) = \sqrt{x^2+25} \quad &\Longrightarrow \quad \frac{dv}{dx} = \frac{x}{\sqrt{x^2+25}} \end{aligned}$$

Apply the quotient rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{v \times \frac{du}{dx} - u \times \frac{dv}{dx}}{v^2} = \frac{\sqrt{x^2+25} - \frac{x^2}{\sqrt{x^2+25}}}{x^2+25} \\ &= \frac{\sqrt{x^2+25}}{x^2+25} - \frac{\frac{x^2}{\sqrt{x^2+25}}}{x^2+25} = \frac{\sqrt{x^2+25}}{x^2+25} - \frac{x^2}{\sqrt{x^2+25}} \times \frac{1}{x^2+25} \\ &= \frac{1}{\sqrt{x^2+25}} - \frac{x^2}{(x^2+25)^{\frac{3}{2}}} \end{aligned}$$

Question 3.5:

(a) Split the function and differentiate:

$$\begin{aligned} u(t) = 4 - t^2, \quad &\Longrightarrow \quad \frac{du}{dt} = -2t, \\ v(x) = 1 + 5t^2 \quad &\Longrightarrow \quad \frac{dv}{dt} = 10t \end{aligned}$$

Apply the product rule:

$$\begin{aligned}\frac{dy}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= 10t(4-t^2) - 2t(1+5t^2) \\ &= 2t(19-10t^2)\end{aligned}$$

Next, we need to know where the function is not changing and so we set the derivative equal to zero and solve. From the factored form of the derivative we can see that the derivative will be zero at:

$$t = 0\ , t = \pm\sqrt{\frac{19}{10}} = \pm 1.3784$$

We need to know is where the derivative is positive (and hence the function is increasing) or negative (and hence the function is decreasing). Because the derivative is continuous we know that the only place it can change sign is where the derivative is zero. We have been given 3 point where the derivative is zero. Lets use these points to define some regions to evaluate the sign of the derivative. This is best illustrated using Figure 3.1. From this we get the following increasing/decreasing information.

$$\begin{aligned}\textit{Increasing}: &\quad -\infty < t < -\sqrt{\frac{19}{10}}, \quad 0 < t < \sqrt{\frac{19}{10}} \\ \textit{Decreasing}: &\quad -\sqrt{\frac{19}{10}} < t < 0, \quad \sqrt{\frac{19}{10}} < t < \infty\end{aligned}$$

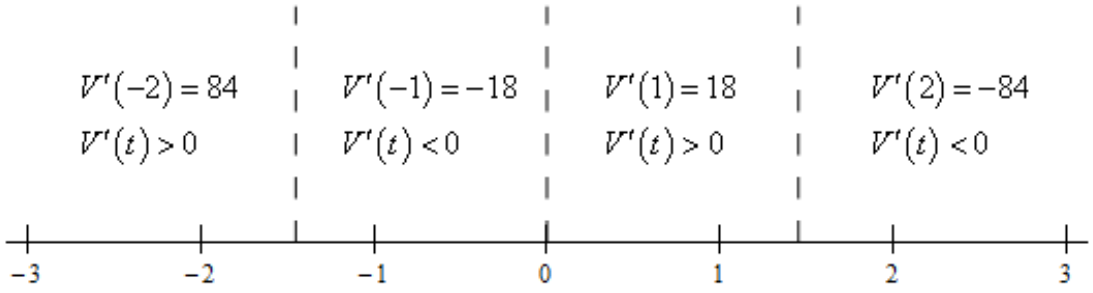


Figure 3.1: Here we identify where along the x axis the function derivative has a value of 0 [dashed lines] and then interrogate the sign of the derivatives either side. From this information we can identify regions where the function is increasing or decreasing.

(b) Split the function and differentiate:

$$\begin{aligned}u(x) = x - x^2, \quad &\Longrightarrow \quad \frac{du}{dx} = 1 - 2x, \\ v(x) = 1 + 8x^2 \quad &\Longrightarrow \quad \frac{dv}{dx} = 16x\end{aligned}$$

Apply the quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{v \times \frac{du}{dx} - u \times \frac{dv}{dx}}{v^2} \\ &= \frac{(1-2x)(1+8x^2) - 16x(x-x^2)}{(1+8x^2)^2} \\ &= \frac{1-2x-8x^2}{(1+8x^2)^2}\end{aligned}$$

Next, we need to know where the function is not changing and so we set the derivative equal to zero and solve. The denominator will never be zero for any real number and so the derivative will only be zero where the numerator is zero. Thus:

$$1-2x-8x^2 = 0$$

When we solve this we arrive at the following roots:

$$x = -\frac{1}{2}, x = \frac{1}{4}$$

We need to know is where the derivative is positive (and hence the function is increasing) or negative (and hence the function is decreasing). Because the derivative is continuous we know that the only place it can change sign is where the derivative is zero. We have been given 3 point where the derivative is zero. Lets use these points to define some regions to evaluate the sign of the derivative. This is best illustrated using the Figure 3.2.From this we get the following increasing/decreasing information.

$$\begin{aligned}\textit{Increasing}: &\quad -\frac{1}{2} < x < \frac{1}{4} \\ \textit{Decreasing}: &\quad -\infty < x < -\frac{1}{2}, \quad \frac{1}{4} < x < \infty\end{aligned}$$

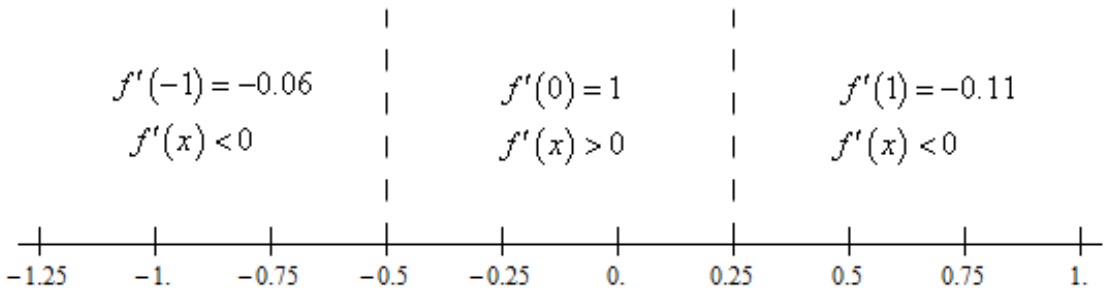


Figure 3.2: Here we identify where along the x axis the function derivative has a value of 0 [dashed lines] and then interrogate the sign of the derivatives either side. From this information we can identify regions where the function is increasing or decreasing.

Question 3.6:

(a) In this Question 3.we need to apply the chain rule twice. To make sure we don't make any mistakes, lets do it part by part. First we differentiate $y = \sin(x^2)$. Split the function and differentiate:

$$\begin{aligned} u(v) = \sin(v), \quad &\Longrightarrow \quad \frac{du}{dv} = \cos(v), \\ v(x) = x^2 \quad &\Longrightarrow \quad \frac{dv}{dx} = 2x \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= 2x\cos(x^2) \end{aligned}$$

Now lets apply the chain rule to $Z = (\sin(x^2))^2$. Split the function and differentiate:

$$\begin{aligned} u(v) = v^2, \quad &\Longrightarrow \quad \frac{du}{dv} = 2v, \\ v(x) = \sin(x^2) \quad &\Longrightarrow \quad \frac{dv}{dx} = 2x\cos(x^2) \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dZ}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= 2sin(x^2) \times 2x\cos(x^2) \\ &= 4xsin(x^2)\cos(x^2) \end{aligned}$$

(b) Split the function and differentiate:

$$\begin{aligned} u(x) = 1 + \sqrt{x^3} = 1 + x^{\frac{3}{2}}, \quad &\Longrightarrow \quad \frac{du}{dx} = \frac{3}{2}x^{\frac{1}{2}}, \\ v(x) = x^{-3} - 2\sqrt[3]{x} = x^{-3} - 2x^{\frac{1}{3}} \quad &\Longrightarrow \quad \frac{dv}{dx} = -3x^{-4} - \frac{2}{3}x^{-\frac{2}{3}} \end{aligned}$$

Apply the product rule:

$$\begin{aligned} \frac{dS}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= \left(\frac{3}{2}x^{\frac{1}{2}}\right)\left(x^{-3} - 2x^{\frac{1}{3}}\right) + \left(1 + x^{\frac{3}{2}}\right)\left(-3x^{-4} - \frac{2}{3}x^{-\frac{2}{3}}\right) \\ &= -3x^{-4} - \frac{3}{2}x^{-\frac{5}{2}} - \frac{2}{3}x^{-\frac{2}{3}} - \frac{11}{3}x^{\frac{5}{6}} \end{aligned}$$

(c) Split the function and differentiate:

$$\begin{aligned} u(w) = 3w + w^4, \quad &\Longrightarrow \quad \frac{du}{dw} = 3 + 4w^3, \\ v(w) = 2w^2 + 1 \quad &\Longrightarrow \quad \frac{dv}{dw} = 4w \end{aligned}$$

Apply the quotient rule:

$$\begin{aligned}\frac{dy}{dw} &= \frac{v \times \frac{du}{dw} - u \times \frac{dv}{dw}}{(2w^2+1)^2} \\ &= \frac{(2w^2+1)(3+4w^3) - 4w(3w+w^4)}{(2w^2+1)^2} \\ &= \frac{4w^5+4w^3-6w^2+3}{(2w^2+1)^2}\end{aligned}$$

Real-world problems

Question 3.7: Imagine we are studying the volume of water in an ecosystem reservoir over time (in hours).

(a) Split the function and differentiate:

$$\begin{aligned} u(v) = V_0 \exp(v), \quad &\Longrightarrow \quad \frac{du}{dv} = V_0 \exp(v), \\ v(t) = -\frac{t^2}{600} \quad &\Longrightarrow \quad \frac{dv}{dt} = -\frac{t}{300}\end{aligned}$$

Apply the chain rule:

$$\begin{aligned}\frac{dV}{dt} &= \frac{du}{dv} \times \frac{dv}{dt} \\ &= -\frac{V_0 \times t}{300} \exp\left(-\frac{t^2}{600}\right)\end{aligned}$$

The drainage rate at $t = 10$ is

$$\begin{aligned}\frac{dV}{dt}(10) &= -\frac{10 \times 10}{300} \exp\left(-\frac{10^2}{600}\right) \\ &= -\frac{1}{3} \exp\left(-\frac{1}{6}\right) \\ &= -0.1829\end{aligned}$$

(b) Split the function and differentiate:

$$\begin{aligned} u(t) = \frac{V_0(t+10)}{10}, \quad &\Longrightarrow \quad \frac{du}{dt} = \frac{V_0}{10}, \\ v(t) = \exp\left(-\frac{t}{5}\right) \quad &\Longrightarrow \quad \frac{dv}{dt} = -\frac{1}{5} \exp\left(-\frac{t}{5}\right)\end{aligned}$$

Apply the product rule:

$$\begin{aligned}
\frac{dV}{dt} &= u \times \frac{dv}{dt} + v \times \frac{du}{dt} \\
&= \frac{V_0(t+10)}{10} \times -\frac{1}{5}\exp\left(-\frac{t}{5}\right) + \exp\left(-\frac{t}{5}\right) \times \frac{V_0}{10} \\
&= -\frac{V_0(t+10)}{50}\exp\left(-\frac{t}{5}\right) + \frac{V_0}{10}\exp\left(-\frac{t}{5}\right) \\
&= V_0 \exp\left(-\frac{t}{5}\right)\left(-\frac{t+10}{50} + \frac{1}{10}\right) \\
&= V_0 \exp\left(-\frac{t}{5}\right)\left(\frac{5-t-10}{50}\right) \\
&= -\frac{V_0(t+5)}{50}\exp\left(-\frac{t}{5}\right)
\end{aligned}$$

The drainage rate at $t = 10$ is

$$\begin{aligned}
\frac{dV}{dt}(10) &= -\frac{10 \times (10+5)}{50}\exp\left(-\frac{10}{5}\right) \\
&= -3\exp(-2) \\
&= -0.406
\end{aligned}$$

(c) Split the function and differentiate:

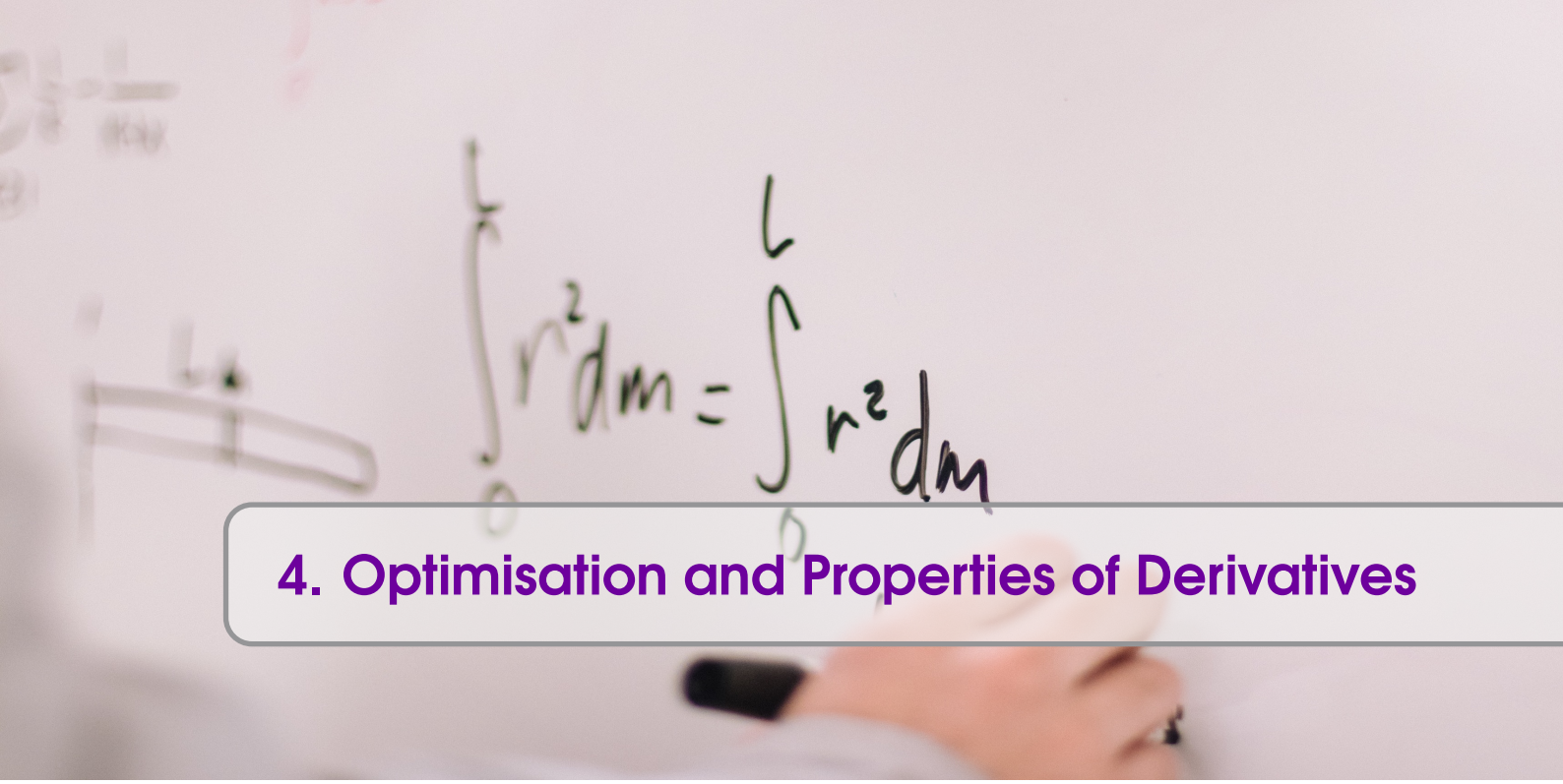
$$\begin{aligned}
u(t) = t + 10, \quad &\Longrightarrow \quad \frac{du}{dt} = 1, \\
v(t) = t^2 + 10 \quad &\Longrightarrow \quad \frac{dv}{dt} = 2t
\end{aligned}$$

Apply the quotient rule:

$$\begin{aligned}
\frac{dV}{dt} &= V_0 \frac{v \times \frac{du}{dt} - u \times \frac{dv}{dt}}{(2w^2+1)^2} \\
&= V_0 \frac{t^2 + 10 - 2t \times (t+10)}{(t^2+10)^2} \\
&= V_0 \frac{-t^2 - 20t + 10}{(t^2+10)^2}
\end{aligned}$$

The drainage rate at $t = 10$ is

$$\begin{aligned}
\frac{dV}{dt}(10) &= 10 \times \frac{-10^2 - 20 \times 10 + 10}{(10^2+10)^2} \\
&= 10 \times \frac{-100 - 200 + 10}{110^2} \\
&= -0.2397
\end{aligned}$$



4. Optimisation and Properties of Derivatives

4.1 Overview

In Chapters 2 and 3, we have been calculating derivatives of functions. Whilst telling us the rate of change of a function, derivatives can also be used to determine certain properties of a function and how it behaves. We now carry on with the topic of differentiation onto the topic of **properties of derviatives** and using them to **find maxima and minima** of functions.

4.2 Properties of a derivative

Remember, the derivative of a function at a given point represents the gradient of a function at that point, indicating the rate of change. Therefore, we can use the derivative to know where a function is increasing, decreasing or neither. We can see that if the rate of change is negative,

$$\frac{dy}{dx} < 0$$

then the function is decreasing and that if the rate of change is positive,

$$\frac{dy}{dx} > 0$$

then the function is increasing. If the derivative is zero,

$$\frac{dy}{dx} = 0$$

then it is likely the function will be changing direction either side. This means that the point on the function where the derivative is either a **maximum value**, **minimum value**

or a **saddle point**. This turns out to be a very useful property and more generally we call these **stationary values** or **stationary points**. Once we know where the these stationary points are for a function, further analysis can help classify them. There will be more on this in the next Chapter, but for now, once you find where the derivatives are 0 you can then explore regions either side to determine if the gradient is positive or negative. Let's consider some examples.

Example 4.1: Let's consider the quadratic function

$$f(x) = \frac{x^2}{2} - 5x + 2$$

This function is shown in the top panel of Figure 4.1, over the range: $-5 < x < 15$. We can see that there is a point on the curve that where the minimum occurs ($x = 5$ in the plot, but note that it is not obvious). Now lets differentiate the function $f(x)$ and look at how it behaves:

$$\frac{df}{dx} = f'(x) = x - 5$$

The derivative function is shown in the bottom panel of Figure 4.1, over the range: $-5 < x < 15$.

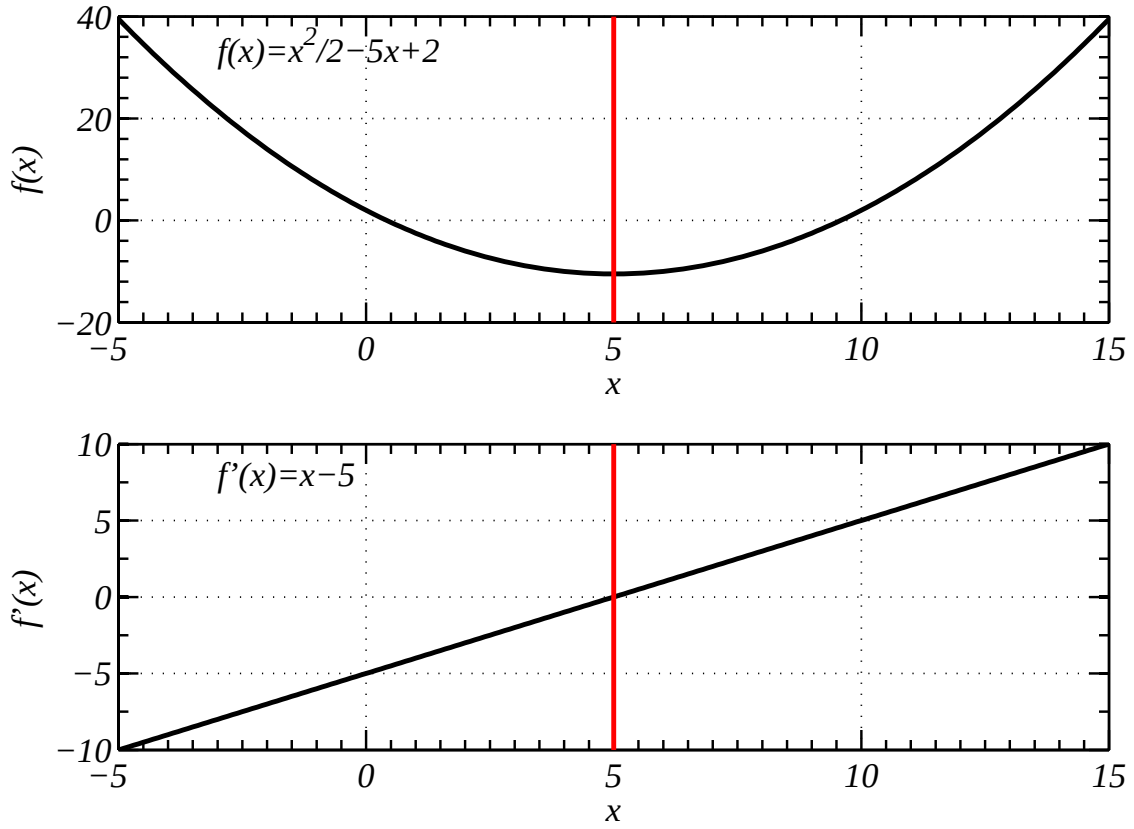


Figure 4.1: Top: y changing as a function of x. Bottom: Derivative value for the same function, also varying as a function of x

The thing to notice about the gradient function, $f'(x)$, is that it is equal to zero at the point where $f(x)$ is a minimum. Therefore we can find the minimum of $f(x)$ by differentiating it and setting it to zero:

$$f'(x) = 0 \implies x - 5 = 0$$

we can then solve $x - 5 = 0$ to get that $x = 5$. This tells us that the minimum value occurs at $x = 5$. We can evaluate the value of f at the stationary value (i.e. maximum and minimum) $f(5) = \frac{5^2}{2} - 5.5 + 2 = -10.5$.

Example 4.2: Sometimes there can be more than one stationary point. Let's consider the cubic function

$$f(x) = \frac{x^3}{3} + 2x^2 - 5x + 1$$

This function is shown in the top panel of Figure 4.2, over the range: $-10 < x < 5$. We can see that there is a point on the curve that where a minimum occurs ($x = 2$) and where a maximum occurs ($x = -5$), again not immediately visible. Now lets differentiate the function $f(x)$ and look at how it behaves:

$$\frac{df}{dx} = f'(x) = x^2 + 4x - 5$$

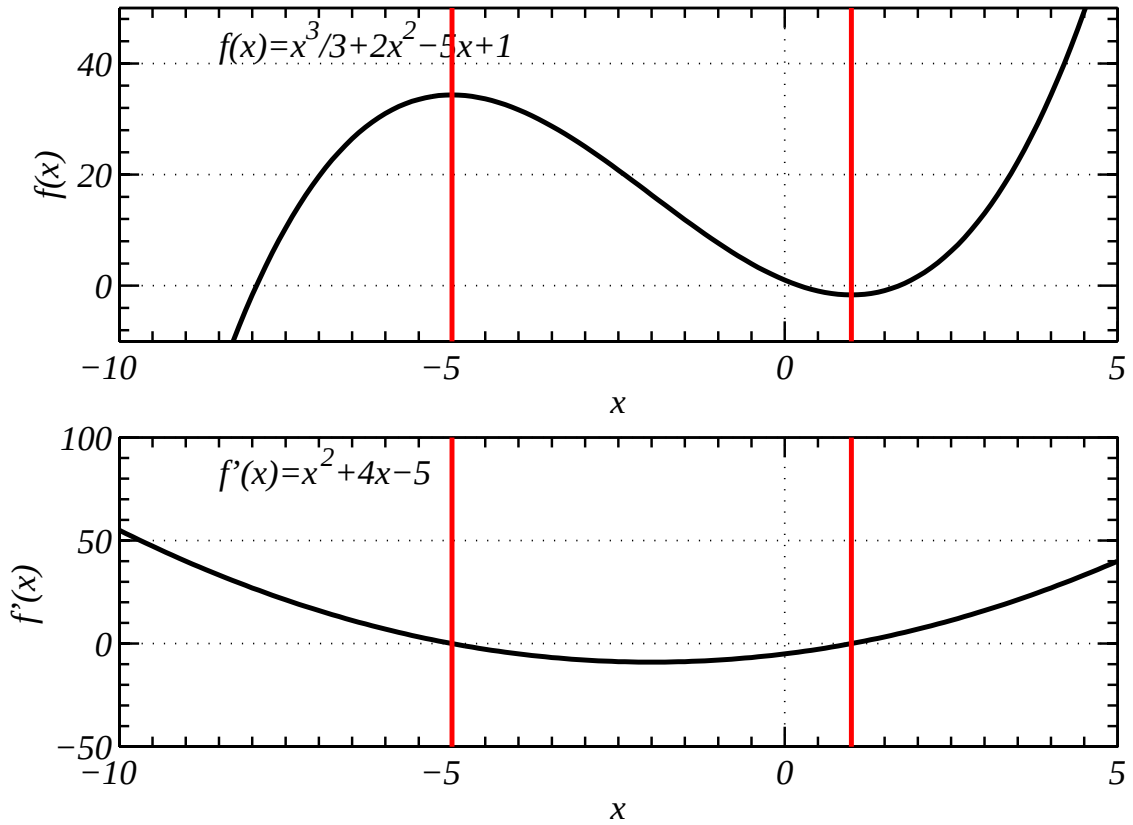


Figure 4.2: Top: y changing as a function of x. Bottom: Derivative value for the same function, also varying as a function of x

The derivative function is shown in the bottom panel of Figure 4.2, over the range: $-10 < x < 5$. If we set the gradient function to zero we have a quadratic equation:

$$f'(x) = 0 \implies x^2 + 4x - 5 = 0$$

which we can solve either by the quadratic formula, or by factorisation:

$$(x+5)(x-1) = 0$$

so that the two solutions are $x = 1$ and $x = -5$ (as shown in the Figure 4.2).

Sometimes it will not always be possible to solve an equation equal to zero by rearranging or factorising, due to the complexity of the function. We then must use an algorithm to find the stationary values. We will explore this topic in the next Chapter.

So, to summarise to find a stationary value we:

1. Differentiate the function, to find the gradient function, $f'(x)$ or $\frac{dy}{dx}$
2. Set the gradient function to zero and solve for x.
3. Substitute x into $f(x)$ to get what the value of the stationary points are.

We could (and strictly should) verify that we have found a minimum and maximum point. We will again see this more in next Chapter, but a quick way of doing this is to evaluate $f(x)$ for x just a little bit more and just a little bit less than the stationary value. Returning to Example 4.1, we can try $f(4.5)$, which is equal to -10.375 and $f(5.5)$ which is also equal to -10.375. Both of these values, are greater than the stationary value, so this tells us that the stationary value is indeed a minimum. If we had found a maximum we would find that $f(x)$ either side of the stationary point to be less than the value of the stationary point. More generally, we can use properties of a derivative. We can see that when $x < 5$, then $f'(x) < 0$ and when $x > 5$, then $f'(x) > 0$, so we see that the stationary value is a minimum.

4.3 Optimisation

Optimisation problems involve finding the best solution, often the maximum or minimum value(s) of a function given set of constraints. They are quite common in areas such as

- Business and Economics: Maximize profit or minimize cost functions.
- Physics and Engineering: Minimize energy or optimize the performance of systems.
- Geometry: Find dimensions that maximize area or volume subject to constraints.

Derivatives and their properties play a crucial role in a wide range of problems such as optimisation. We can study a function to find the maxima and minima which "optimises" a function, which can play a vital role in decision making.

Exercises

Algebraic problems

Question 4.1: Find the stationary point of the following functions and verify whether they are maxima or minima either by evaluating the function, or using the properties of the derivative.

(a) $f(x) = x^2 + x$
(b) $f(x) = 2x - x^2$
(c) $f(x) = (x-2)^2$
(d) $f(x) = (x-2)(3-x)$
(e) $f(x) = \frac{x^3}{3} - 4x^2 + 15x + 5$
(f) $f(x) = ln(x) - x$

Question 4.2: Find two non-negative numbers whose sum is 9 and so that the product of one number and the square of the other number is a maximum. To solve this, first construct the two equations that define these conditions, using variables x and y. To start, you can write $x + y = 9$.

Real-world problems

Question 4.3: Amazon wants to construct a box with length l, width w and height h which has volume:

$$V = l \times w \times h$$

and only has 10 m^2 of material to use in construction of the box:

$$10 = 2lw + 2wh + 2lh$$

If the box will have a square base (i.e. $l = w$) and assuming that all the material is used in the construction process, determine the maximum volume that the box can have. You will need to rearrange the constraints and substitute into the volume equation to find a function with respect to one variable before optimising.

Question 4.4: Amazon now wants to construct an open rectangular box with square base is to be made from 48 m^2 of material. What dimensions will result in a box with the largest possible volume?

Question 4.5: A cylindrical aluminium can of cola is to hold 330 ml of liquid. Coca-Cola would like to know what the dimensions of the can that uses the least amount of aluminium. The volume for a cylinder of base radius r and height h is given by

$$V = \pi r^2 h$$

The formula for surface area of a cylinder is given by:

$$A = 2\pi rh + 2\pi r^2$$

The volume is constrained so for whatever value of r we pick, we can calculate h and therefore estimate the surface area of the can. Rewrite the volume equation to get:

$$h = \frac{V}{\pi r^2} = \frac{330 \times 10^{-6}}{\pi r^2}$$

Substitute this into the surface area equation:

$$A = 2\pi rh + 2\pi r^2 = 2\pi r \times \frac{330 \times 10^{-6}}{\pi r^2} + 2\pi r^2 = \frac{660 \times 10^{-6}}{r} + 2\pi r^2$$

If you choose $r = 0.01$m, then, since the volume has to be 330×10^{-6}, the height, h, and surface area, A, has to be:

$$h = \frac{330 \times 10^{-6}}{\pi \times 0.01^2} = 1.050m \qquad A = \frac{660 \times 10^{-6}}{0.01} + 2\pi \times 0.01^2 = 0.067$$

(a) Complete the table below for surface area for different values of the radius r. Which is the best value to select?

Radius	Height for volume to equal 330 ml	Surface area of aluminium
0.010	1.050	0.067
0.020		
0.030		
0.040		
0.050		
0.060		
0.070		

(b) Use differentiation to find the minimum amount, what value do you arrive at?

Question 4.6: A gold seam optimisation problem. You decide to mine a gold seam and want to decide on how long to mine it for to maximise your profits. The cost of gold you can mine per day, G, is a function of the number of days, n, that you mine for. It becomes more difficult to mine the gold as time goes on:

$$G = 1 \times 10^5 - \frac{1 \times 10^5}{n+1}$$

In addition the cost of renting the equipment to mine the gold, C, increases with the number of days:

$$C = 1000 \times n$$

So that your profit is $P = G - C$. Graphs of P, G and C are shown in Figure 4.3.

(a) Work out how long you should mine the gold to optimise (find the stationary value of) your profit, P. (Hint: you can either use *chain or quotient rules* to perform the differentiation. To find the stationary points you will need to solve a quadratic equation).

(b) In the project a "snag" occurs that causes no profit to be made on one day (see Figure 4.4). After the snag is overcome mining can proceed at a faster rate. A function describing the cost of gold mined as the project goes on is:

$$G = 1 \times 10^5 \left[1 - \left(0.01n^2 + 1\right) \exp\left(-0.1n\right)\right]$$

Use this function to find the day where the amount of gold extracted hits a snag (you will need to make use of the product rule).

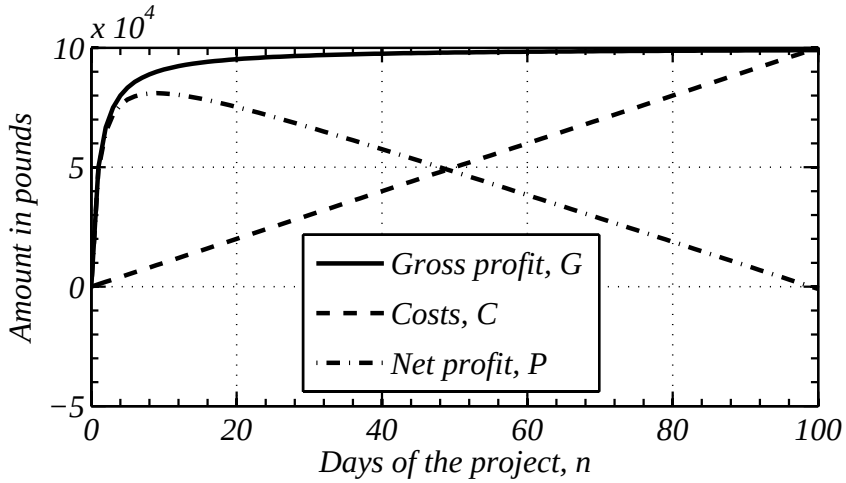


Figure 4.3: Hypothetical plot of profit as a function of time

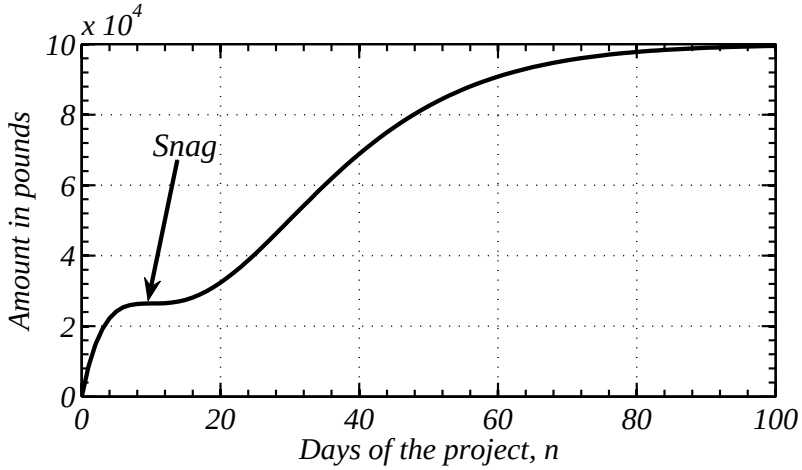


Figure 4.4: Hypothetical plot of profit as a function of time

Question 4.7: Geoengineering. In this problem economists investigating the *economic cost to society*, E, of an increase in global temperature, ΔT (e.g. see the Stern report) estimate that:

$$E = 500 \times \Delta T \tag{4.1}$$

with E in billions of dollars (see Figure 4.5). In addition climate scientists investigating the *marine cloud brightening* proposal for geoengineering estimate that the technique does have the capability to offset warming by rising CO_2. They estimate spending money, C, on a program to implement geoengineering can result in a lower increase in temperature:

$$\Delta T = 10 \exp(-dC) \tag{4.2}$$

where C is the amount spent on the scheme and d is a constant that has some uncertainty (see Figure 4.6).

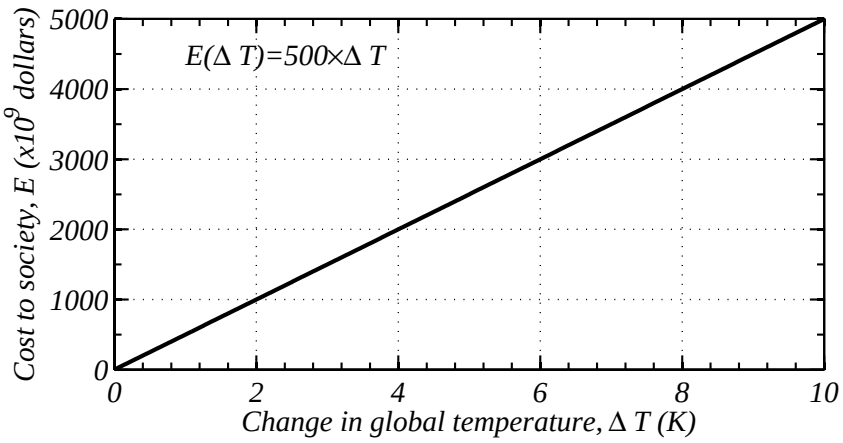


Figure 4.5: Hypothetical cost to society as a function of changes in global temperatures

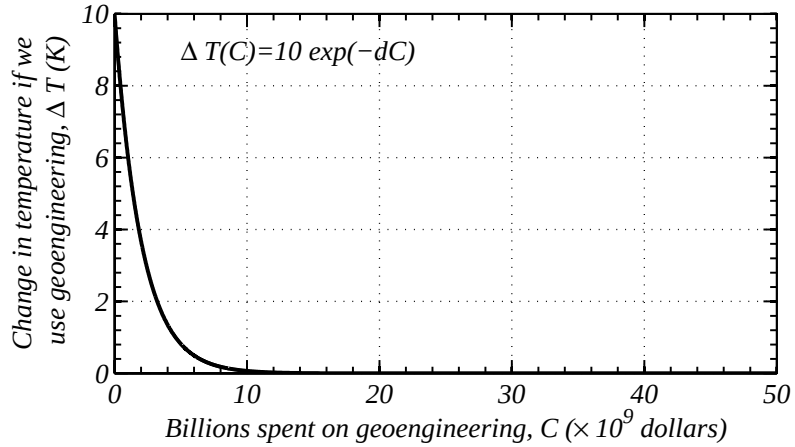


Figure 4.6: Hypothetical change in global temperatures as a function of investment in geo-engineering

In order to decide whether the scheme is worth implementing we need to get an idea of the *total cost* that implementing the scheme has (i.e. the benefit plus cost of scheme) vs the amount spent on implementing it. To do this rearrange Equation 4.1 to give $\Delta T = \frac{E}{500}$ and substitute into Equation 4.2 to get an equation for the economic cost to society, E, vs the amount spent on geoengineering, C. Then add on the cost of the scheme, C, to get the *total cost*, $T = E + C$:

$$T = 500 \times 10 \exp(-dC) + C \tag{4.3}$$

This function can be seen in Figure 4.7 and we can see there is a minimum in the *total cost*.

Your goal is to optimise / find the stationary value of Equation 4.3 (i.e. find how much we should spend on the scheme so that we can minimise the total cost). Choose the parameter d to be equal to 2×10^{-4}; how much should be spent? What is the meaning of your answer (i.e. what should the advice to policy makers be?)

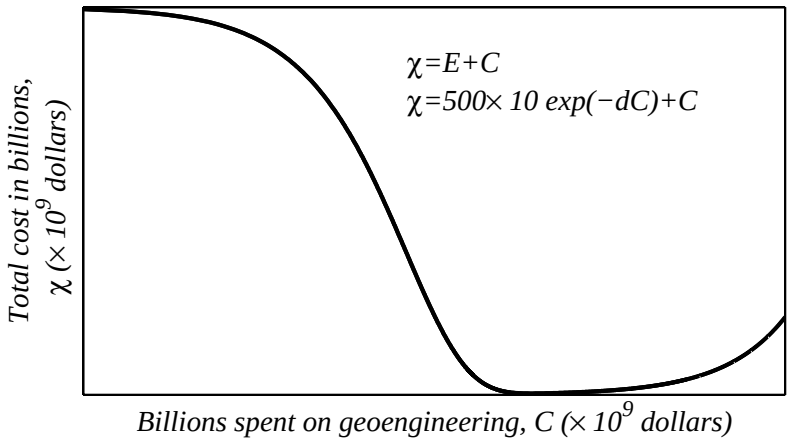


Figure 4.7: Hypothetical total cost to society as a function of investment in geo-engineering

Solutions

Question 4.1: Find the stationary point of the following functions and verify whether they are maxima or minima either by evaluating the function, or using the properties of the derivative.

(a) $f(x) = x^2 + x \implies \frac{df}{dx} = f'(x) = 2x + 1 \implies 2x + 1 = 0$
$x = -\frac{1}{2}$ is a stationary value and this is a minimum.

(b) $f(x) = 2x - x^2 \implies \frac{df}{dx} = f'(x) = 2 - 2x \implies 2 - 2x = 0$
$x = 1$ is a stationary value and this is a maximum.

(c) $f(x) = (x-2)^2 \implies \frac{df}{dx} = f'(x) = 2(x-2) \implies 2(x-2) = 0$
$x = 2$ is a stationary value and this is a minimum.

(d) $f(x) = (x-2)(3-x) \implies \frac{df}{dx} = f'(x) = (3-x) - (x-2) = 5 - 2x \implies 5 - 2x = 0$
$x = 2.5$ is a stationary value and this is a maximum.

(e) $f(x) = \frac{x^3}{3} - 4x^2 + 15x + 5 \implies \frac{df}{dx} = f'(x) = x^2 - 8x + 15 = (x-3)(x-5) \implies (x-3)(x-5) = 0$
$x = 3$ and $x = 5$ are stationary values, and they are a maximum and a minimum respectively.

(f) $f(x) = ln(x) - x \implies \frac{df}{dx} = f'(x) = \frac{1}{x} - 1 \implies \frac{1}{x} - 1 = 0$
$x = 1$ is a stationary value and this is a maximum.

Question 4.2: Write down an equation describing the problem: $x + y = 9$, where $x \times y^2$ is a maximum. Using the first equation we can write: $x = 9 - y$, sub this in the second to give:

$$f(y) = (9-y)y^2$$

so expanding, differentiating and setting to zero gives

$$f'(y) = 18y - 3y^2 = 0$$

Factorise to get

$$3y(6-y) = 0$$

which gives us $y = 0$ or $y = 6$ as stationary values. We see that $y = 0$ is a minimum and $y = 6$ is a maximum. Choosing $y = 6$ gives $x = 9 - 6 = 3$ and the maximum value is $3 \times 6^2 = 108$.

Question 4.3: We should write down the relevant equations and constraints. Since the box has a square base (i.e $l = w$) we can rewrite the equations as

$$V = lwh = w^2 h$$
$$10 = 2lw + 2wh + 2lh = 2w^2 + 4wh$$

Since we want to maximise V and use a derivative, can we re-arrange the expressions to express V with regards to w? We can. An expression for h is given as:

$$h = \frac{10 - 2w^2}{4w} = \frac{5 - w^2}{2w}$$

Substitute this in to the expression for V to arrive at the following expression that we will need to maximise:

$$V(w) = w^2 \left(\frac{5 - w^2}{2w} \right) = \frac{1}{2}(5w - w^3)$$

The derivative is given by:

$$\frac{dV}{dw} = \frac{1}{2}(5 - 3w^2)$$

Set this to 0 and rearrange to find we have one position solution as per the following:

$$w = \pm\sqrt{\frac{5}{3}} = \pm 1.2910$$

Since we cannot get negative width, we see the volume is maximised at

$$w = \sqrt{\frac{5}{3}} = 1.2910m$$

$$h = \frac{5 - w^2}{2w} = \frac{5 - \sqrt{\frac{5}{3}}^2}{2\sqrt{\frac{5}{3}}} = 1.2910m$$

so the optimum box is a cube. The maximum volume is:

$$V = w^2 h = \left(\sqrt{\frac{5}{3}} \right)^3 = 2.1517m^3$$

Question 4.4: This is very similar to the previous question. We should write down the relevant equations and constraints. The rectangular box has a square base (i.e $l = w$) and it is open so we can write the equations as

$$V = lwh = w^2 h$$

$$48 = lw + 2wh + 2lh = w^2 + 4wh$$

Note that we're leaving one square side off the constraint. Since we want to maximise V and use a derivative, can we re-arrange the expressions to express V with regards to w? We can. An expression for h is given as:

$$h = \frac{48 - w^2}{4w} = \frac{48 - w^2}{4w}$$

Substitute this in to the expression for V to arrive at the following expression that we will need to maximise:

$$V(w) = w^2\left(\frac{48-w^2}{4w}\right) = \frac{1}{4}(48w - w^3)$$

The derivative is given by:

$$\frac{dV}{dw} = \frac{1}{4}(48 - 3w^2)$$

Set this to 0 and rearrange to find we have one position solution as per the following:

$$w = \pm 4$$

Since we cannot get negative width, we see the volume is maximised at

$$w = 4m$$
$$h = \frac{48-w^2}{4w} = \frac{48-4^2}{4\times 4} = 2m$$

The maximum volume is:

$$V = w^2 h = 4\times 4\times 2 = 32m^3$$

Question 4.5:

(a) See table below: $r = 0.04$ m seems to minimise the surface area required.

Radius	Height for volume to equal 330 ml	Surface area of aluminium
0.010	1.050	0.067
0.020	0.263	0.036
0.030	0.117	0.028
0.040	0.066	0.027
0.050	0.042	0.029
0.060	0.029	0.034
0.070	0.021	0.040

(b) Since we need to minimise the surface area, we need to rearrange the volume to get an expression for h

$$h = \frac{330\times 10^{-6}}{r^2\pi}$$

Substitute this into the surface area equation to get a function in terms of r

$$A(r) = 2\pi r\left(\frac{330\times 10^{-6}}{r^2\pi}\right) + 2\pi r^2$$
$$= \frac{660\times 10^{-6}}{r} + 2\pi r^2$$

The derivative is given by:

$$\frac{dA}{dw} = -\frac{660 \times 10^{-6}}{r^2} + 4\pi r$$

Set this to 0 and rearrange to find we have one position solution as per the following:

$$\frac{660 \times 10^{-6}}{r^2} = 4\pi r \implies r = \left(\frac{660 \times 10^{-6}}{4\pi}\right)^{\frac{1}{3}}$$

we see the surface area is minimised at

$$w = r = \left(\frac{660 \times 10^{-6}}{4\pi}\right)^{\frac{1}{3}} = 0.0374$$
$$h = \frac{330 \times 10^{-6}}{r^2\pi} = 0.0749$$

Question 4.6: A gold seam optimisation problem.

(a) Firstly, write down the equation for profit, $P = G - C$:

$$P = 1 \times 10^5 - \frac{1 \times 10^5}{n+1} - 1000n$$

Find $\frac{dP}{dn}$:

$$\frac{dP}{dn} = -\frac{d}{dn}\left(\frac{1 \times 10^5}{n+1}\right) - 1000$$

Use the chain rule to differentiate $f = \frac{1\times10^5}{n+1}$ call $u = n+1$, so $f = \frac{1\times10^5}{u}$ and then

$$\frac{df}{dn} = \frac{df}{du} \times \frac{du}{dn} = -1 \times 10^5 u^{-2}$$

so we see that

$$\frac{dP}{dn} = \frac{1 \times 10^5}{(n+1)^2} - 1000$$

Now set to zero and solve for n to find the stationary values:

$$\frac{1 \times 10^5}{(n+1)^2} - 1000 = 0$$
$$(n+1)^2 = 100$$
$$n^2 + 2n - 99 = 0$$
$$(n+11)(n-9) = 0$$

We see that this function has two stationary values:

$$n = -11, 9$$

So the answer is that the profit is maximised after 9 days. Substitute this into the formula for P to get $P = 80,000$ profit after 9 days.

(b) To find the snag, we differentiate G with respect to n:

$$G = 1\times 10^5\left[1-\left(0.01n^2+1\right)\exp\left(-0.1n\right)\right]$$
$$\frac{dG}{dn} = -1\times 10^5\times\frac{d}{dn}\left[\left(0.01n^2+1\right)\exp\left(-0.1n\right)\right]$$

Note that the first term is a constant so, when differentiated it goes to zero. To differentiate $\left(0.01n^2+1\right)\exp\left(-0.1n\right)$ we use the product rule. Split the function and differentiate:

$$u(n) = 0.01n^2+1, \quad \Longrightarrow \quad \frac{du}{dn} = 0.02n,$$
$$v(n) = \exp\left(-0.1n\right) \quad \Longrightarrow \quad \frac{dv}{dn} = -0.1\exp\left(-0.1n\right)$$

Apply the product rule:

$$\begin{aligned}\frac{dy}{dn} &= u\times\frac{dv}{dx}+v\times\frac{du}{dx}\\ &= -0.1\exp\left(-0.1n\right)\times\left(0.01n^2+1\right)+0.02n\times\exp\left(-0.1n\right)\\ &= -\exp\left(-0.1n\right)\left[-0.1\times\left(0.01n^2+1\right)-0.02n\right]\\ &= -\exp\left(-0.1n\right)\left[0.001n^2-0.02n+0.1\right]\end{aligned}$$

Now set to zero and solve for n to find the stationary values:

$$\begin{aligned}-\exp\left(-0.1n\right)\left[0.001n^2-0.02n+0.1\right] &= 0\\ 0.001n^2-0.02n+0.1 &= 100\\ n^2-20n+100 &= 0\\ (n-10)^2 &= 0\end{aligned}$$

We see that this function has a stationary value:

$$n = 10$$

So the answer is that the snag occurs after 10 days. This is an example of a saddle point.

Question 4.7: Geoengineering. For the problem you should first find $\frac{dT}{dC}$,

$$T = 5000\exp\left(-dC\right)+C$$
$$\frac{dT}{dC} = -5000d\exp\left(-dC\right)+1$$

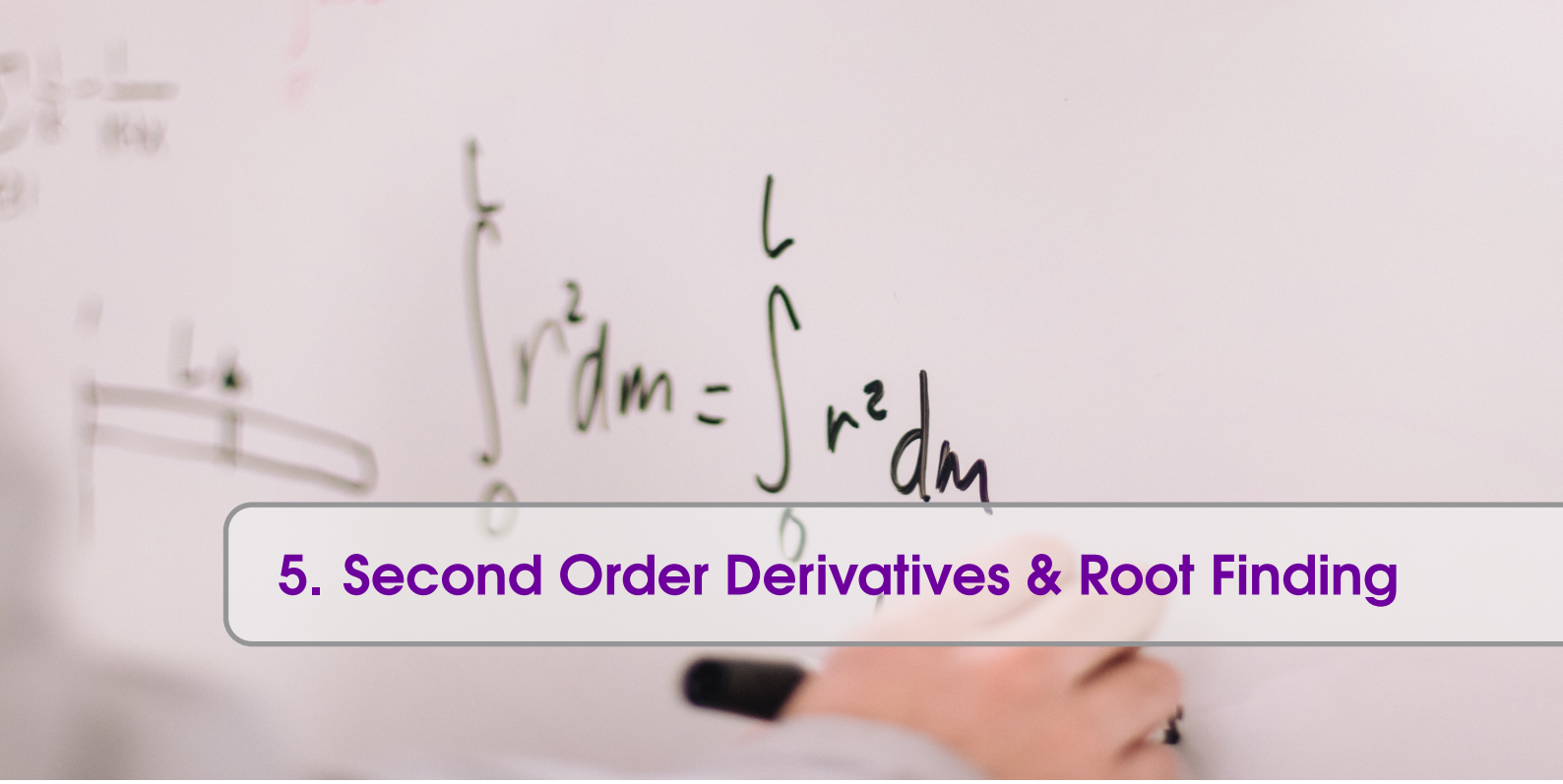
Now set to zero and solve for C to find the stationary values:

$$\begin{aligned} -5000d\exp(-dC)+1 &= 0 \\ d\exp(-dC) &= \frac{1}{5000} \\ \exp(dC) &= 5000d \\ dC &= \ln(5000d) \\ C &= \frac{1}{d}\ln(5000d) \end{aligned}$$

Substitute $d = 2\times10^{-4}$ to find the amount to be spent on geoengineering, C, that minimises the total cost, χ:

$$\begin{aligned} C &= \frac{1}{d}\ln(5000d) \\ C &= \frac{1}{2\times10^{-4}}\ln\left(5000\times2\times10^{-4}\right) \\ C &= 0 \end{aligned}$$

therefore this means that we can only *increase* the temperature by implementing the scheme! It should not go ahead.



5. Second Order Derivatives & Root Finding

5.1 Overview

In Chapters 2–4, we learned how to calculate derivatives of functions, which can be used tells us a rate of change and help us identify various properties and behaviors of functions. In this Chapter, we conclude our study of differentiation by examining the further properties of functions through the **second-order derivative** and exploring how derivatives can be applied to locate the **roots of functions**.

5.2 Second-order derivatives

In the previous Chapter, we explored the **properties of derivatives** and how to use them to identify the **stationary values** of functions. These properties are invaluable for optimising functions or processes. However, to determine whether a stationary value represents a maximum or a minimum, we relied on either (a) looking the function's plot or (b) evaluating the original function on either side of the stationary value. There is, however, a formal method for verifying this, which involves something called the **second-order derivative**.

Up to this point, we have only differentiated functions once. For a given function $f(x)$, its derivative is expressed as:

$$f'(x) = \frac{df}{dx}$$

This is referred to as the **first-order derivative**. However, differentiation can be carried out multiple times. By differentiating a function again, we obtain the **second-order derivative**,

which is defined as:

$$f''(x) = \frac{d}{dx}\left(\frac{df}{dx}\right) = \frac{d^2f}{dx^2}$$

The second-order derivative has some very interesting and useful properties. Previously we saw that the first-order derivative, $f'(x)$ tells us about the rate of change of a function, $f(x)$. The second-order derivative, $f''(x)$, goes a step further by telling us how the rate of change, $f'(x)$, itself is changing. This has many applications in physics, such as investigating stationary values or determining steady state, but to understand why its useful, imagine you're driving:

1. Distance is how far you've traveled over time (call this $x = f(t)$)
2. Speed tells us how fast you're traveling at any given moment. It's the rate at which your distance changes over time. Mathematically, it's the first-order derivative of distance:
$$v = \frac{dx}{dt}$$
3. Acceleration measures how quickly your speed is increasing or decreasing. It's the rate of change of velocity over time. Mathematically, this is the second derivative of distance (or the first derivative of velocity):
$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

Stationary values

These properties of second-order derivative make them very useful for understanding the nature of stationary points, which are the points where $f'(x) = 0$. Here's how $f''(x)$ helps us classify these points:

- If $f''(x) < 0$, then $f(x)$ has a local maximum at the stationary value x
- If $f''(x) > 0$, then $f(x)$ has a local minimum at the stationary value x
- If $f''(x) = 0$, then it is not possible to conclude anything about the point x. It could be saddle point (neither a maximum nor a minimum) so we need to either visually inspect the stationary value or or further evaluation of $f(x)$ is needed.

Let's return to an example from the previous Chapter.

Example 5.1: Let's consider the cubic function

$$f(x) = \frac{x^3}{3} + 2x^2 - 5x + 1$$

This function is shown in the top panel of Figure 4.2, over the range: $-10 < x < 5$. In the previous Chapter, we differentiated this function to find:

$$\frac{df}{dx} = f'(x) = x^2 + 4x - 5$$

By solving the quadratic equation, we determined the stationary points at $x = -5$ and $x = 1$. From the graph, these points appear to be a maximum and a minimum, respectively. To confirm this, let's examine the second derivative:

$$\frac{d^2f}{dx^2} = f''(x) = 2x + 4$$

Now, we evaluate the second derivative at the stationary points:

$$f''(-5) = 2 \times (-5) + 4 = -6 \text{ and } f''(-5) = 2 \times 1 + 4 = 6$$

Since $f''(x) < 0$ at $x = -5$ and $f''(x) > 0$ at $x = 1$, we can confirm these stationary values are a local maximum and minimum respectively.

Steady state

You may not be familiar with the concept of **steady-state**. When a system is in steady-state, the key variables that describe the system's behavior do not change over time. In a chemical system, this means that even though processes are happening that could cause change, the system's variables stay the same. For this to happen, there must be a continuous flow of materials through the system.

Let's look at an example using Figure 5.1. Imagine water is being poured into a container that has an outlet on the side. At first, when the water level is below the outlet, the water level in the container rises over time, as shown in (a). But once the water level reaches a certain height, the rate at which water is entering the container equals the rate at which it is leaving through the outlet. At this point, the water level stops rising and stays constant, as shown in (b). This is when the system has reached a steady state.

To describe the total volume of water in the system, let's denote it as w. In our differential equation, we aim to express how this total volume changes. We can say that the rate of change of the water volume remains constant, either with respect to time or space. This rate of change can be written as:

$$\frac{dw}{dt} = f(t, x_1, x_2, \ldots)$$

This equation means that the rate at which the total volume of water w is a function of time and multiple sources and sinks of water which I have expressed as x_1,x_2. Using the above definition of steady state, here the rate of change does not change over time, we can take the derivative of the equation above:

$$\frac{d}{dt}\left(\frac{dw}{dt}\right) = \frac{d^2w}{dt^2} = w'' = 0$$

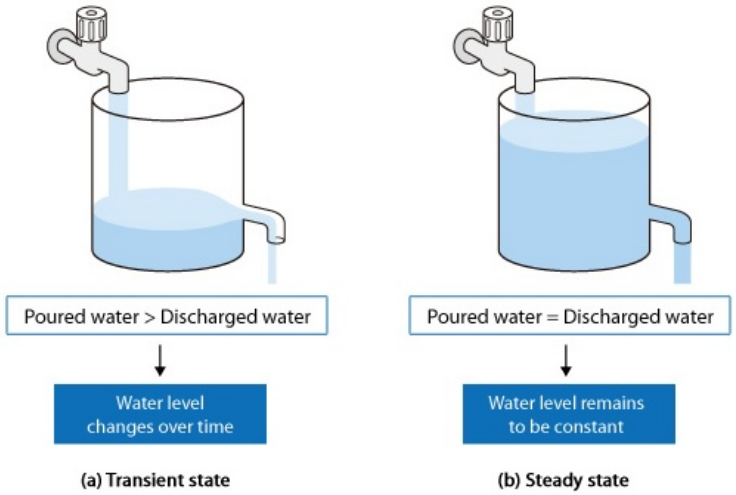


Figure 5.1: Diagram showing a system in (a) transient state and (b) steady state.

This equation tells us that the second derivative of w with respect to time is zero. In other words, the rate of change of the volume of water is not changing—it remains constant. This implies that there is a steady flow, and the total amount of water in the system remains constant. Let us work through an example.

Example 5.2: Imagine an object moving under constant acceleration. Mathematically, we can express the fact that acceleration is constant by saying:

$$\frac{da}{dt} = 0$$

This means the rate of change of acceleration is zero. Since acceleration is the rate of change of velocity (see above), we can also write this as:

$$\frac{da}{dt} = \frac{d^2v}{dt^2} = 0$$

This tells us that the velocity is not changing in a way that would cause acceleration to vary. If we had an expression for velocity as a function of time, we could differentiate that expression and check if this is true. Take a hypothetical example where the velocity is given by

$$v = \alpha t + 10$$

where t is time (in seconds) and a is a constant (in m/s^{-2}. We can show that the second-order derivative is 0 through the following steps:

$$a = \frac{dv}{dt} = \alpha \implies \frac{da}{dt} = \frac{d^2v}{dt^2} = 0$$

Since α is a constant, the acceleration does not change over time. Differentiating confirms that the acceleration is constant, and the second derivative of velocity with respect to time is zero, as expected and the system is in steady state.

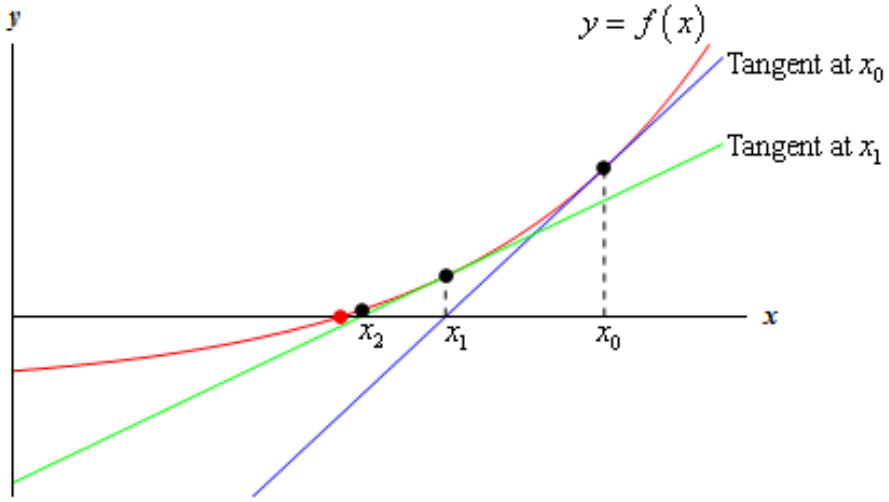


Figure 5.2: Function that varies non-linearly with x and example tangents at different points. Credit to https://tutorial.math.lamar.edu/Classes/CalcI/NewtonsMethod.aspx

5.3 Finding roots of equations: Newton-Raphson method

We've been discussing how derivatives help us understand change. So far, most of our examples have been a combination of hypothetical and real-world examples on using derivatives to calculate the gradient (or slope) of a line. While this is a useful concept, it can sometimes be difficult to see how it applies to real-world situations. When we start learning about integration, we'll see that we can reverse the process of differentiation. This allows us to calculate actual values (or "absolute values") from the past and predict future values.

But derivatives have other practical uses too. For example, one important application is **finding the roots** of a function, which is where a **function equals zero**. You have encountered these when finding stationary values! Take a look at Figure 5.2, which shows a typical function $y = f(x)$. Our function line clearly shows that when $x = 0$, y has a negative value and as x increases, y increases. At some point, the function crosses the x-axis and becomes zero. How do we find out when this occurs?

In some cases, we might be able to solve the equation directly to find the root (see above). However, consider the function:

$$f(x) = 2x^2 + 5 - e^x$$

In this case, it's difficult to find where $f(x) = 0$ using simple algebra. Instead, we could iteratively insert values for x and gradually work towards smaller and smaller values for y. By doing this repeatedly, we could gradually get closer to the root. However, this approach

is not only time-consuming and laborious, but also impossible to visualise for complex functions. What if there were a better way? What if we could use the information provided by the function's gradient (or slope), that is, the tangent line to help us find the root more efficiently?

Imagine our first value for x is denoted as x_0. At this point, the value of the function $f(x_0) > 0$, and we can calculate the slope (or gradient) at x_0, which we write as $f'(x_0)$. This slope represents the tangent line to the curve at x_0. Now, follow the tangent line from x_0 until it crosses the x-axis. The tangent crosses the x-axis and arrives at another value for x which we shall denote as x_1. We can write the equation of the tangent line as follows:

$$y = f(x_0) + f'(x_0)(x - x_0)$$

To find where the tangent line crosses the x-axis, we set $y = 0$:

$$0 = f(x_0) + f'(x_0)(x_1 - x_0)$$

Rearranging to get x_1:

$$x_1 - x_0 = -\frac{f(x_0)}{f'(x_0)} \implies x_1 = x_0 - \frac{f(x_0}{f'(x_0)}$$

So, x_1 is the new estimate for the root of the function. Next, to find x_2, we repeat the same process, using x_1 as the new starting point and get:

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

As we continue this process, the values of x will get closer and closer to the true root. However, we may never reach an exact value of $y = 0$. Instead, we define a tolerance level for what we consider as $y = 0$ or the percentage change in x. For example, we may stop the iterations if y is smaller than a certain threshold (e.g., $y < 0.001$) or if the change in x is very small (e.g., less than 0.01%). This depends on the nature of the problem you are trying to solve. There is no guarantee this method will converge to a root. Indeed there are cases where each iteration will get worse and worse. However it is good and simple method for root finding in cases where we cannot easily access an analytical solution. The general formula for the Newton-Raphson method is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

This formula is used iteratively to find increasingly accurate approximations to the root of the function. Let's consider an example.

Example 5.3: Suppose we want to find the root of this equation to an accuracy of 6 decimal places:

$$f(x) = 2x^2 + 5 - e^x$$

Since it is not feasible to solve $f(x) = 0$ through simple rearrangement, we employ Newton's method. This requires the derivative of $f(x)$, which is:

$$f'(x) = \frac{df}{dx} = 4x - e^x$$

To arrive at the value for x at the root, we need an initial guess. This can be important. We are told, on this occasion, the root exists somewhere between $x = 3$ and $x = 4$. If we start at $x_0 = 3.5$, what is the value for x_1? Using Newton's method, the next approximation, x_1, is computed as:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 3.5 - \frac{-3.615}{-19.115} = 3.310862 \tag{5.1}$$

In the class examples, you are asked to proceed with this problem until you arrive at an approximation for the root which is ≈ 3.275601.

Exercises

Algebraic problems

Question 5.1: Find the second order derivative of the following functions (we have seen two of these formulas when we practised using the product and quotient rule):

(a) $y = \sqrt{x^2 + 25}$
(b) $y = \frac{x^5}{4} + x^3 + 12$
(c) $f(x) = x\exp(x)$
(d) $P(z) = P_0 \exp\left(-\frac{g}{R_a T} z\right)$ which is the same simplified form of the hydrostatic equation we have met before, where P_0, g, R_a and T are constants.

Question 5.2: Find the stationary values of the following functions:

(a) $f(x) = \frac{x^3}{3} + \frac{x^2}{2} - 30x + 5$
(b) $f(x) = x^3$

Verify whether the stationary values are maxima, minima or saddle points.

Question 5.3: Complete Chapter 5 Exercise 3, and use Newtons Method to find the root (accurate to 6 decimal places) of following function:

$$f(x) = 2x^2 + 5 - e^x$$

Assume the root is between $x = 3$ and $x = 4$. Newtons method is used for finding the root, so you need to move all terms to one side. Use a first guess of 3.5.

Question 5.4: Using Newtons method, find the second guess of the root (i.e. x_2) for the following functions:

(a) $f(x) = x^3 - 7x^2 + 8x - 3$, with $x_0 = 5$
(b) $f(x) = sin(x) - ln(x)$, with $x_0 = 2$
(c) $f(x) = xcos(x) - x^2$, with $x_0 = 1$

Question 5.5: Use Newtons method to find all the roots of the following equation to 3 decimal places. You may use a starting guess of $x_0 = -3.5$ and $x_0 = 4.5$ to find the left and right sided roots respectively:

$$x^3 - x^2 - 15x + 1 = 0$$

Take a look at Figure 5.3. We seem to have 3 roots, so use this information to confirm the values of these roots to within 3 decimal places.

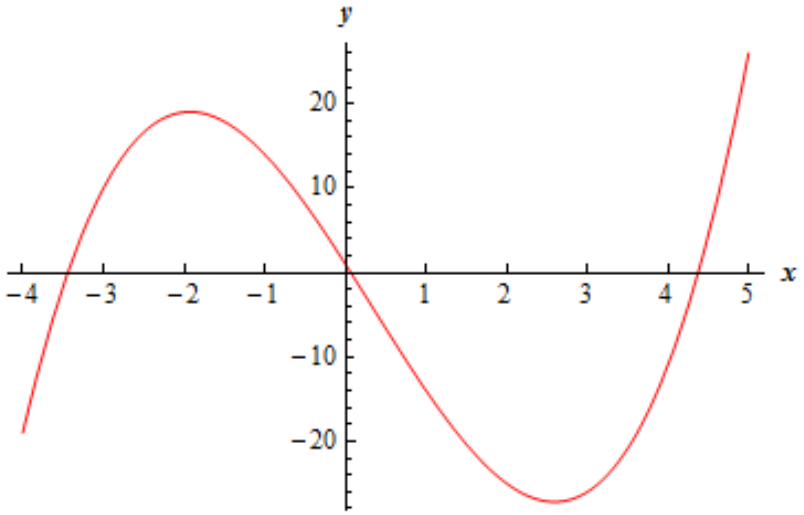


Figure 5.3: Function that varies according to a cubic function

Real-world problems

Question 5.6: An industrial incinerator produces a steady output of particulate matter having a concentration of $C_0 = 1.0$ mg m^{-3} of air at the chimney. 1 km away from the incinerator plant the concentration is $C_{1\,km} = 0.1$ mg m^{-3}. Some houses are a distance of 500 m away from the plant. The equation that lets us calculate the concentration as a function of distance from the incinerator is provided by the following expression.

$$C(x) = C_0 + \frac{C_{1\,km} - C_0}{1000}x$$

Prove that the flux is in steady state.

Question 5.7: At 3km below the surface there is a hot granite bed, with a temperature of 250 °C. The temperature just below the surface the temperature stays relatively constant at 10°C. Fourier's law in 1-d has been used to construct the following expression.

$$T(x) = 10 + \frac{250 - 10}{3000}x$$

Is the system in steady state?

Question 5.8: Imagine you are a business analyst for a company that sells a product. The company wants to model its gross profit, $P(x)$, based on the number of units sold, where x represents the number of units sold in thousands. The profit function is given by:

$$P(x) = e^x - 3x^2$$

(a) Determine the number of units sold where the gross profit is at its minimum. *HINT: You may need to apply Newton's method to find the root of the derivative in order to locate the stationary point. Use a starting guess of $x_0 = 3$ and find the root to 3 decimal places.*

(b) Confirm that the gross profit is minimised at this value.

Solutions

Question 5.1:

(a) For the first-order derivative (from Chapter 3 Question 5.4(a)), we split the function and differentiated using the chain rule:

$$\begin{aligned} u(v) = \sqrt{(v)}, \quad &\Longrightarrow \quad \frac{du}{dv} = \frac{1}{2\sqrt{v}}, \\ v(x) = x^2 + 25 \quad &\Longrightarrow \quad \frac{dv}{dx} = 2x \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= \frac{2x}{2\sqrt{x^2+25}} \\ &= \frac{x}{\sqrt{x^2+25}} \end{aligned}$$

For the second-order derivative (from Chapter 3 Question 5.4(d)), we split the function and differentiate using the quotient rule:

$$\begin{aligned} u(x) = x, \quad &\Longrightarrow \quad \frac{du}{dx} = 1, \\ v(x) = \sqrt{x^2+25} \quad &\Longrightarrow \quad \frac{dv}{dx} = \frac{x}{\sqrt{x^2+25}} \end{aligned}$$

Apply the quotient rule:

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{v \times \frac{du}{dx} - u \times \frac{dv}{dx}}{v^2} \\ &= \frac{\sqrt{x^2+25} - \frac{x^2}{\sqrt{x^2+25}}}{x^2+25} \\ &= \frac{1}{\sqrt{x^2+25}} - \frac{x^2}{(x^2+25)^{\frac{3}{2}}} \end{aligned}$$

(b) The first-order derivative is provided by the following expression:

$$\frac{dy}{dx} = \frac{5x^4}{4} + 3x^2$$

Therefore the second-order derivative is provided by the following expression:

$$\frac{d^2y}{dx^2} = 5x^3 + 6x$$

(c) For the first-order derivative (from Chapter 3 Question 5.2(a)), we split the function and differentiated using the product rule:

$$\begin{aligned} u(x) = x, \quad &\Longrightarrow \quad \frac{du}{dx} = 1, \\ v(x) = \exp(x) \quad &\Longrightarrow \quad \frac{dv}{dx} = \exp(x) \end{aligned}$$

Apply the product rule:

$$\begin{aligned}\frac{df}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= \exp(x) + x\exp(x)\end{aligned}$$

To compute the second-order derivative, note that the first term, $exp(x)$, remains unchanged due to the exponential derivative rule. For the second term, $xexp(x)$, the product rule must be applied again. Interestingly, this term is the same structure as the original function, allowing us to substitute it back and simplify further, yielding:

$$\begin{aligned}\frac{d^2f}{dx^2} &= \exp(x) + \exp(x) + x\exp(x) \\ &= (2+x)\exp(x)\end{aligned}$$

(d) Our final equation is the hydrostatic equation. For the first-order derivative (from Chapter 2 Question 5.8), we that this is an exponential function with multiple constants so:

$$\frac{dP}{dz} = P_0 \frac{g}{R_a T} \exp\left(-\frac{g}{R_a T} z\right)$$

To obtain the second derivative, $\frac{d^2P}{dz^2}$, we again recognise there are multiple constants and retain those that multiple an expression involving z:

$$\frac{d^2P}{dz^2} = P_0(-\frac{g}{R_a T})(-\frac{g}{R_a T}) \exp\left(-\frac{g}{R_a T} z\right)$$

which we can further simplify to the following:

$$\frac{d^2P}{dz^2} = P_0(\frac{g}{R_a T})^2 \exp\left(-\frac{g}{R_a T} z\right)$$

Alternatively, notice that we can write

$$\frac{dP}{dz} = -\frac{g}{R_a T} P$$

so then

$$\frac{d^2P}{dz^2} = -\frac{g}{R_a T}\frac{dP}{dz} = \left(\frac{g}{R_a T}\right)^2 P$$

Question 5.2: Find the stationary values of the following functions:

(a) We find the first derivative

$$f'(x) = \frac{d}{dx}\left(\frac{x^3}{3} + \frac{x^2}{2} - 30x + 5\right) = x^2 + x - 30$$

and solve $f'(x) = 0$ to find stationary points

$$x^2 + x - 30 = 0$$
$$(x-5)(x+6) = 0$$

So we see two stationary values at $x = -6$ and $x = -5$. We then use the second derivative to classify the stationary values. The second derivative is:

$$f''(x) = \frac{d}{dx}(x^2 + x - 30) = 2x + 1$$

At the stationary values, the second derivative is:

$$f''(-6) = 2(-6) + 1 = -11 < 0$$
$$f''(5) = 2(5) + 1 = 11 > 0$$

so $x = -6$ is a maximum and $x = 5$ is a minimum.

(b) We find the first derivative

$$f'(x) = \frac{d}{dx}(x^3) = 3x^2$$

and solve $f'(x) = 0$ to find stationary points

$$3x^2 = 0 \implies x = 0$$

So we see a stationary value at $x = 0$. We then use the second derivative to classify the stationary point. The second derivative is:

$$f''(x) = 6x$$

At the stationary values, the second derivative is:

$$f''(0) = 0$$

Since $f''(0) = 0$, we cannot classify the stationary point using the second derivative test. Instead, consider the behaviour of $f(x) = x^3$:

- For $x < 0$, $f'(x) > 0$, indicating $f(x)$ is increasing.
- For $x > 0$, $f'(x) > 0$, indicating $f(x)$ is still increasing.

Thus, $x = 0$ is a saddle point.

Question 5.3: In this Question 5.we are asked to find a root within a specified accuracy, assuming it is between $x = 4$ and $x = 6$ for the following function:

$$f(x) = 2x^2 + 5 - e^x$$

As we saw in Chapter 5, we need to first derive our derivative function which, in this case, is:

$$\frac{df}{dx} = 4x - e^x$$

We iterate through the Newton method:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

until our only changes at an accuracy greater than 6 decimal places (i.e. we can stop at the chosen level of tolerance).

$$x_1 = 3.500000 - \frac{-3.615451}{-19.115451} = 3.310862$$
$$x_2 = 3.310862 - \frac{-0.485131}{-14.165301} = 3.276614$$
$$x_3 = 3.276614 - \frac{-0.013546}{-13.379492} = 3.275601$$
$$x_4 = 3.275601 - \frac{-1.15205 \times 10^{-5}}{-13.356740} = 3.275601$$

so the root of $f(x)$ to 6 decimal places is 3.275601.

Question 5.4: In this Question 5.we are asked to find a root after two iterations of Newtons method. However, you may choose your own stopping criteria for practice

(a) We need to derive the derivative of the function

$$f(x) = x^3 - 7x^2 + 8x - 3$$

which takes the form:

$$\frac{df}{dx} = 3x^2 - 14x + 8$$

We then apply the Newton method twice:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

with a starting guess of $x_0 = 5$:

$$x_1 = 5 - \frac{-13}{13} = 6$$
$$x_2 = 6 - \frac{9}{32} = 5.71875$$

so the root of $f(x)$ after two iterations is 5.71875.

(b) We need to derive the derivative of the function

$$f(x) = sin(x) - ln(x)$$

which takes the form:

$$\frac{df}{dx} = cos(x) - \frac{1}{x}$$

We then apply the Newton method twice:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

with a starting guess of $x_0 = 2$:

$$x_1 = 2.000000 - \frac{0.216150}{-0.916146} = 2.235934$$
$$x_2 = 2.235934 - \frac{-0.017827}{-1.064407} = 2.219185$$

so the root of $f(x)$ after two iterations is 2.219185.

(c) We need to derive the derivative of the function

$$f(x) = xcos(x) - x^2$$

To differentiate $h(x) = xcos(x)$ we split the function and differentiate using the product rule:

$$u(x) = x, \quad \Longrightarrow \quad \frac{du}{dx} = 1,$$
$$v(x) = \cos(x) \quad \Longrightarrow \quad \frac{dv}{dx} = -\sin(x)$$

Apply the product rule:

$$\begin{aligned}\frac{dh}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= \cos(x) - x\sin(x)\end{aligned}$$

so we see that

$$\frac{df}{dx} = \cos(x) - x\sin(x) - 2x$$

We then apply the Newton method twice:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

with a starting guess of $x_0 = 1$:

$$x_1 = 1.000000 - \frac{-0.459697}{-2.301168} = 0.800232$$
$$x_2 = 0.800232 - \frac{-0.082978}{-1.478108} = 0.744094$$

so the root of $f(x)$ after two iterations is 0.744094.

Question 5.5: In this Question 5.we are asked to find all the roots to 3 decimal places for the following function:

$$f(x) = x^3 - x^2 - 15x + 1 = 0$$

In this problem we have been given a figure to help us decide on where to start our initial guess for each root. Since we have 3 roots (according to what we can see), it makes sense to select values near these points. We have been given $x_0 = -3.5$ and $x_0 = 4.5$ as starting values but there is also a root near zero so lets also use $x_0 = 0$ as the final starting value to check this. We need to derive the derivative of $f(x)$, which takes the form:

$$\frac{df}{dx} = 3x^2 - 2x - 15 = 0$$

We then apply the Newton method:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

for each starting guess until the root is accurate within 3 decimal places:

- For $x_0 = -3.5$:

$$x_1 = -3.500 - \frac{-1.625}{28.75} = -3.443$$
$$x_2 = -3.443 - \frac{-0.036}{27.459} = -3.442$$
$$x_3 = -3.442 - \frac{-2.008 \times 10^{-5}}{27.429} = -3.442$$

- For $x_0 = 0$:

$$x_1 = 0.067$$
$$x_2 = 0.066$$
$$x_3 = 0.066$$

- For $x_0 = 4.5$:

$$x_1 = 4.381$$
$$x_2 = 4.376$$
$$x_3 = 4.376$$

so the three roots of $f(x)$ to 3 decimal places are -3.442, 0.066 and 4.376. Please note that the following solution is dependent on the starting guess so your results may change. You may even find your value grows! So care is needed when there are multiple roots.

Question 5.6: We need to prove that an industrial incinerator emissions

$$C(x) = C_0 + \frac{C_{1km} - C_0}{1000}x$$

are in steady state. To do this we differentiate, the

$$\frac{dC}{dx} = \frac{C_{1km} - C_0}{1000}$$

which is constant, so the flux is in steady state. Verify this by taking the second derivative

$$\frac{d^2C}{dx^2} = 0$$

As the second derivative is zero this verifies that the flux is in steady state.

Question 5.7: We need to prove that a system

$$T(x) = 10 + \frac{250 - 10}{3000}x$$

are in steady state. To do this we differentiate, the

$$\frac{dT}{dx} = \frac{240}{3000}$$

which is constant, so the system is in steady state. Verify this by taking the second derivative

$$\frac{d^2T}{dx^2} = 0$$

As the second derivative is zero this verifies that the system is in steady state.

Question 5.8:

(a) We need to determine where the profit

$$P(x) = e^x - 3x^2$$

is at its minimum, which corresponds to a stationary point. To find this, we differentiate $P(x)$ and set the derivative equal to zero:

$$\frac{dP}{dx} = e^x - 6x = 0$$

Since this equation cannot be solved algebraically, we use Newton's method to approximate the root. It's important to note that we are solving for the root of $\frac{dP}{dx}$, not $P(x)$. Let us define $f(x) = \frac{dP}{dx}$, and compute its derivative:

$$f(x) = e^x - 6x \implies f'(x) = e^x - 6$$

Using Newton's method, the iterative formula is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Starting with an initial guess of $x_0 = 3$, we iterate until the values converge to an accuracy of three decimal places:

$$x_1 = 3 - \frac{2.0855}{14.0855} = 2.8519$$
$$x_2 = 2.8519 - \frac{0.2096}{11.3213} = 2.8334$$
$$x_3 = 2.8334 - \frac{0.0029}{11.0034} = 2.8331$$

Thus, the root of $f(x)$ to three decimal places is $x = 2.833$. Therefore, the profit is at its minimum when approximately 2833 units are sold.

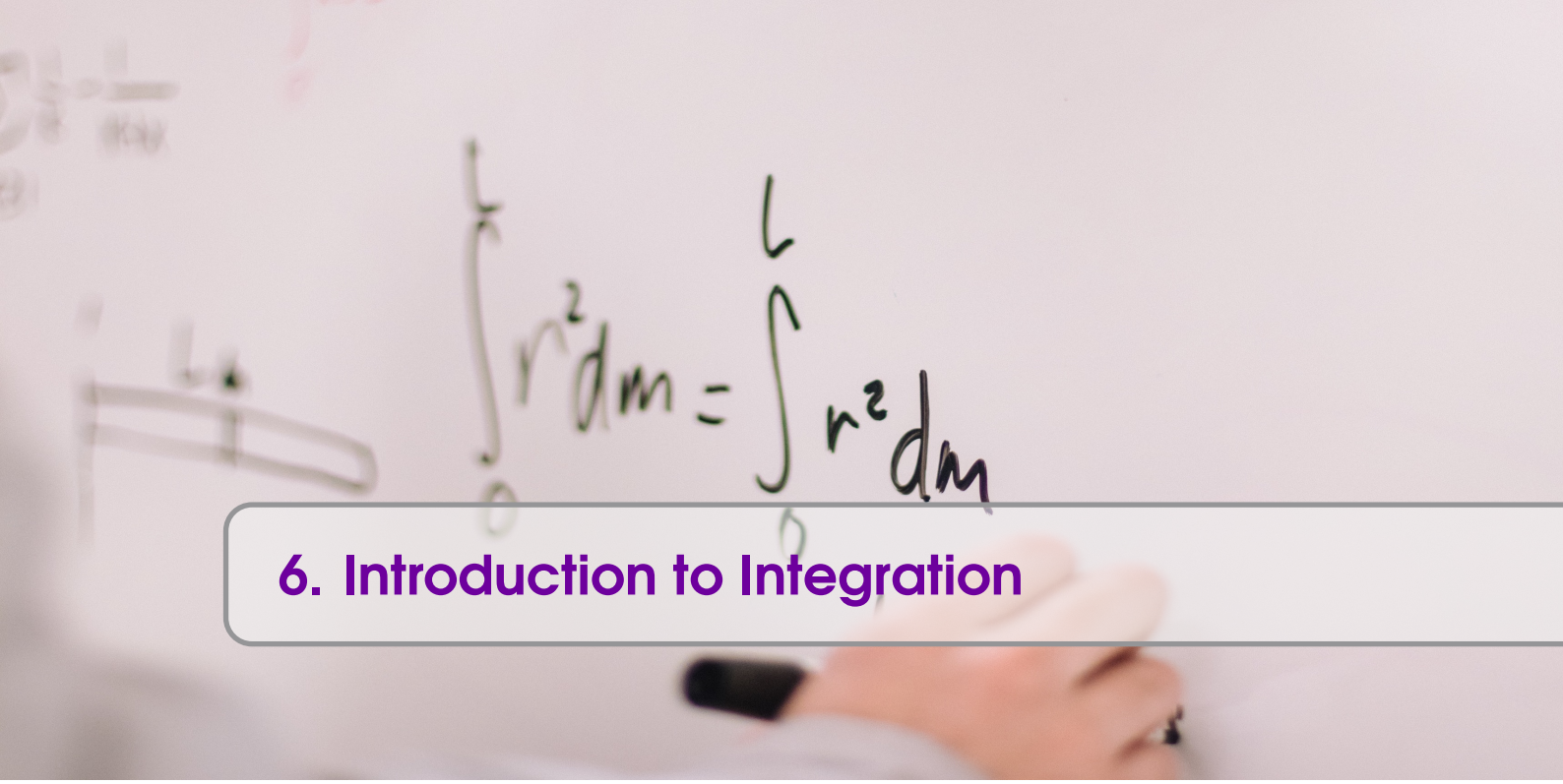
(b) We use the second derivative to classify stationary values. We already calculated the second derivative in the last part through Newton's method:

$$P(x) = e^x - 3x^2$$
$$\frac{dP}{dx} = f(x) = e^x - 6x$$
$$\frac{d^2P}{dx^2} = f(x) = e^x - 6$$

so at the 2.833 the second derivative is

$$\frac{d^2P}{dx^2}(2.833) = e^2.833 - 6 = 10.99889 > 0$$

Since the second derivative is positive, then the profit is definitively minimised at 2.833.



6. Introduction to Integration

6.1 Overview

Throughout Chapters 1–5, we considered at differentiation as a topic. This is the process of finding a rate of change (or gradients) of functions or processes. However, there may be times where we want to do the opposite, i.e. we know the rate of change but want to know the volume or total change over an interval. For example, we might know how fast a car is traveling and we want to know how far that car has traveled over time. This is where **integration** comes in.

6.2 Integration as the reverse of differentiation

Integration is the **reverse** of differentiation (Figure 6.1). Since the derivative function provides us with a gradient, integration therefore provides us with a function that can be used to calculate absolute values. For example, if a derivative function tells us how much the volume of water in a tank is changing over time, integrating that function tells us how much water is in the tank at a given time or how much the volume of water has changed over a specific time period.

To integrate a differential equation therefore requires us to reverse the steps taken to get to that point. At first glance, this seems like an impossible task since it is likely we did not generate the differential equation. However you will start to recognise forms of equations and what steps you therefore need to carry out.

Let us keep the water example for now to explore this using a graph. If we consider the

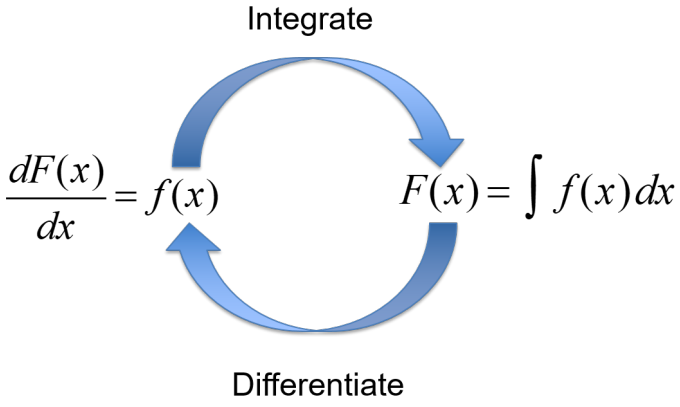


Figure 6.1: Interplay between differentiation and integration throughout this course.

flow rate of water through a hole we might obtain a graph similar to Figure 6.2. The top graph plots the rate of change as a function of time, thus the derivative function. Look at the middle part, between 30 and 70 seconds. Here the rate of change is constant. So, after 1 second, the water has increased by a fixed amount. In the next second, the water increases by the same amount. Now we can consider working out the area under the curve to arrive at the total amount. Indeed, if we want to describe how the volume of water in the system changes with time, t, then we can calculate it from the above graph by finding the area under the graph from $Time = 0$ to any time, t. For Fig. 6.2 we can find the area by adding up areas of triangles and rectangles.

Often flow rate curves are not as simple as in Fig. 6.2. Fig. 6.3 shows the curve $f(x) = -0.1x^2 + 10x$ and a schematic of how to estimate the area under the curve. Basically it shows that we can split the area into lots of smaller rectangles, with a base of length Δx and a height equal to $f(x)$, the value of the function at x. The area of each rectangle is then $f(x) \times \Delta x$ and the area of the curve is the sum of all of these rectangles. This is a **numerical approximation** method, similar to central differences method for differentiation, representing a manual integration of the function.

We can use (analytical) *integration* to find a function that describes the area under the curve. In this case the function $\frac{-0.1x^3}{3} + 5x^2$ describes the area under the curve between $x = 0$ and x. This is shown in the bottom graph of Figure 6.3.

Because integration can be seen as a *summing up* of lots of smaller rectangles we write it formally with an S-shaped symbol:

$$\int_0^x \left(-0.1x^2 + 10x\right) dx = \frac{-0.1x^3}{3} + 5x^2$$

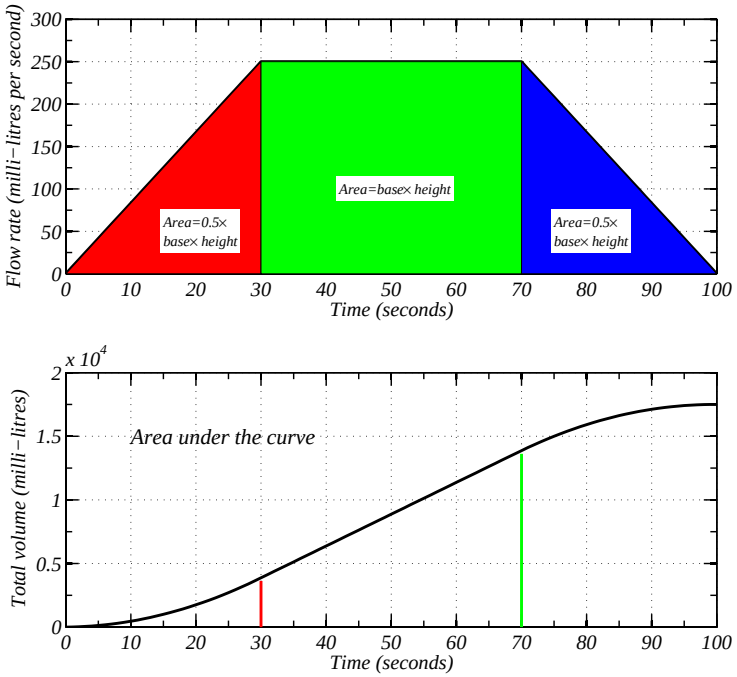


Figure 6.2: Top shows a graph of flow rate of a substance vs time; bottom shows the volume of that substance vs time (i.e. the area under the top curve). We see that the volume increases at a non-constant rate at first, but between 30 and 70 seconds the volume increases according to a straight line. After 70 seconds the volume increases at a non-constant rate, before leveling off

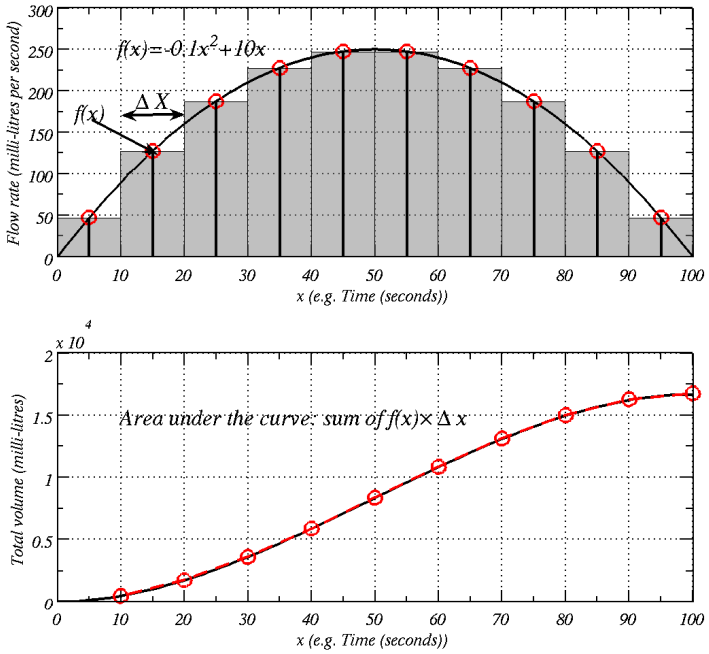


Figure 6.3: Top shows the function $f(x) = -0.1x^2 + 10x$. The schematic shows a way of estimating the area under the curve by dividing the area into lots of rectangles. Bottom shows a function describing the area from $Time = 0$ to a later time, t.

here the $\int$ symbol represents a 'summing up'. The dx represents the length of the base of each rectangle (i.e. Δx) and the sub and super-scripts on the $\int$ are known as *limits of integration*. In this example it means find the area under the curve between 0 and x.

The *integral* of some simple functions of x are shown below:

Function	Integral
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
$3x^2$	$x^3 + d$
x^3	$\frac{x^4}{4} + d$
$2x^4$	$\frac{2x^5}{5} + d$
$\vdots$	$\vdots$
x^n	$\frac{x^{n+1}}{n+1} + d$

where d is a constant. So we see that given these simple functions of x we can calculate the integral of them by **raising the power of x by 1** and **dividing by the new power that x is raised**. Why do we add the constant, d? Well, integration is the opposite of differentiation (remember differentiation of a constant is zero) and differentiation "loses" any constant that might have been present in the original function so we add this in. For standard integration we cannot say what the constant is: it can take on any value so we just call it a **constant**. For this reason it is called **indefinite integration**.

To find the **area under a curve** we have to perform **definite integration** between limits. This means that we calculate the integral (as in the table above) then evaluate the integral for two values of x, and calculate the difference.

Example 6.1: Let's find the area under the curve $y = x^2$ from $x = 1$ to $x = 2$. We write the integration out as follows:

$$\begin{aligned}
\int_1^2 x^2 dx &= \left[\frac{x^3}{3}\right]_1^2 \\
&= \left[\frac{2^3}{3}\right] - \left[\frac{1^3}{3}\right] \\
&= \left[\frac{8}{3}\right] - \left[\frac{1}{3}\right] \\
&= \frac{7}{3}
\end{aligned}$$

note that we calculate the integral and then evaluate the function by putting in both limits and calculating the difference.

Note that when calculating a difference the constant in the integration, d, disappears so we do not have to consider it.

In the table below you can find a list of common integrals we will use through this course:

Differential equation $\frac{dy}{dx}$	Integral, y
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
$3x^2$	$x^3 + d$
x^3	$\frac{x^4}{4} + d$
$\vdots$	$\vdots$
x^n	$\frac{x^{n+1}}{n+1} + d$
ax^n	$\frac{ax^{n+1}}{n+1} + d$
ke^{kx}	$e^{kx} + d$
$\frac{1}{x}$	$ln(x) + d$
$cos(x)$	$sin(x) + d$
$sin(x)$	$-cos(x) + d$

6.3 Solving Ordinary Differential Equations (ODEs)

Now we can change our narrative a little with regards to solving problems in Physics. So far, we have been given $f(x)$ and asked to find $\frac{df}{dx}$, but now we are starting with $\frac{df}{dx}$ instead. We often refer to $\frac{df}{dx}$ as **Ordinary Differential Equations (ODEs)**.

The theoretical base of many of the processes we are interested in are written in the form of ODEs. To predict values or properties in the past or future, we can integrate ODEs either without limits, or between specific limits. In this Chapter, we solve the ODEs that describe radioactive decay and Newtons laws of motion, following some algebraic exercises.

The radioactive decay and hydrostatic equation have the same algebraic formulation. They are both equations that describe a logarithmic change in one variable as a function of another. Why is that? By looking at the **ODE** that describes a rate of change:

$$\frac{dy}{dx} = \frac{1}{x}$$

then we know the solution to this general equation will take the form of the following equation:

$$y = ln(x) + d$$

In the following, we relate the derivative with the analytical integrals. For more information you may refer to Chapter 11 that covers all of the physical theories we cover in this course.

Example 6.2: The radioactive decay equation. For α and β decay we can apply the radioactive decay equations. The law states that the number of decays per unit time is proportional to the number of unstable nuclei:

$$\frac{dN}{dt} = -\lambda N$$

where λ is a constant for a specific isotope and N is the number of unstable nuclei. This kind of law is quite common in physics. It is to be expected since if you have more unstable nuclei you would expect to observe more decays per second and vice versa. Another statement of this law is that the number of unstable nuclei decays away with time:

$$N(t) = N_0 \exp(-\lambda t)$$

where N_0 is the number of unstable nuclei at time $t = 0$. The constant λ is used to characterise how quickly the unstable nucleus decays. To get this solution, we use a technique called **separation of variables**. We rearrange the decay equation so that all N's are on one side and all t's are on the other and then integrate both sides:

$$\frac{dN}{dt} = -\lambda N$$

We first divide both sides by N

$$\frac{1}{N}\frac{dN}{dt} = -\lambda \frac{\cancel{N}}{\cancel{N}}$$

Then multiply both sides by dt

$$\frac{1}{N}\frac{dN}{\cancel{dt}} \times \cancel{dt} = -\lambda dt$$

Then integrate both sides

$$\int \frac{1}{N}\, dN = -\lambda \int dt$$

The integral $\int \frac{1}{x} dx$ is equal to $\ln x$ so we have

$$\ln N = -\lambda t + c$$

and if we say at $t = 0$, $N = N_0$ we get that $c = \ln N_0$, we therefore obtain

$$N = N_0 \exp(-\lambda t)$$

as the solution to Newton's radioactive equations.

Example 6.3: The hydrostatic equation. The density of air depends on the pressure, P, and temperature of the air, T, (through the ideal gas law): $\rho = \frac{P}{R_a T}$, where $R_a = 287$ J kg^{-1} K^{-1}. So we can write:

$$\frac{dP}{dz} = -\frac{P}{R_a T} g$$

We can use separation of variables and integration again to show that the solution to this ODE is:

$$P(z) = P_0 \exp\left(-\frac{g}{R_\alpha T} z\right)$$

In actual fact the assumption that the temperature is constant is not true as it usually decreases with height in the atmosphere at about 10°C km^{-1} (i.e. a drop of 10 °C for every km). When we take into account that temperature drops with altitude the hydrostatic equation becomes:

$$\frac{dP}{dz} = -\frac{P}{R_a (T_0 - 0.01z)} g$$

which has an analytical solution of:

$$P(z) = P_0 \exp\left(-\frac{100g}{R_a} \ln\left[\frac{T_0}{T_0 - 0.01z}\right]\right)$$

but we need some fancier integration techniques, which we will see later, to be able to show this!

Exercises

Algebraic problems

Question 6.1: Find the following indefinite integrals:

(a) $\int x^2 dx$
(b) $\int 12x^4 dx$
(c) $\int sin(x) dx$
(d) $\int cos(x) dx$
(e) $\int e^x dx$
(f) $\int e^{2x} dx$
(g) $\int x^2 + 2x + 3 dx$
(h) $\int sin(x) + cos(x) dx$
(i) $\int e^{2x} + 2x dx$
(j) $\int e^x + cos(x) dx$

Question 6.2: Find the integrals between the limits specified:

(a) $\int_0^{10} x^2 dx$
(b) $\int_{-10}^{10} x^2 dx$
(c) $\int_{-1}^{1} x^3 dx$
(d) $\int_0^{80} \left(-0.1x^2 + 10x\right) dx$
(e) $\int_0^{50} \left(-0.1x^2 + 10x\right) dx$
(f) $\int_0^1 \exp(3x)\, dx$

Question 6.3: Find the integrals between the limits specified:

(a) $\int_0^b \frac{h}{b} x dx$ (i.e. the area of a triangle with base b and height h)
(b) $\int_0^b h dx$
(c) $\int_0^T \exp(-\lambda t)\, dt$

Question 6.4: Use the fact that $\int ax^n = \frac{ax^{n+1}}{n+1} + d$ to integrate the following:

(a) $f(x) = \sqrt{x}$
(b) $f(x) = \frac{1}{x^2}$
(c) $f(x) = \frac{1}{\sqrt{x}}$

Question 6.5: Use separation of variables to integrate the following ODE:

$$\frac{dN}{dx} = 1/N$$

Real-world problems

Question 6.6: Water flows down a river at a rate $\frac{dW}{dt}$ (m^3 s^{-1}) described by:

$$\frac{dW}{dt} = 1000t^{0.5}$$

Determine how much water (m^3) flows down the river between $t = 100$ and $t = 700$ s?

Question 6.7: The rate (in mm/hr) that rain falls from a cloud between time $t = 3.5$ and $t = 6.5$ hours can be approximated by $-3t^2 + 30t - 65$. Calculate the total precipitation, in mm, that falls between $t = 3.5$ and $t = 6.5$ hours.

Question 6.8: The radioactive decay equation. Solve the radioactive decay equation by separation of variables and integration. Recall that it states that the number of decays per unit time is proportional to the number of unstable nuclei:

$$\frac{dN}{dt} = -\lambda N$$

where N is the number of unstable nuclei, t is time and λ is called the decay constant, $\lambda = \frac{\ln 2}{t_{1/2}}$ where $t_{1/2}$ is the half-life of the sample.

Question 6.9: Imagine a sheep was allowed to graze in a field that had been contaminated by ^{137}Cs released at Chernobyl. About a kilogram of ^{137}Cs was released, but only a small fraction of this was atomised and subsequently spread over a very wide area. The sheep had been eating grass that contained a surface density of radioactive nuclei. That density changes over time as the radioactive material decays. If we assume that the the surface density of atoms starts at a value of $2.16 \times 10^9 m^{-2}$, and the surface area A is 7.47 m^2, integrate the following equation to calculate the number of atoms ingested over 1 year:

$$A \int_{t=0}^{t=1} N(t)dt$$

where $N(t)$ is the radioactive decay equation. Note that, ^{137}Cs is a beta emitter with a half life of 30 years (it decays to ^{137}Ba).

Question 6.10: The hydrostatic equation. Solve the hydrostatic equation by separation of variables and integration. Recall that it states the density of air at altitude z depends on the pressure, P, and temperature of the air, T, (through the ideal gas law): $\rho = \frac{P}{R_a T}$, where $R_a = 287$ J kg^{-1} K^{-1}:

$$\frac{dP}{dz} = -\frac{P}{R_a T} g$$

Solutions

Question 6.1:

(a) $\int x^2 dx = \frac{x^3}{3} + d$

(b) $\int 12x^4 dx = \frac{12x^5}{5} + d$

(c) $\int sin(x) dx = -\cos(x) + d$

(d) $\int cos(x) dx = \sin(x) + d$

(e) $\int e^x dx = e^x + d$

(f) $\int e^{2x} dx = \frac{e^{2x}}{2} + d$

(g) $\int x^2 + 2x + 3 dx = \frac{x^3}{3} + x^2 + 3x + d$

(h) $\int sin(x) + cos(x) dx = -\cos(x) + \sin(x) + d$

(i) $\int e^{2x} + 2x dx = \frac{e^{2x}}{2} + x^2 + d$

(j) $\int e^x + cos(x) dx = e^x + \sin(x) + d$

Question 6.2:

(a) $\int_0^{10} x^2 dx = \left[\frac{x^3}{3}\right]_0^{10} = \frac{10^3}{3} - 0 = 333.33$

(b) $\int_{-10}^{10} x^2 dx = \left[\frac{x^3}{3}\right]_{-10}^{10} = \frac{10^3}{3} - \frac{(-10)^3}{3} = 666.67$

(c) $\int_{-1}^{1} x^3 dx = \left[\frac{x^4}{4}\right]_{-1}^{1} = \frac{1^4}{4} - \frac{(-1)^4}{4} = 0$

(d) $\int_0^{80} \left(-0.1x^2 + 10x\right) dx = \left[-0.1\frac{x^3}{3} + \frac{10x^2}{2}\right]_0^{80} = -0.1\frac{80^3}{3} + 5(80)^2 = 14933.33$

(e) $\int_0^{50} \left(-0.1x^2 + 10x\right) dx = -0.1\frac{50^3}{3} + 5(50)^2 = 8333.33$ (See above)

(f) $\int_0^1 \exp(3x)\, dx = \left[\frac{1}{3}\exp(3x)\right]_0^1 = \frac{1}{3}(\exp(3) - \exp(0)) = 6.3618$

Question 6.3: Find the integrals between the limits specified:

(a) $\int_0^b \frac{h}{b} x dx = \frac{h}{b}\left[\frac{x^2}{2}\right]_0^{10} = \frac{1}{2} \times \frac{h}{b} \times b^2 - 0 = \frac{1}{2}hb$ (half base times height—area of a triangle)

(b) $\int_0^b h dx = h[x]_0^b = h(b-0) = h \times b$ (base times height—area of a rectangle)

(c) $\int_0^T \exp(-\lambda t)\, dt = \left[-\frac{1}{\lambda}\exp(-\lambda t)\right] = -\frac{1}{\lambda}(-\exp(-\lambda t) + \exp(0)) = \frac{1}{\lambda}(1 - \exp(-\lambda T))$

Question 6.4: Use the fact that $\int ax^n = \frac{ax^{n+1}}{n+1} + d$ to integrate the following:

(a) Rewrite $\sqrt{x}$ as $x^{1/2}$ then $\int \sqrt{x} = \frac{2}{3}x^{3/2} + d$

(b) Rewrite $\frac{1}{x^2}$ as x^{-2} then $\int \frac{1}{x^2} = \frac{x^{-1}}{-1} + d = -\frac{1}{x} + d$

(c) Rewrite $\frac{1}{\sqrt{x}}$ as $x^{-1/2}$ then $\int \frac{1}{\sqrt{x}} = 2x^{1/2} + d = 2\sqrt{x} + d$

Question 6.5:We rearrange the decay equation so that all N's are on one side and all t's are on the other and then integrate both sides:

$$\frac{dN}{dt} = \frac{1}{N}$$

We first multiply both sides by N

$$N\frac{dN}{dt} = \frac{\cancel{N}}{\cancel{N}}$$

Then multiply both sides by dt

$$N\frac{dN}{\cancel{dt}} \times \cancel{dt} = dt$$

Then integrate both sides

$$\int N \; dN = \int dt$$

The integral $\int x \; dx$ is equal to $\frac{x^2}{2}$ so we have

$$\frac{N^2}{2} = t + c$$

Rearranging this equation gives us

$$N(t) = \sqrt{2t + c}$$

as the solution to the ODE.

Question 6.6: Perform definite integration of the function to get (and using Question 6.4(b)):

$$\begin{aligned}
\int_{100}^{700} 1000t^{0.5} dt &= 1000 \times \left[t^{3/2}\right]_{100}^{700} \\
&= 1000 * \left(700^{3/2} - 100^{3/2}\right) \\
&= 1.17 \times 10^7 m^3
\end{aligned}$$

Question 6.7: Perform definite integration of the function to get:

$$\begin{aligned}
\int_{3.5}^{6.5} -3t^2 + 30t - 65 dt &= 1000 \times \left[-t^3 + 15t^2 - 65t\right]_{3.5}^{6.5} \\
&= (-(6.5)^3 + 15(6.5)^2 - 65(6.5)) \\
&\qquad -((3.5)^3 + 15(3.5)^2 - 65(3.5)) \\
&= -63.375 + 86.625 \\
&= 23.25
\end{aligned}$$

Question 6.8: We use a technique called *Separation of variables*. We rearrange the decay equation so that all N's are on one side and all t's are on the other and then integrate both sides:

$$\frac{dN}{dt} = -\lambda N$$

We first divide both sides by N

$$\frac{1}{N}\frac{dN}{dt} = -\lambda\frac{\cancel{N}}{\cancel{N}}$$

Then multiply both sides by dt

$$\frac{1}{N}\frac{dN}{\cancel{dt}} \times \cancel{dt} = -\lambda dt$$

Then integrate both sides

$$\int \frac{1}{N}\, dN = -\lambda \int dt$$

The integral $\int \frac{1}{x} dx$ is equal to $\ln x$ so we have

$$\ln N = -\lambda t + c$$

and if we say at $t = 0$, $N = N_0$ we get that $c = \ln N_0$, we therefore obtain

$$N = N_0 \exp(-\lambda t)$$

as the solution to Newton's radioactive equations.

Question 6.9: We want to perform the integral

$$A\int_{t=0}^{t=1} N(t)dt$$

where

$$N = N_0 \exp(-\lambda t)$$

We integrate this equation we follow the steps below:

$$\begin{aligned}
A\int_0^1 N(t)dt &= A\int_0^1 N_0 \exp(-\lambda t)dt \\
&= AN_0\int_0^1 \exp(-\lambda t)dt \\
&= AN_0\left[-\frac{1}{\lambda}\exp(-\lambda t)\right]_0^1 \\
&= \frac{AN_0}{\lambda}\left(-\exp(-\lambda\times 1)+\exp(-\lambda\times 0)\right) \\
&= \frac{AN_0}{\lambda}\left(1-\exp(-\lambda\times 1)\right)
\end{aligned}$$

Plugging in that $t_{1/2} = 30$, $\lambda = \frac{\ln(2)}{t_{1/2}} = \frac{\ln(2)}{30} = 0.0231$, $A = 7.47$, $N_0 = 2.16\times 10^9$ we see that:

$$A\int_0^1 N(t)dt \cong 1.595\times 10^{10}$$

so 1.595×10^{10} atoms we eaten in that time.

Question 6.10: We use a technique called *Separation of variables*. We rearrange the hydrostatic equation so that all P's are on one side and all z's are on the other and then integrate both sides:

$$\frac{dP}{dz} = -\frac{P}{R_a T} g$$

We first divide both sides by P

$$\frac{1}{P}\frac{dP}{dz} = -\frac{1}{R_a T} g$$

Then multiply both sides by dz

$$\frac{1}{P}\frac{dP}{\cancel{dz}} \times \cancel{dz} = -\frac{1}{R_a T} g dz$$

Then integrate both sides

$$\int \frac{1}{P} dP = \int -\frac{g}{R_a T} dz$$

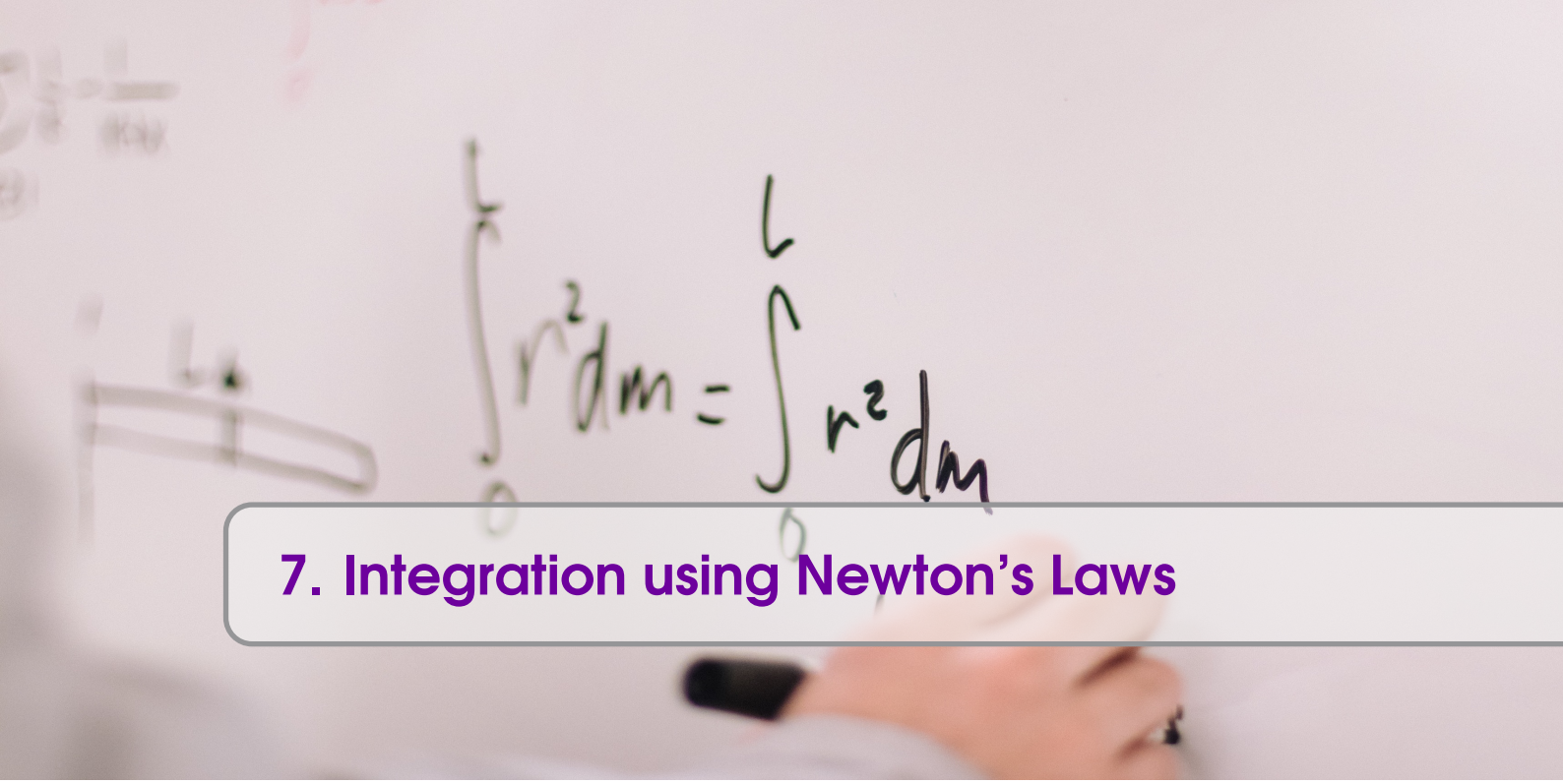
The integral $\int \frac{1}{x} dx$ is equal to $\ln x$ so we have

$$\ln P = -\frac{g}{R_a T} z + c$$

and if we say at $t = 0$, $P = P_0$ we get that $c = \ln P_0$, we therefore obtain

$$P(z) = P_0 \exp\left(-\frac{g}{R_a T} z\right)$$

as the solution to the hydrostatic equation.



7. Integration using Newton's Laws

7.1 Overview

In Chapter 6, we presented Integration as the reverse of differentiation. We discussed **definite** and **indefinite** integration and solving **ordinary differential equations (ODEs)**. We continue that here but focus on Newtons laws of motion as a special case.

7.2 Newtons laws of motion

Before the 20th century, our understanding of the physical world was built entirely on classical mechanics. More specifically - **Newtonian Physics**. This framework emerged from the collective efforts of the "Scientific Revolution", spanning figures like Nicolaus Copernicus, Galileo Galilei, and Isaac Newton.

Newton's laws of motion, a cornerstone of classical mechanics, were first presented in `Philosophiae Naturalis Principia Mathematica (1687)`, commonly known as the `Principia`. In 1543, Nicolaus Copernicus proposed a heliocentric model of the universe, placing the Sun rather than Earth at its center. This radical idea laid the groundwork for subsequent contributions by Galileo, Johannes Kepler, and René Descartes, who collectively challenged and replaced the Aristotelian worldview.

In the `Principia`, Newton formulated **three laws** to explain planetary motion, in order to explain why the orbits of the planets are ellipses rather than circles. These laws, however, extend far beyond planetary motion, offering a universal framework for understanding force and motion. Newton's laws are as follows:

1. **Law of Inertia:** A body at rest or moving at a constant speed in a straight line will remain in that state unless acted upon by an external force.
2. **Law of Force and Momentum:** The rate of change of a body's momentum is directly proportional to the force applied, with the change occurring in the same direction as the force (i.e. if a force acts on a body, that body will experience a change in momentum). This is captured in equations you see throughout this course.
3. **Law of Action and Reaction:** When two bodies interact, they exert forces on each other that are equal in magnitude and opposite in direction. You will have likely come across this law before; also known as the law of action and reaction.

In a number of our problems, we solve Newtons second law to predict how fast or far a body will travel down an incline. Since gravity is pulling the body downwards, why doesn't the body move through the surface? Well, in very simplistic terms the surface applies an equal and opposite force to the body resting on it. In reality, molecular forces that dictate the properties of both the object and surface also play a role but these laws are adequate to explain our observed motion.

7.3 Newtons second law of motion

For the remainder of this Chapter, we focus on Newton's second laws. In this law, we see that acceleration of an object depends on the mass of the object and the amount of force applied. This concerns a change in momentum. Newton's second law of motion is given by the following equation:

$$F = ma$$

where F is the net force acting on a body that has a mass m and therefore acceleration a.

In Chapter 5, we discussed second-order derivatives and observed that acceleration (a) is the rate of change of velocity (v), while velocity (v) is the rate of change of position or distance (x).Newton's second law can therefore be expressed as an ordinary differential equation:

$$F = m\frac{dv}{dt} \quad F = m\frac{d}{dt}\left(\frac{dx}{dt}\right) \quad F = m\frac{d^2x}{dt^2} \quad F = mv\frac{dv}{dx}$$

where these expressions are equal, and we use the fact that $a = v\frac{dv}{dx}$ for the final expression (because $v = \frac{dx}{dt}$, so $v\frac{dv}{dx} = \frac{dv}{dt}$, which is equal to a).

To help understand how this works, imagine a plane moving through the atmosphere. The plane has a constant mass m and its momentum depends on its velocity V, which is controlled solely by the plane (i.e., there are no external forces acting on it). The change in velocity between any two time points is $m(V_1 - V_0)$. If we define force as the rate of

change of momentum with respect to time, then:

$$F = m\frac{V_1 - V_0}{t_1 - t_0}$$

This resembles the calculation of a gradient from Chapter 1! As seen in Chapters 2, taking the limit as $t_1 - t_0 \to 0$, this becomes us the derivative:

$$F = m\,\frac{dv}{dt}$$

By formulating the total forces acting on an object, we can then solve the ODE to as a function of time to tell us something about the processes.

In multiple cases we are presented with problems where an object experiences multiple forces, acting in multiple directions. We will always need to ensure we understand the balance of each for as an algebraic formulation. Let's work through two examples which you can see Figure 7.1.

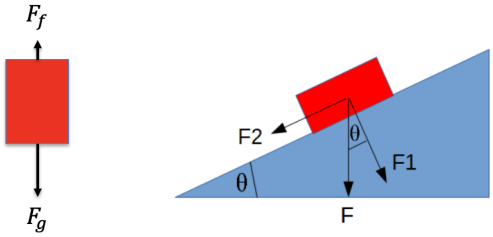


Figure 7.1: Schematic illustrating forces acting on a body (left) falling downwards and (right) placed on an incline

Example 7.1: Object Falling in a Vacuum . Consider an object falling in a vacuum, as shown on the left side of Figure 7.1. In this scenario, the object experiences only Earth's gravitational force, which acts vertically from the center of the object toward the Earth's center. If we assume an object is falling in a vacuum, can we prove that a brick fall the same distance as a feather in the same amount of time? Yes! In a vacuum, there is no air resistance, so the only force acting on the object is gravity, proportional to its mass (mg). This allows us to model the motion using a first-order ordinary differential equation (ODE) for velocity:

$$mg = m\,\frac{dv}{dt}$$

Canceling the mass (m) gives:

$$\frac{dv}{dt} = g$$

Using separation of variables, we can re-write our expression as an integral so we can calculate velocity as a function of time. We integrate to find the velocity, assuming the

object starts from rest ($v = 0$) at $t = 0$:

$$\int_0^v dv = \int_0^t g\, dt \quad \Longrightarrow \quad [v]_0^v = [gt]_0^t \quad \Longrightarrow v = gt$$

Thus, the velocity of any object in a vacuum increases linearly with time, regardless of its mass, therefore, all objects fall at the same velocity with respect to time. However, we want to show that all objects fall the same distance with respect to time. Remember that, velocity is the rate of change for the distance

$$\frac{dx}{dt} = gt$$

Integrating again, and assuming the object starts from $x = 0$ at $t = 0$, we get:

$$\int_0^x dx = \int_0^t gt\, dt \quad \Longrightarrow \quad [x]_0^x = \left[\frac{1}{2}gt\right]_0^t \quad \Longrightarrow x = \frac{1}{2}gt^2$$

This final expression for distance shows that all objects fall the same distance (quadratically) in the same amount of time in a vacuum, regardless of their mass.

Example 7.2: Object on an Incline. Consider an object resting on an incline, as shown on the right side of Figure 7.1. Again, the object experiences only Earth's gravitational force, which acts vertically from the center of the object toward the Earth's center. Assuming the triangle is a hard object, the forces can be resolved into two components: (i) the force dragging the block down the slope (labeled as $F2$) and (ii) the force acting perpendicular to the slope known as the normal force (labeled as $F1$). If the gravitational force is given by mg (where m is the object's mass and g is the gravitational constant), what is the magnitude of both $F1$ and $F2$? Using trigonometry, we can write these as follows:

$$F1 = mg\cos(\theta) \quad F2 = mg\sin(\theta)$$

We can see that there is an obvious directional component to the motion of this object ($F2$). Assuming that there is no friction acting against the object, we expect the block to travel down the incline. This means the force $F2$ can be used to formulate our ODE:

$$m\frac{dv}{dt} = mg\sin(\theta)$$

Canceling the mass (m) gives:

$$\frac{dv}{dt} = g\sin(\theta)$$

Using separation of variables, we can re-write our expression as an integral so we can calculate velocity as a function of time. We integrate to find the velocity, assuming the object starts from rest ($v = 0$) at $t = 0$:

$$\int_0^v dv = \int_0^t g\sin(\theta)\, dt \quad \Longrightarrow \quad [v]_0^v = [gt\sin(\theta)]_0^t \quad \Longrightarrow v = gt\sin(\theta)$$

Thus, the velocity of an object on an incline increases linearly with time. The rate of increase depends on the incline's angle (θ) but is independent of the object's mass.

Both of these examples are simplified. We could derive a general expression for v as a function of t? What if the object was moving at $t = 0$? What if there was another force acting in the opposite direction? You will see examples of this in the exercises.

Example 7.3: Drag forces on a droplet. We can apply Newton's second law of motion to a case where we have an object moving through a fluid. As that object moves and displaces the fluid, the object will experience varying drag forces that can reduce the objects velocity. We experience this ourselves as we move faster through the air. Fluid dynamics is a very large area of research in itself, but we can at least use some suggested formulations that capture this drag as a function of velocity and environmental conditions. Figure 7.2 shows streamlines of fluid flow around a sphere as it moves through a fluid. As the difference in speed between the object and fluid increases, so will the nature of the forces that dictate motion. These forces include contributions from shear viscosity, eddy viscosity and stagnation pressure. In Physics, we often use dimensionless variables that give us an idea of when these change and by how much. In fluid dynamics, the Reynolds number (often denoted R_e), captures this change. We do not cover that in our course. For now we can assume that, for a droplet moving through the atmosphere, the drag force on a sphere which is moving through air is approximately equal to

$$F = \pi r^2 \frac{\rho}{2} v_{rel}^2$$

where r is the sphere's radius, v_{rel} is the velocity it is moving through the air and $\rho = 1.2$ kg m^{-3} is the density of air. How might we balance this force with the force acting on the droplet due to gravity?

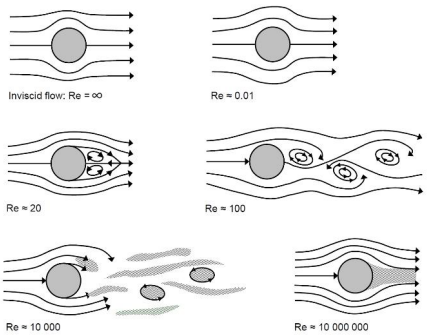


Figure 7.2: Schematic illustrating forces acting on a a sphere moving through a fluid with varying velocity and thus Reynolds number. Taken from https://cfdflowengineering.com/cfd-modeling-of-flow-over-a-cylinder/

Exercises

In some of the following questions, an object resting on an incline, as shown on the right side of Figure 7.1. The object experiences Earth's gravitational force, which acts vertically from the center of the object toward the Earth's center. Assuming the triangle is a hard object, the forces can be resolved into two components: (i) the force dragging the block down the slope (labeled as $F2$) and (ii) the force acting perpendicular to the slope known as the normal force (labeled as $F1$). If the gravitational force is given by mg (where m is the object's mass and g is the gravitational constant), what the magnitude of $F1$ and $F2$ are as follows:

$$F1 = mg\cos(\theta) \quad F2 = mg\sin(\theta)$$

We can also write Newtons second law using three ordinary differential equations (ODE) as follows:

$$\text{(i) } F = m\frac{dv}{dt} \quad \text{(ii) } F = mv\frac{dv}{dx}$$

All are equal but allow you to calculate speed as a function of time OR speed as a function of distance.

Question 7.1: An object, mass m, is at rest on a plane, which is inclined at 30° to the horizontal. Assuming there is no friction how fast will the object be moving after it has moved down the slope a distance of 10 metres? HINT: Use an ODE that relates both distance and velocity.

Question 7.2: An object, mass m, is at rest on a plane, which is inclined at 30° to the horizontal. Assuming friction is equal to 0.1 times the contact force (i.e. the force of gravity perpendicular to the slope) how fast will the object be moving after it has slid down the slope distance of 10 metres? HINT: Alter the total forces acting on the object.

Question 7.3: An object, mass m, is at rest on a plane, which is inclined at 30° to the horizontal. Assuming friction is equal to 0.1 times the contact force (i.e. the force of gravity perpendicular to the slope) how fast will the object be moving after it has slid down the slope after 10 seconds? HINT: Alter the ODE to relate velocity and time.

Question 7.4: An object, mass m, is on on a plane, which is inclined at 30° to the horizontal. Assuming there is no friction and and an initial velocity of $2m/s$, how fast will the object be moving after it has moved down the slope a distance of 10 metres? HINT: Use an ODE that relates both distance and velocity. For this question, as you integrate the appropriate form of Newtons law, think about how to set your limits of integration. You can pair the starting speed and distance with the end speed and distance. Once you integrate with these

limits, you re-arrange to access the final distance.

Question 7.5: The drag force on a sphere which is moving through air is approximately equal to

$$\pi r^2 \frac{\rho}{2} v_{rel}^2$$

where r is the sphere's radius, v_{rel} is the velocity it is moving through the air and $\rho = 1.2$ kg m^{-3} is the density of air. A sphere, radius 0.5 m, and mass $m = 1$ kg, is pushed along a horizontal plane at an initial velocity of $v = 10\ m/s$; the wind speed is zero. How far does the sphere move before slowing down to 0.01 m/s?

Question 7.6: Derive an equation for the velocity versus distance of an object moving with an initial velocity v_0 and end velocity v_1, starting position $x = 0$ and experiencing a constant acceleration. Start by using the following form of Newtons law:

$$F = ma = mv\frac{dv}{dx}$$

noting that you can separate terms dv and dx and then integrate.

Question 7.7: Derive an equation for the velocity versus time if an object moving at an initial velocity v_0 is experiencing a constant acceleration. Start by using the following form of Newtons law:

$$F = m\frac{dv}{dt}$$

Question 7.8: A ball is thrown into the air with an initial velocity of $10m/s$. It's acceleration due to gravity (ignoring air resistance) is -9.8m/s^2. How high will the ball reach before falling back to earth?

Question 7.9: A ball is thrown into the air with an initial velocity of $10m/s$. It's acceleration is as before. How long does it take to reach it's maximum height?

Solutions

Question 7.1: Assuming that there is no friction acting against the object, the component of force parallel to the plane ($F2$) is the only force acting on the object, which is $mg\sin(30)$. As we want to estimate the velocity after it has moved $10m$ down the incline, we use ODE (ii) from above:

$$mv\frac{dv}{dx} = mg\sin(30)$$

Cancel the mass (m) and use separation of variables to integrate. In terms of the limits, we want to see estimate the velocity after it has rolled $10m$ down the incline from rest so we integrate x between 0 and 10, and integrate v from 0 to v:

$$\begin{aligned}
\int_0^v v\,dv &= g\sin(30)\int_0^{10} dx \\
\left[\frac{1}{2}v^2\right]_0^v &= g\sin(30)\times[x]_0^{10} \\
\frac{1}{2}v^2 &= 10*g\sin(30) \\
v^2 &= \sqrt{20*g\sin(30)} = 9.9
\end{aligned}$$

So the object was moving at $9.9m/s$ after it had moved a distance of 10 metres.

Question 7.2: This one is very similar to the above question, but now we have some force of friction. The force parallel to the plane ($F2$) is acting on the object and friction of 0.1 times the normal force ($F1$) so the total forces acting on the object is:

$$\begin{aligned}
F &= F2 - 0.1F1 \\
&= mg\sin(30) - 0.1mg\cos(30)
\end{aligned}$$

Again, as we want to estimate the velocity after it has moved $10m$ down the incline, we use ODE (ii) from above:

$$mv\frac{dv}{dx} = mg\sin(30)$$

Cancel the mass (m) and use separation of variables to integrate x between 0 and 10, and integrate v from 0 to v (since it was at rest at $x = 0$):

$$\begin{aligned}
\int_0^v v\,dv &= (g\sin(30) - 0.1g\cos(30))\int_0^{10} dx \\
\left[\frac{1}{2}v^2\right]_0^v &= (g\sin(30) - 0.1g\cos(30))\times[x]_0^{10} \\
\frac{1}{2}v^2 &= 10(g\sin(30) - 0.1g\cos(30)) \\
v^2 &= \sqrt{20(g\sin(30) - 0.1g\cos(30))} = 9.0
\end{aligned}$$

So the object was moving at $9.0m/s$ after it had moved a distance of 10 metres.

Question 7.3: This one is very similar to the above question, but we need to calculate velocity in terms of time instead of distance. Like the last question, the force parallel to the plane ($F2$) is acting on the object and friction of 0.1 times the normal force ($F1$) so the total forces acting on the object is:

$$\begin{aligned} F &= F2 - 0.1F1 \\ &= mg\sin(30) - 0.1mg\cos(30) \end{aligned}$$

As we want to estimate the velocity 10 seconds after it moved down the incline, we use ODE (i) from above:

$$m\frac{dv}{dt} = mg\sin(30)$$

Cancel the mass (m) and use separation of variables to integrate. In terms of the limits, we want to estimate the velocity 10 seconds after it has moved down the incline from rest so we integrate t between 0 and 10, and integrate v from 0 to v:

$$\begin{aligned} \int_0^v dv &= (g\sin(30) - 0.1g\cos(30))\int_0^{10} dt \\ [v]_0^v &= (g\sin(30) - 0.1g\cos(30)) \times [t]_0^{10} \\ v &= 10(g\sin(30) - 0.1g\cos(30)) = 40.5129 \end{aligned}$$

So the object was moving at $40.51m/s$ after 10 seconds.

Question 7.4: This Question 7.is similar to Question 7.1 but instead of starting from rest the object is moving at $2m/s$. As we want to estimate the velocity after it has rolled $10m$ down the incline, we use ODE (ii) from above:

$$mv\frac{dv}{dx} = mg\sin(30)$$

Cancel the mass (m) and use separation of variables to integrate. In terms of the limits, we want to estimate the velocity of the object which had an initial velocity of $2m/s$ after it has rolled $10m$ down the incline from rest so we integrate x between 0 and 10, and integrate v from 2 to v:

$$\begin{aligned} \int_2^v v\,dv &= g\sin(30)\int_0^{10} dx \\ \left[\frac{1}{2}v^2\right]_2^v &= g\sin(30) \times [x]_0^{10} \\ \frac{1}{2}v^2 - \frac{1}{2}(2)^2 &= 10 * g\sin(30) \\ \frac{1}{2}v^2 - 2 &= 10 * g\sin(30) \\ v^2 &= \sqrt{20 * g\sin(30) + 4} = 10.1 \end{aligned}$$

So the object with an initial velocity of $2m/s$ was travelling at $10.1m/s$ after it had rolled down a distance of 10 metres.

Question 7.5: Since the sphere is moving horizontally, there is no force acting on the object in the direction of travel. As the wind speed is zero, the total forces on the object is:

$$\frac{\rho v^2}{2}\pi r^2$$

We want to estimate the distance the sphere with an initial velocity of $10m/s$ travels before slowing down to $0.01m/s$, we use ODE (ii) from above:

$$mv\frac{dv}{dx} = -\frac{\rho v^2}{2}\pi r^2$$

Using separation of variables we can integrate this ODE. In terms of the limits, we want to estimate the distance travelled as the sphere goes from $10m/s$ and $0.01m/s$, so we integrate x between 0 and x, and integrate v from 10 to 0.01:

$$\begin{aligned}
\frac{m}{v}\,dv &= -\frac{\rho}{2}\pi r^2\,dx \\
\int_{10}^{0.01}\frac{m}{v}dv &= -\int_0^x \frac{\rho}{2}\pi r^2 dx \\
m\left[\ln v\right]_{10}^{0.01} &= -\frac{\rho}{2}\pi r^2\left[x\right]_0^x \\
m\left[\ln 10 - \ln 0.01\right] &= \frac{\rho}{2}\pi r^2 x \\
x &= \frac{2\times 6.91m}{\rho\pi r^2} \cong 14.66\text{m}
\end{aligned}$$

So the sphere with an initial velocity of $10m/s$ had moved $14.66m$ when it reached a velocity of $0.01m/s$.

Question 7.6: We can relate an object moving with initial velocity v_0 and end velocity v_1 and a starting position $x = 0$ through:

$$mv\frac{dv}{dx} = ma$$

Since the object is moving with constant acceleration then we can just use separation of variables and integrate. In terms of the limits we integrate x between 0 and x and integrate

v between v_0 and v_1:

$$\begin{aligned} v\frac{dv}{dx} &= a \\ \int_{v_0}^{v_1} v\,dv &= \int_0^x a dx \\ \left[\frac{1}{2}v^2\right]_{v_0}^{v_1} &= a\,[x]_0^x \\ \frac{1}{2}(v_1^2 - v_0^2) &= 2ax \\ v_1^2 - v_0^2 &= 2ax \end{aligned}$$

Question 7.7: We can relate an object moving with initial velocity v_0 at time $t = 0$ through:

$$m\frac{dv}{dt} = ma$$

Since the object is moving with constant acceleration then we can just use separation of variables and integrate. In terms of the limits we integrate t between 0 and t and integrate v between v_0 and v_1:

$$\begin{aligned} \frac{dv}{dt} &= a \\ \int_{v_0}^{v} dv &= \int_0^t a dt \\ v - v_0 &= at \end{aligned}$$

Question 7.8: We need to find the highest point a ball thrown into the air reaches. The ball has an initial velocity of $10m/s$ and will reach its highest point where it has a velocity of $0m/s$. Since the object has constant acceleration we can use the general formula we developed in Question 7.7 with $v_0 = 10$, $v_1 = 0$, and $a = -9.8$ we see that

$$\begin{aligned} 2ax &= v_1^2 - v_0^2 \\ x &= \frac{v_1^2 - v_0^2}{2a} \\ x &= \frac{0^2 - 10^2}{2 * (-9.8)} \\ x &= \frac{-100}{-19.6} = 5.101 \end{aligned}$$

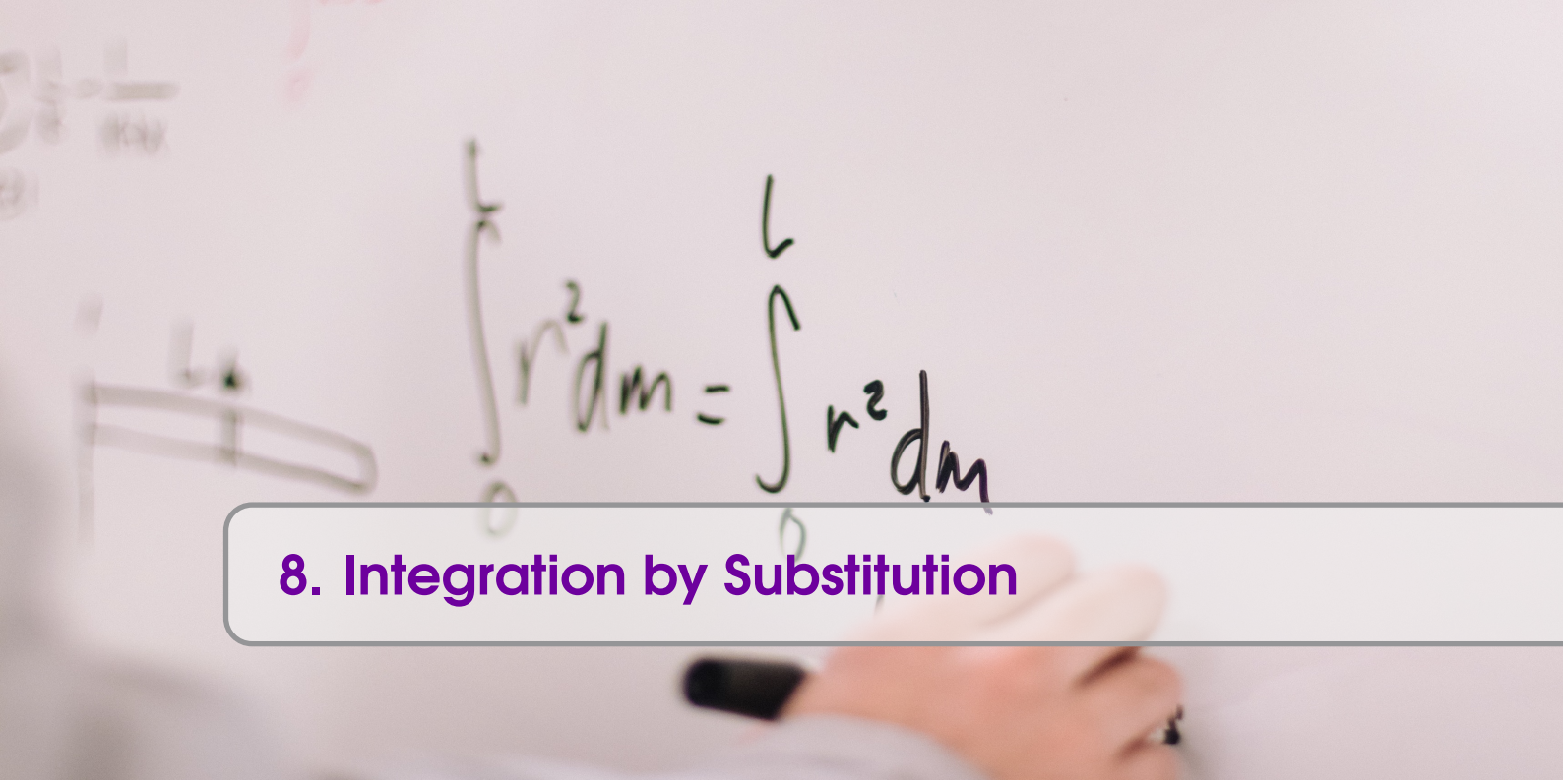
so the ball reached $5.1m$ high.

Question 7.9: This is similar to the last question, but we need to find the time it takes to find its maximum height. The ball has an initial velocity of $10m/s$ and will reach its highest point where it has a velocity of $0m/s$. Since the object has constant acceleration

we can use the general formula we developed in Question 7.8 with $v_0 = 10$, $v_1 = 0$, and $a = -9.8$ we see that

$$\begin{aligned} at &= v - v_0 \\ t &= \frac{v - v_0}{a} \\ t &= \frac{0 - 10}{-9.8} \\ t &= \frac{-10}{-9.8} = 1.02 \end{aligned}$$

so the ball reached its maximum height after $1.02s$.



8. Integration by Substitution

8.1 Overview

In Chapters 6 and 7, we introduced rules that allow us to directly integrate certain basic functions if we can recognize their forms. These can be found in this table:

Differential equation $\frac{dy}{dx}$	Integral, y
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
$3x^2$	$x^3 + d$
x^3	$\frac{x^4}{4} + d$
$\vdots$	$\vdots$
x^n	$\frac{x^{n+1}}{n+1} + d$
ax^n	$\frac{ax^{n+1}}{n+1} + d$
ke^{kx}	$e^{kx} + d$
$\frac{1}{x}$	$ln(x) + d$
$cos(x)$	$sin(x) + d$
$sin(x)$	$-cos(x) + d$

There may be a need to rearrange the equation, often by using techniques like separation of variables (e.g., rearranging N and t) before we can use them, but they are there for us to use. Furthermore, like differentiation for a sum of functions, the **integral of a sum** is equal to the **sum of the integrals** of the individual functions.

$$\frac{dy}{dx} = u(x) + v(x) \quad \Longrightarrow \quad y = \int u(x)dx + \int v(x)dx$$

In some cases, we cant directly integrate by using these rules, as we are faced with more complicated combinations of functions such as **functions of functions**. For example, what happens if we want to integrate

$$y = \int \frac{dy}{dx} = \int (3x+2)^4 dx$$

This is where **integration by substitution** comes in.

8.2 Integration by substitution

Integration by substitution is essentially the reverse process of the **chain rule** in differentiation. Let's return to the example where we want to evaluate

$$y = \int (3x+2)^4 dx$$

At first glance, it may appear that this resembles the derivative of some function, allowing us to integrate directly. What if we replace the expression in the brackets with another variable such that we can integrate using the rules we already know? Making a **substitution** can often simplify things and enables us to perform what seems to be an otherwise difficult integration.

Lets use $u = 3x+2$ so the integral becomes

$$y = \int u^4 dx$$

Now, the integral is expressed in terms of u, but we need to address the dx. This is asking us to integrate u with respect to x. Since dx relates to x, we need to rewrite it in terms of u. This step ensures that the integral is entirely in terms of the new variable. When the integral contains a factor like $ax+b$, substituting $u = ax+b$ often simplifies the problem. Along with the substitution, we must also determine dx in terms of u to complete the process.

Let's show how **integration by substitution** works in full:

1. Write out the integral in its simplest form:

$$y = \int (3x+2)^4 dx$$

2. Find a new variable (u) as a function of the old variable (x) as a substitution:

$$u = 3x+2$$

3. Rearrange to get an expression for the old variable (x) in terms of the new variable (u):

$$x = \frac{1}{3}(u-2)$$

4. Differentiate with respect to the new variable u:

$$\frac{dx}{du} = \frac{1}{3}$$

5. Get an expression for dx:

$$dx = \frac{du}{3}$$

6. Make the substitution in the original integral

$$y = \int (3x+2)^4 dx \implies y = \int u^4 \frac{du}{3} \implies y = \frac{1}{3}\int u^4 \, du$$

7. Integrate as usual:

$$y = \frac{1}{3}\int u^4 \, du \implies \frac{1}{15}u^5 + d$$

8. Substitute the original expression back in to get the final expression:

$$y = \frac{1}{15}(3x+2)^5 + d$$

Let's see some more examples of this.

Example 8.1: Lets evaluate the integral

$$y = \int (x+4)^5 \, dx$$

Let's use $u = x+4$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = x+4 \implies x = u-4 \implies \frac{dx}{du} = 1 \implies dx = du$$

Use these to make the substitution:

$$\int (x+4)^5 \, dx \implies \int u^5 du$$

Now we can evaluate using the basic laws of integration:

$$\int u^5 du = \frac{1}{6}u^6 + c$$

and substituting $u = x+4$ back in gives us:

$$y = \frac{1}{6}(x+4)^6 + c$$

You can differentiate using the chain rule to verify that this works.

Example 8.2: Lets evaluate the integral

$$y = \int \cos(3x+4) \, dx$$

Let's use $u = 3x+4$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 3x+4 \implies x = \frac{1}{3}(u-4) \implies \frac{dx}{du} = \frac{1}{3} \implies dx = \frac{1}{3}du$$

Use these to make the substitution:

$$\int \cos(3x+4)\,dx \implies \frac{1}{3}\int \cos(u)\,du$$

Now we can evaluate using the basic laws of integration:

$$\frac{1}{3}\int \cos(u)\,du = \frac{1}{3}\sin(u)+c$$

and substituting $u = 3x+4$ back in gives us:

$$y = \frac{1}{3}\sin(3x+4)+c$$

You can differentiate using the chain rule to verify that this works.

So we see that we can integrate more complicated functions using substitution. We want to use a substitution so that the integration becomes something we can use our table of standard integrals for. However, this is integration by substitution for **indefinite integrals**. In many cases, we want to find the **area under a curve** so the integral has limits. For example,

$$y = \int_{-1}^{1} (3x+2)^4 dx$$

therefore we have to perform **definite integration**. This means that we calculate the integral (using integration by substitution) then evaluate the integral but for the new limits calculated under the substitution.

Let's show how definite **integration by substitution** works in full:

1. Write out the integral in its simplest form:

$$y = \int_{-1}^{1} (3x+2)^4 dx$$

2. Find a new variable (u) as a function of the old variable (x) as a substitution:

$$u = 3x+2$$

3. Rearrange to get an expression for the old variable (x) in terms of the new variable (u):

$$x = \frac{1}{3}(u-2)$$

4. Differentiate with respect to the new variable u:

$$\frac{dx}{du} = \frac{1}{3}$$

5. Get an expression for dx:

$$dx = \frac{du}{3}$$

6. Find the new limits of the integral:

$$\text{When } x = -1, \;\; u = 3 \times (-1) + 2 = -1$$
$$\text{When } x = 1, \;\; u = 3 \times 1 + 2 = 5$$

7. Make the substitution in the original integral

$$y = \int_{-1}^{1} (3x+2)^4 dx \implies y = \int_{-1}^{5} u^4 \frac{du}{3} \implies y = \frac{1}{3}\int_{-1}^{5} u^4 \, du$$

8. Integrate as usual:

$$\begin{aligned} \frac{1}{3}\int_{-1}^{5} u^4 \, du &= \frac{1}{3}\left[\frac{1}{5}u^5\right]_{-1}^{5} \\ &= \frac{1}{3} \times \frac{1}{5} \times (5^5 - 1^5) \\ &= \frac{3124}{15} = 208.267 \end{aligned}$$

Let's see some more examples of this.

Example 8.3: Lets evaluate the integral

$$y = \int_{-4}^{6} (x+4)^5 \, dx$$

Let's use $u = x+4$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = x+4 \implies x = u-4 \implies \frac{dx}{du} = 1 \implies dx = du$$

Find the new limits:

$$\text{When } x = -4, \;\; u = -4+4 = 0$$
$$\text{When } x = 6, \;\; u = 6+4 = 10$$

Use these to make the substitution:

$$\int_{-4}^{6} (x+4)^5 \, dx \implies \int_{0}^{10} u^5 du$$

Now we can evaluate using the basic laws of integration:

$$\begin{aligned}\int_0^{10} u^5\,du &= \left[\frac{1}{6}u^6\right]_0^{10} \\ &= \frac{1}{6}\times(10^6-0^6) \\ &= \frac{1,000,000}{6} = 166,666.67\end{aligned}$$

Example 8.4: Lets evaluate the integral

$$y=\int_{-9}^{1}\frac{1}{x+10}dx$$

Let's use $u=x+10$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u=x+10 \implies x=u-10 \implies \frac{dx}{du}=1 \implies dx=du$$

Find the new limits:

$$\begin{aligned}&\text{When } x=-9,\ \ u=-9+10=1\\ &\text{When } x=1,\ \ u=1+10=11\end{aligned}$$

Use these to make the substitution:

$$\int_{-9}^{1}\frac{1}{x+10}dx \implies \int_1^{11}\frac{1}{u}du$$

Now we can evaluate using the basic laws of integration:

$$\begin{aligned}\int_1^{11}\frac{1}{u}du &= [\ln(u)]_1^{11} \\ &= \ln(11)-\ln(1) \\ &= \ln(11)=2.398\end{aligned}$$

Exercises

Algebraic problems

Question 8.1: Using *integration by substitution*, find the indefinite integral of the following:

(a) $\int 4\,(5x+6)^3\,dx$

(b) $\int (x-2)^3\,dx$

(c) $\int (x+5)^4\,dx$

(d) $\int (1-2x)^7\,dx$

Question 8.2: Using *integration by substitution*, find the definite integral of the following:

(a) $\int_0^1 4\,(5x+6)^3\,dx$

(b) $\int_0^1 (x-2)^3\,dx$

(c) $\int_0^1 (x+5)^4\,dx$

(d) $\int_{-1}^1 (1-2x)^7\,dx$

Question 8.3: Using *integration by substitution*, find the indefinite integral of the following:

(a) $\int \sin(7x-3)\,dx$

(b) $\int \exp(5x+2)\,dx$

(c) $\int \cos(1-x)\,dx$

(d) $\int \frac{1}{7x+5}dx$

Question 8.4: Using *integration by substitution*, find the definite integral of the following:

(a) $\int_0^{\pi/2} \sin(7x-3)\,dx$

(b) $\int_0^1 \exp(5x+2)\,dx$

(c) $\int_{-\pi/4}^{\pi/4} \cos(1-x)\,dx$

(d) $\int_0^1 \frac{1}{7x+5}dx$

Question 8.5: Using *integration by substitution*, find the definite integral of the following:

$$\int_{-1}^{1} \frac{4}{(2x+1)^3}$$

Question 8.6: We can see that from the table of basic integration laws that

$$\frac{dy}{dx} = x^n \implies y = \frac{x^{n+1}}{n+1}$$

Use integration by substitution to find a general expression for y when

$$\frac{dy}{dx} = (ax+b)^n$$

where a, b and n are constants.

Real-world problems

Question 8.7: The density of air changes with pressure and temperature (because of the ideal gas law). Density is equal to $\rho = \frac{P}{R_a T}$, where $R_a = 287$ J kg^{-1}K^{-1} is the gas constant for air and gravity is equal to $g = 9.8$.. So the differential equation becomes:

$$\frac{dP}{dz} = -\frac{P}{R_a T} g$$

We saw that in Chapter 6 that the solution to this problem is:

$$P(z) = P_0 \exp\left(-\frac{g}{R_a T} z\right)$$

However, for simplicity, we assumed that temperature was constant which isn't a very good assumption. Temperature T is actually a function of height in the atmosphere: $T(z) = T_{surface} - 0.01 \times z$, where $T_{surface}$ is the surface temperature. So the final differential equation to solve can be written:

$$\frac{dP}{dz} = -\frac{P}{R_a \left(T_{surface} - 0.01z\right)} g$$

During a day in the summer, the surface temperature at sea level over Tenerife is 290 K and the surface pressure is 100000 Pa. Clouds are observed to form at a height of 1000 m. What are the pressure and temperature at this height? HINT: Rearrange this ODE and use integration by substitution.

Question 8.8: The density of air changes with pressure and temperature (because of the ideal gas law). Density is equal to $\rho = \frac{P}{R_a T}$, where $R_a = 287$ J kg^{-1}K^{-1} is the gas constant for air and gravity is equal to $g = 9.8$.Use integration by substitution to show that the general solution of the hydrostatic equation:

$$\frac{dP}{dz} = -\frac{P}{R_a \left(T_{surface} - 0.01z\right)} g$$

is of the form

$$P(z) = P_0 \exp\left(\frac{100 \times 9.8}{287} \ln\left(\frac{T_{surface} - 0.01z}{T_{surface}}\right)\right)$$

where P_0 is the pressure at $z = 0$.

Question 8.9: In the previous Chapter, we saw the drag force on a sphere which is moving through air is approximately equal to

$$\pi r^2 \frac{\rho}{2} v^2$$

where r is the sphere's radius, v is the velocity it is moving through the air and $\rho = 1.2$ kg m^{-3} is the density of air. A sphere, radius 0.5 m, and mass $m = 1$ kg, is pushed along a

horizontal plane at an initial velocity of $v = 10$ m s^{-1}. However, this time the wind speed is $10m/s$ in the opposite direction of the motion of the sphere. We need to re-write out the above equation slightly to account for the wind speed. If we let w be the wind speed, then the drag force

$$\pi r^2 \frac{\rho}{2}(v+w)^2$$

How far does the sphere move before slowing down to 0.01 m s^{-1}? HINT: You will need to use the appropriate version of Newtons second law and then use integrate by substitution to do this.

Solutions

Algebraic problems

Question 8.1:

(a) Let's use $u = 5x+6$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 5x+6 \implies x = \frac{1}{5}(u-6) \implies \frac{dx}{du} = \frac{1}{5} \implies dx = \frac{1}{5}du$$

Use these to make the substitution:

$$\int 4\,(5x+6)^3\,dx \implies \frac{4}{5}\int u^3 du$$

Now we can evaluate using the basic laws of integration:

$$\frac{4}{5}\int u^3 du = \frac{4}{5}\times\frac{1}{4}\times u^4 + c$$

and substituting $u = 5x+6$ back in gives us:

$$y = \frac{1}{5}u^4 + c$$

You can differentiate using the chain rule to verify that this works.

$$u = x-2 \implies x = u+2 \implies \frac{dx}{du} = 1 \implies dx = du$$

Use these to make the substitution:

$$\int (x-2)^3\,dx \implies \int u^3 du$$

Now we can evaluate using the basic laws of integration:

$$\int u^3 du = \frac{1}{4}u^4 + c$$

and substituting $u = x-2$ back in gives us:

$$y = \frac{1}{4}(x-2)^4 + c$$

You can differentiate using the chain rule to verify that this works.

(c) Let's use $u = x+5$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = x+5 \implies x = u-5 \implies \frac{dx}{du} = 1 \implies dx = du$$

Use these to make the substitution:

$$\int (x+5)^4\,dx \implies \int u^4 du$$

Now we can evaluate using the basic laws of integration:

$$\int u^4 du = \frac{1}{5}u^5 + c$$

and substituting $u = x+5$ back in gives us:

$$y = \frac{1}{5}(x+5)^5 + c$$

You can differentiate using the chain rule to verify that this works.

(d) Let's use $u = 1-2x$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 1-2x \implies x = \frac{1}{2}(1-u) \implies \frac{dx}{du} = -\frac{1}{2} \implies dx = -\frac{1}{2}du$$

Use these to make the substitution:

$$\int (1-2x)^7 dx \implies -\frac{1}{2}\int u^7 du$$

Now we can evaluate using the basic laws of integration:

$$-\frac{1}{2}\int u^7 du = -\frac{1}{16}u^8 + c$$

and substituting $u = 1-2x$ back in gives us:

$$y = -\frac{1}{16}(1-2x)^8 + c$$

You can differentiate using the chain rule to verify that this works.

Question 8.2:

(a) Let's continue from Question 8.1(a) where we used a substitution of $u = 5x+6$. Find the new limits:

$$\text{When } x = 0,\ u = 5\times 0 + 6 = 6$$
$$\text{When } x = 1,\ u = 5\times 1 + 6 = 11$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_0^1 4(5x+6)^3 dx &= \frac{4}{5}\int_6^{11} u^3 du \\ &= \frac{4}{5}\left[\frac{1}{4}u^4\right]_6^{11} \\ &= \frac{1}{5}(11^4 - 6^4) = 2669\end{aligned}$$

(b) Let's continue from Question 8.1(b) where we used a substitution of $u = x - 2$. Find the new limits:

$$\text{When } x = 0,\ u = 0 - 2 = -2$$
$$\text{When } x = 1,\ u = 1 - 2 = -1$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_0^1 (x-2)^3\,dx &= \int_{-2}^{-1} u^3 du \\ &= \left[\frac{1}{4}u^4\right]_{-2}^{-1} \\ &= \frac{1}{4}((-1)^4 - (-2)^4) = -3.75\end{aligned}$$

(c) Let's continue from Question 8.1(c) where we used a substitution of $u = x + 5$. Find the new limits:

$$\text{When } x = 0,\ u = 0 + 5 = 5$$
$$\text{When } x = 1,\ u = 1 + 5 = 6$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_0^1 (x+5)^4\,dxdx &= \int_5^6 u^4 du \\ &= \left[\frac{1}{5}u^5\right]_5^6 \\ &= \frac{1}{5}(6^5 - 5^5) = 930.2\end{aligned}$$

(d) Let's continue from Question 8.1(d) where we used a substitution of $u = 1 - 2x$. Find the new limits:

$$\text{When } x = -1,\ u = 1 - 2 \times (-1) = 3$$
$$\text{When } x = 1,\ u = 1 - 2 \times 1 = -1$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_{-1}^1 (1-2x)^7\,dx &= -\frac{1}{2}\int_3^{-1} u^7 du \\ &= -\frac{1}{16}\left[u^8\right]_3^{-1} \\ &= \frac{1}{16}(3^8 - 1^8) = 410\end{aligned}$$

Question 8.3: Using *integration by substitution*, find the indefinite integral of the following:

(a) Let's use $u = 7x - 3$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 7x - 3 \implies x = \frac{1}{7}(u+3) \implies \frac{dx}{du} = \frac{1}{7} \implies dx = \frac{1}{7}du$$

Use these to make the substitution:

$$\int \sin(7x-3)\,dx \implies \frac{1}{7}\int \sin(u)\,du$$

Now we can evaluate using the basic laws of integration:

$$\frac{1}{7}\int \sin(u)\,du = \frac{1}{7}\sin(u) + c$$

and substituting $u = 7x - 3$ back in gives us:

$$y = \frac{1}{7}\sin(7x-3) + c$$

You can differentiate using the chain rule to verify that this works.

(b) Let's use $u = 5x + 2$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 5x + 2 \implies x = \frac{1}{5}(u-2) \implies \frac{dx}{du} = \frac{1}{5} \implies dx = \frac{1}{5}du$$

Use these to make the substitution:

$$\int \exp(5x+2)\,dx \implies \frac{1}{5}\int \exp(u)\,du$$

Now we can evaluate using the basic laws of integration:

$$\frac{1}{5}\int \exp(u)\,du = \frac{1}{5}\exp(u) + c$$

and substituting $u = 5x + 2$ back in gives us:

$$y = \frac{1}{5}\exp(5x+2) + c$$

You can differentiate using the chain rule to verify that this works.

(c) Let's use $u = 1 - x$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 1 - x \implies x = 1 - u \implies \frac{dx}{du} = -1 \implies dx = -du$$

Use these to make the substitution:

$$\int \cos(1-x)\,dx \implies -\int \cos(u)\,du$$

Now we can evaluate using the basic laws of integration:

$$-\int \cos(u)\,du = -\sin(u) + c$$

and substituting $u = 1 - x$ back in gives us:

$$y = -\sin(1-x) + c$$

(d) Let's use $u = 7x + 5$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 7x+5 \implies x = \frac{1}{7}(u-5) \implies \frac{dx}{du} = \frac{1}{7} \implies dx = \frac{1}{7}du$$

Use these to make the substitution:

$$\int \frac{1}{7x+5}dx \implies \frac{1}{7}\int \frac{1}{u}du$$

Now we can evaluate using the basic laws of integration:

$$\frac{1}{7}\int \frac{1}{u}du = \frac{1}{7}\ln(u) + c$$

and substituting $u = 7x + 5$ back in gives us:

$$y = \frac{1}{7}\ln(7x+5) + c$$

Question 8.4: Using *integration by substitution*, find the definite integral of the following:

(a) Let's continue from Question 8.3(a) where we used a substitution of $u = 7x - 3$. Find the new limits:

$$\text{When } x = 0,\ u = 7\times 0 - 3 = 4$$
$$\text{When } x = \frac{\pi}{2},\ u = 7\times\frac{\pi}{2} - 3$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_0^{\pi/2} \sin(7x-3)\,dx &= \frac{1}{7}\int_4^{\frac{7\pi}{2}-3} \sin(u)\,du \\ &= -\frac{1}{7}\left[\sin(u)\right]_4^{\frac{7\pi}{2}-3} \\ &= \frac{1}{7}\left(\sin\left(\frac{7\pi}{2}-3\right) - \sin(4)\right) = 0.25\end{aligned}$$

(b) Let's continue from Question 8.3(b) where we used a substitution of $u = 5x + 2$. Find the new limits:

$$\text{When } x = 0,\ u = 5\times 0 + 2 = 2$$
$$\text{When } x = 1,\ u = 5\times 1 + 2 = 7$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_0^1 \exp(5x+2)\,dx &= \frac{1}{5}\int_2^7 \exp(u)\,du \\ &= \frac{1}{5}\left[\exp(u)\right]_2^7 \\ &= \frac{1}{5}(\exp(7)-\exp(2)) = 217.849\end{aligned}$$

(c) Let's continue from Question 8.3(c) where we used a substitution of $u = 1-x$. Find the new limits:

$$\begin{aligned}&\text{When } x = -\frac{\pi}{4},\ u = 1-(-\frac{\pi}{4}) == 1+\frac{\pi}{4} \\ &\text{When } x = \frac{\pi}{4},\ u = 1-\frac{\pi}{4} = 1-\frac{\pi}{4}\end{aligned}$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_{-\pi/4}^{\pi/4} \cos(1-x)\,dx &= -\int_{1+\frac{\pi}{4}}^{1-\frac{\pi}{4}} \sin(u)du \\ &= -\left[\sin(u)\right]_{1+\frac{\pi}{4}}^{1-\frac{\pi}{4}} \\ &= -\left(\sin\left(1-\frac{\pi}{4}\right)-\sin\left(1+\frac{\pi}{4}\right)\right) = 0.764\end{aligned}$$

(d) Let's continue from Question 8.3(d) where we used a substitution of $u = 7x+5$. Find the new limits:

$$\begin{aligned}&\text{When } x = 0,\ u = 7\times 0+5 = 3 \\ &\text{When } x = 1,\ u = 7\times 1+5 = 12\end{aligned}$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_0^1 \frac{1}{7x+5}dx &= \frac{1}{7}\int_3^{12} \frac{1}{u}du \\ &= -\frac{1}{7}\left[\ln(u)\right]_3^1 \\ &= \frac{1}{7}(\ln(12)-\ln(3)) = 0.198\end{aligned}$$

Question 8.5: Let's use $u = 2x+1$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 2x+1 \implies x = \frac{1}{2}(u-1) \implies \frac{dx}{du} = \frac{1}{2} \implies dx = \frac{1}{2}du$$

Find the new limits:

$$\begin{aligned}&\text{When } x = -1,\ u = 2\times(-1)+1 = -1 \\ &\text{When } x = 1,\ u = 2\times 1+1 = 3\end{aligned}$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_{-1}^{1} \frac{4}{(2x+1)^3}dx &= \frac{1}{2}\int_{-1}^{3} \frac{1}{u^3}du \\ &= \frac{1}{2}\left[-\frac{1}{2}\times u^{-2}\right]_{-1}^{3} \\ &= -\frac{1}{4}(3^{-2}-(-1)^{-2}) = 0.222\end{aligned}$$

Question 8.6: We want to integrate the following:

$$\frac{dy}{dx} = (ax+b)^n$$

Let's use $u = ax+b$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = ax+b \implies x = \frac{1}{a}(u-b) \implies \frac{dx}{du} = \frac{1}{a} \implies dx = \frac{1}{a}du$$

Use these to make the substitution:

$$\int (ax+b)^n dx \implies \frac{1}{a}\int u^n du$$

Now we can evaluate using the basic laws of integration:

$$\frac{1}{a}\int u^n du = \frac{1}{a(n+1)}u^{n+1} + c$$

and substituting $u = ax+b$ back in gives us:

$$y = \frac{(ax+b)^{n+1}}{a(n+1)} + c$$

You can differentiate using the chain rule to verify that this works.

Real-world problems

Question 8.7: We want to solve the hydrostatic equation to find the pressure P at $z = 1000$ when the surface pressure is $100,000$Pa (i.e. $z = 0$) and $T_{surface} = 290$. We rearrange and integrate pressure, P, between 100000 and P and z between 0 and 1000:

$$\begin{aligned}\frac{dP}{dz} &= -\frac{Pg}{R(T_{surface}-0.01z)} \\ \int_{100000}^{P} \frac{1}{P}dP &= -\frac{g}{R}\int_{0}^{1000} \frac{1}{290-0.01z}dz\end{aligned}$$

Let's use $u = 290-0.01z$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 290-0.01z \implies x = 100(290-u) \implies \frac{dx}{du} = -100 \implies dx = -100du$$

Find the new limits

$$\begin{aligned}&\text{When } x=0,\ \ u=290-0.01\times 0=290\\&\text{When } x=1000,\ \ u=290-0.01\times(1000)=280\end{aligned}$$

Make the substitution with limits and integrate:

$$\begin{aligned}\int_{100000}^{P}\frac{1}{P}dP &= -\frac{g}{R}\int_{0}^{1000}\frac{1}{290-0.01z}dz\\ \int_{100000}^{P}\frac{1}{P}dP &= \frac{100g}{R}\int_{290}^{280}\frac{1}{u}du\\ [\ln P]_{100000}^{P} &= \frac{100g}{R}[\ln u]_{290}^{280}\\ \ln(P)-\ln(100000) &= \frac{100g}{R}(\ln(280)-\ln(290))\\ \ln\left(\frac{P}{100000}\right) &= \frac{100g}{R}\ln\left(\frac{280}{290}\right)\end{aligned}$$

Rearranging this and substituting in $g=9.8$ and $R=287$, gives us

$$P=100000\times\exp\left(\frac{100\times 9.8}{287}\ln\frac{280}{290}\right)\cong 88708\text{Pa}$$

and thus the pressure at 1000m is 88708Pa.

Question 8.8: We can also demonstrate how to integrate the hydrostatic equation between general limits:

$$\begin{aligned}\frac{dP}{dz} &= -\frac{Pg}{R(T_{surface}-0.01z)}\\ \int_{P_0}^{P}\frac{1}{P}dP &= -\frac{g}{R}\int_{0}^{z}\frac{1}{T_{surface}-0.01z}dz\end{aligned}$$

Let's use $u=T_{surface}-0.01z$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u=T_{surface}-0.01z\implies x=100(T_{surface}-u)\implies\frac{dx}{du}=-100\implies dx=-100du$$

Find the new limits

$$\begin{aligned}&\text{When } x=0,\ \ u=T_{surface}-0.01\times 0=T_{surface}\\&\text{When } x=1000,\ \ u=T_{surface}-0.01z\end{aligned}$$

Make the substitution with limits and integrate:

$$\begin{aligned}
\int_{P_0}^{P} \frac{1}{P} dP &= -\frac{g}{R} \int_0^z \frac{1}{290 - 0.01z} dz \\
\int_{P_0}^{P} \frac{1}{P} dP &= \frac{100g}{R} \int_{T_{surface}}^{T_{surface} - 0.01z} \frac{1}{u} du \\
[\ln P]_{P_0}^{P} &= \frac{100g}{R} [\ln u]_{T_{surface}}^{T_{surface} - 0.01z} \\
\ln(P) - \ln(P_0) &= \frac{100g}{R} (\ln(T_{surface} - 0.01z) - \ln(T_{surface})) \\
\ln\left(\frac{P}{P_0}\right) &= \frac{100g}{R} \ln\left(\frac{T_{surface} - 0.01z}{T_{surface}}\right)
\end{aligned}$$

Rearranging this and substituting in $g = 9.8$ and $R = 287$, gives us

$$P(z) = P_0 \exp\left(\frac{100 \times 9.8}{287} \ln\left(\frac{T_{surface} - 0.01z}{T_{surface}}\right)\right)$$

the desired expression.

Question 8.9: This Question 8.extends Question 8.6 from Chapter 7, however, the sphere has wind speed as well as the speed of the object. This makes the integration a little more difficult. The total forces on the object are

$$\pi r^2 \frac{\rho}{2} (v + w)^2$$

We want to estimate the distance the sphere with an initial velocity of $10m/s$ travels before slowing down to $0.01m/s$ with a wind speed of $w = 10$:

$$\begin{aligned}
mv\frac{dv}{dx} &= -\pi r^2 \frac{\rho}{2} (v + w)^2 \\
m\frac{v}{(v+w)^2}\frac{dv}{dx} &= -\pi r^2 \frac{\rho}{2} \\
m\frac{v}{(v+10)^2}\frac{dv}{dx} &= -\pi r^2 \frac{\rho}{2}
\end{aligned}$$

Using integration by substitution we can integrate this ODE. In terms of the limits, we want to estimate the distance travelled as the sphere goes from $10m/s$ and $0.01m/s$, so we integrate x between 0 and x, and integrate v from 10 to 0.01:

$$m \int_{10}^{0.01} \frac{v}{(v+10)^2} dv = -\int_0^x \pi r^2 \frac{\rho}{2} dx$$

Let's use $z = v + 10$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$z = v + 10 \implies v = z - 10 \implies \frac{dv}{dz} = 1 \implies dv = dz$$

Find the new limits:

$$\text{When } x = 10,\ \ z = 10 + 10 = 20$$
$$\text{When } x = 0.01,\ \ z = 0.01 + 10 = 10.01$$

Make the substitution with limits and integrate:

$$\begin{aligned}
m\int_{10}^{0.01} \frac{v}{(v+w)^2}dv &= -\int_0^x \pi r^2 \frac{\rho}{2}dx \\
m\int_{20}^{10.01} \frac{z-10}{z^2}dz &= -\int_0^x \pi r^2 \frac{\rho}{2}dx \\
m\int_{20}^{10.01} \frac{1}{z} - \frac{10}{z^2}dz &= -\left[\pi r^2 \frac{\rho}{2}\right]_0^x \\
m\left[\ln(z) + \frac{10}{z}\right]_{20}^{10.01} &= -\pi r^2 \frac{\rho}{2}x \\
m\left(\ln(10.01) + \frac{10}{10.01} - \ln(20) - \frac{10}{20}\right) &= -\pi r^2 \frac{\rho}{2}x \\
-0.193m &= -\pi r^2 \frac{\rho}{2}x
\end{aligned}$$

Substituting $m = 1$, $\rho = 1.2$, and $r = 0.5$ gives us:

$$\begin{aligned}
-0.193 &= -0.471x \\
x &= \frac{0.193}{0.471} = 0.41
\end{aligned}$$

So we see that the total distance travelled has reduced from 14.66 to 0.41m by adding in wind speed, so the wind causes the sphere to slow down much quicker.



9. Integration by Parts

9.1 Overview

In Chapters 6 and 7,, we introduced rules for directly integrating certain basic functions by recognising their forms. However, techniques like separation of variables are often insufficient when dealing with more complex combinations of functions, such as **functions of functions** and thus we introduced **integration by substitution** in Chapter 8. In this Chapter, we look at we focus on more difficult combinations, particularly **products of functions**. For example, what happens if we want to integrate:

$$y = \int xe^x dx$$

This is where **integration by parts** comes in.

9.2 Integration by parts

Integration by parts is essentially the reverse process of the **product rule** in differentiation. The product rule states that when we have a product of two functions $u(x)v(x)$ then we can differentiate this as follows:

$$\frac{d}{dx}(u(x)v(x)) = u \times \frac{dv}{dx} + v \times \frac{du}{dx}$$

Let's integrate this expression:

$$\int \frac{d}{dx}(u(x)v(x)) = \int u \times \frac{dv}{dx} + \int v \times \frac{du}{dx}$$

Simplifying this gives us

$$u(x)v(x) = \int u \times \frac{dv}{dx} + \int v \times \frac{du}{dx}$$

and doing a final rearranging gives

$$\int u \times \frac{dv}{dx} = u(x)v(x) - \int v \times \frac{du}{dx}$$

This is the formula we need to use to perform integration by parts! It may look counterintuitive, as this would involve splitting the function, integrating one part and differentiating another, but it does indeed work!

To prove that it does, let's return to the integral

$$y = \int xe^x dx$$

to show how **integration by parts** works in full:

1. Write out the integral in its simplest form.

$$y = \int xe^x \, dx$$

2. Identify and assign u and differentiate to get $\frac{du}{dx}$. You should select the term that simplifies upon differentiation, so lets choose $u(x) = x$ here:

$$u(x) = x \implies \frac{du}{dx} = 1$$

3. Identify the other term and assign $\frac{dv}{dx}$ and integrate to get v.

$$\frac{dv}{dx} = e^x \implies \int dv = \int e^x \, dx \implies v = e^x$$

4. Substitute these values into the integration by parts formula:

$$\begin{aligned} y &= \int u\frac{dv}{dx} = uv - \int v\frac{du}{dx}dx \\ &= xe^x - \int e^x \, dx \end{aligned}$$

5. Integrate the remaining term:

$$y = xe^x - e^x + d$$

6. Simplify to get the final expression:

$$y = e^x(x-1) + d$$

Example 9.1:

Example 9.2:

So we see that we can integrate products of functions using integration by parts. However, this is integration by substitution for **indefinite integrals**. In many cases, we want to find the **area under a curve** so the integral has limits. For example,

$$y = \int_0^1 xe^x dx$$

therefore we have to perform **definite integration**. This means that we calculate the integral (using integration by parts) then evaluate the integral but for the new limits calculated under the substitution.

Let's show how definite **integration by parts** works in full:

1. Write out the integral in its simplest form.
$$y = \int_0^1 xe^x \, dx$$
2. Identify and assign u and differentiate to get $\frac{du}{dx}$. You should select the term that simplifies upon differentiation.
$$u = x \implies \frac{du}{dx} = 1$$
3. Identify the other term and assign $\frac{dv}{dx}$ and integrate to get v.
$$\frac{dv}{dx} = e^x \implies \int dv = \int e^x \, dx \implies v = e^x$$
4. Substitute these values into the integration by parts formula:
$$\begin{aligned} y &= \int_0^1 u\frac{dv}{dx} = uv - \int v\frac{du}{dx}dx \\ &= [xe^x]_0^1 - \int e^x \, dx \\ &= [xe^x]_0^1 - \int_0^1 e^x \, dx \end{aligned}$$
5. Integrate the remaining term:
$$\begin{aligned} y &= e - [e^x]_0^1 \\ &= e - (e-1) \\ &= 1 \end{aligned}$$

Example 9.3:

Example 9.4:

Exercises

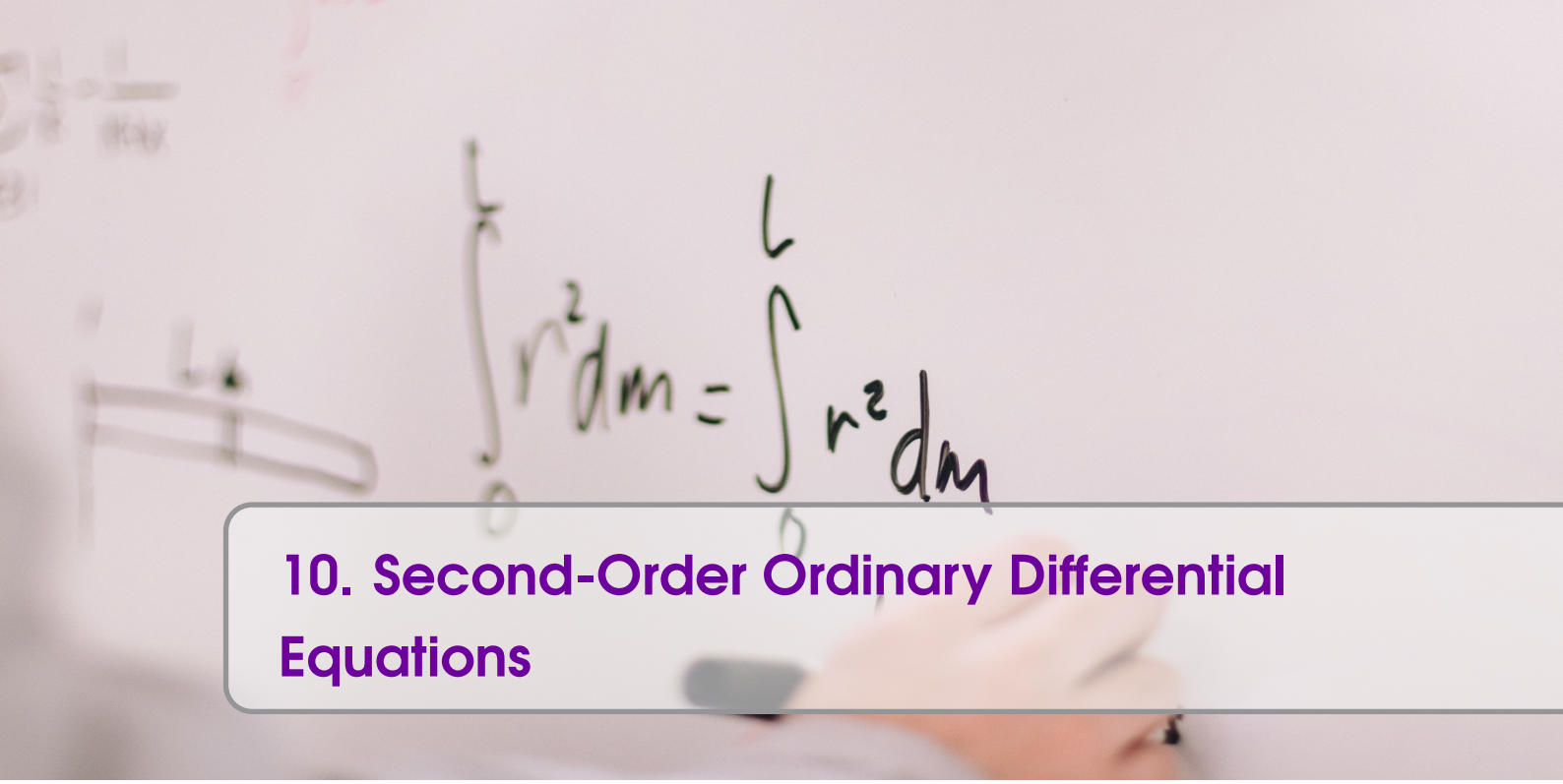
Algebraic problems

Real-world problems

Solutions

Algebraic problems

Real-world problems



10. Second-Order Ordinary Differential Equations

10.1 Overview

In Chapters 6–8, we explored the process of integration, which is the inverse of differentiation and a powerful tool for calculating the area under curves. This has been really useful in understanding various properties and behaviors of systems, such as those described by Newton's Laws of Motion. However, our focus so far has been primarily on first-order derivatives (i.e., $\frac{dy}{dx}$). In this Chapter, we conclude our study of integration by examining **second-order ordinary differential equations**.

10.2 Second-order Differential Equations

*S*econd-order ordinary differential equations are ordinary differential equations (ODEs) that involve a *second-order derivative*. We could devote an entire course looking at solving different types of second-order ODEs, a general form is given by:

$$a(x)\frac{d^2y}{dx^2} + b(x)\frac{dy}{dx} + c(x)y = d(x)$$

Here, $a(x)$, $b(x)$, $c(x)$ and $d(x)$ are general functions of the variable x, referred to as coefficients. In fact, we can simplify the above expression by writing $a(x)$, $b(x)$ and $c(x)$ as constant coefficients (which we will only ever look at in this course):

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = d(x)$$

This equation is classified as a **linear second-order ODEs** because the second derivative, $\frac{d^2y}{dx^2}$, is raised to the power of 1. If the value of $d(x) = 0$ we can also say the equation is **homogeneous**, whilst if $d(x) \neq 0$ the equation is calles **non-homogeneous**. In this course

we will briefly look at methods for solving **homogeneous, linear, second-order ODEs** which resemble the following expression:

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

Please note that we can write the exact same expression using alternative ODE notation as per the following equation:

$$ay'' + by' + cy = 0$$

where a, b and c remain the same constants. In every problem so far, we have wanted to arrive at a general analytical solution from integrating a differential equation. How do we deal with a second-order ODE as above?

Fortunately, there is a general form we can use if the equation to be solved has the same form. Notice how it looks like we might be able to use the approach to solving quadratic equations, but how do we find a value for y. Lets just jump to an educated guess and assume the expression for y is going to be:

$$y = e^{\lambda x}$$

where λ is a constant with an unknown value. If we differentiate (twice) this expression we can arrive at two expressions:

$$\frac{dy}{dx} = \lambda e^{\lambda x} \quad \frac{d^2y}{dx^2} = \lambda^2 e^{\lambda x}$$

Lets now substitute these expressions back into our general differential equation as follows:

$$a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} = 0$$

Because $e^{\lambda x}$ cant be zero, it is safe to divide through to acquire a quadratic in λ:

$$a\lambda^2 + b\lambda + c = 0$$

This is called the **auxiliary** equation or **characteristic** equation. When you encounter a **homogeneous second-order ODE**, you can jump straight to this point and follow the procedure below.

Looks like we have a quadratic equation. If can solve this, we can get to a value of λ which we can use to find a solution to the second-order ODE. Remember that we can solve (or find roots of) quadratic equations using the following formula:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

When we solve a quadratic expression we will arrive at 2 roots, which we can call λ_1 and λ_2.

Since there are two roots to the auxiliary equation, the general solution to the second-order ODE will be a combination of the two function. If λ_1 and λ_2 are real and not equal ($\lambda_1 \neq \lambda_2$ then we can write a general solution as:

$$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$$

If our two roots are real and equal (i.e. $\lambda_1 = \lambda_2$) then we use the formula:

$$y = (A + Bx)e^{\lambda_1 x}$$

We don't consider what happens when the two roots are not real (i.e. $b^2 - 4ac < 0$) in this course. In both general solutions, we have added two new constants A and B as we are combining multiple solutions we need to know how much each will contribute to the final solution. To calculate their values, we need some initial conditions or boundary conditions in y (we'll see this later).

Example 9.1: Imagine we are asked to solve the following equation:

$$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 5y = 0$$

We can see that this is a linear, homogeneous, second-order ODE. Firstly, we write out the auxiliary equation:

$$a\lambda^2 + b\lambda + c = 0$$

Secondly, we need to check what the values for λ_1 and λ_2 are to understand what solution is needed. To do this, by using the **auxiliary** equation we know we have a quadratic as follows. Find the roots (λ_1 and λ_2) of this equation:

$$\begin{aligned}
\lambda &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
&= \frac{6 \pm \sqrt{6^2 - 4 * 1 * 5}}{2 * 1} \\
&= \frac{6 \pm \sqrt{36 - 20}}{2} \\
&= \frac{6 \pm 4}{2} \\
&= 3 \pm 2
\end{aligned}$$

so we have roots $\lambda_1 = 1$ and $\lambda_2 = 5$. Since they are real and distinct then the solution will have form

$$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$$

Thirdly, we plug the values of λ_1 and λ_2

$$y = Ae^x + Be^{5x}$$

and this is our general solution to the problem.

The next question is how to calculate the values for constants A and B? We need more information to constrain these values such as $y = 0$ and $y = 1$ when $x = 0$. These are called **initial conditions**, and they tell us the value of the solution and/or its derivatives at a specific point, allowing for a unique solution to the differential equation. For example, for $\frac{dy}{dx} = f(x)$, an initial condition might be $y = y_0$ at $x = x_0$. You will have seen examples when solving the radioactive decay equation, hydrostatic equation and Newton's Laws of Motion. This will lead us to solving simultaneous equations. Do you recall how you solve simultaneous equations? If we want to calculate two unknowns we need to formulate two expressions that provide us with an ability to re-arrange and isolate.

Example 9.2: Let's return to the previous example:

$$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 5y = 0$$

We wrote out the auxiliary equation and found the roots to arrive at the general solution our general solution to the problem:

$$y = Ae^x + Be^{5x}$$

Suppose that we are told that $y = 0$ and $y' = 1$ when $x = 0$ then we use this information to find A and B. Write out y and y':

$$\begin{aligned} y &= Ae^x + Be^{5x} \\ y' &= Ae^x + 5Be^{5x} \end{aligned}$$

Plug in the initial conditions in to y and y' to get:

$$y(0) = A + B = 0 \tag{10.1}$$

$$y'(0) = A + 5B = 1 \tag{10.2}$$

This leaves us with a system of simultaneous equations which we can solve. Take Equation 10.1 from Equation 10.2 gives us:

$$A + 5B - (A + B) = 1 - 0 \implies 4B = 1 \implies B = \frac{1}{4}$$

Substitute $B = \frac{1}{4}$ into Equation 10.1 gives us:

$$A + \frac{1}{4} = 0 \implies A = -\frac{1}{4}$$

Then the full general solution is given by:

$$y = \frac{1}{4}e^{5x} - \frac{1}{4}e^x$$

10.3 Applied boundary value problems and steady-state

In the last section, we dived into finding general solutions of second-order ODEs and captured some definitions and an approach to solve special cases. In this section, we can ground some of this is reality and take a look look at how these types of equations are used in real-world physical theories, therefore, then when we solve second-order ODEs, it tells us about the state of the system we are interested in. We also get back to the process of integration rather than working through a known solution as the previous section.

Examples of second-order ordinary differential equations are:

- **Kinematics**: Acceleration (a) as the rate of change of the rate of change of location (x) in time (t), $\frac{d^2x}{dt^2}$.
- **Forces and motion**: Newton's second law of motion: $F = ma$.
- **Fick's law**: Steady-state mass diffusion: $\frac{d^2C}{dt^2} = 0$.
- **Fourier's law**: Steady-state heat transfer: $\frac{d^2T}{dt^2} = 0$.

These equations look different to the previous section. In some ways, they look easier because they offer us a starting point to begin the process of integration. Indeed, to solve second-order ODEs as we have here, we often have to integrate the equation twice. Firstly to get the first gradient and secondly to get the solution. Let's look at some examples.

Example 9.3: Suppose we want to use Newton's second law of motion to derive the equation of motion (distance versus time) of an object moving down a ramp inclined at 30°, starting with velocity $v = 0$. We will not consider air resistance or friction. Remember that (see Chapter 7 Example 2) if the gravitational force is given by mg (where m is the object's mass and g is the gravitational constant) then the force dragging the block down the slope ($F2$) and the normal force ($F1$) can be written as:

$$F1 = mg\cos(\theta) \quad F2 = mg\sin(\theta)$$

Furthermore, remember that when using Newton's second law of motion, we need to determine the net forces that pull or push a body along a surface. Why are second-order ODEs relevant here? Remember, we can write Newtons second law of motion as follows:

$$F = ma \qquad F = m\frac{dv}{dt}$$
$$F = m\frac{d^2x}{dt^2} \qquad F = mv\frac{dv}{dx}$$

Each expression is equivalent to the other. If we integrate the second equation ($F = m\frac{dv}{dt}$), we can arrive at an expression of velocity (v) as a function of time (t). However, we have been asked to derive an expression of distance (x) as a function of time. This means we

would need to integrate our expression for v again, given that $v = \frac{dx}{dt}$. To solve this problem we have to integrate twice. Lets see this in action. As there is no friction acting on object the total forces on the object are $F = mg\sin(30°)$. We integrate to get the velocity (v) of this object (v from 0 to v and t from 0 to t):

$$\cancel{m}\frac{dv}{dt} = \cancel{m}g\sin(30°) \implies \int_0^v dv = \frac{g}{2}\int_0^t dt \implies [v]_0^v = \frac{g}{2}[t]_0^t \implies v = \frac{g}{2}t$$

We integrate again to get the distance travelled (x) of this object (x from 0 to x and t from 0 to t):

$$v = \frac{dx}{dt} = \frac{g}{2}t \int_0^x dx = \int_0^t \frac{g}{2}t dt \implies [x]_0^x = \frac{g}{2}\left[\frac{t^2}{2}\right]_0^t \implies x = \frac{g}{4}t^2$$

So the velocity increases linearly and the distance quadratically in this problem.

The ODEs that have been covered so far are called **initial value problems**. This is where we specify some initial values (i.e. when $x = 0$, $v = 0$, $t = 0$), and we solve the ODE to find something about our process in the future or a certain distance away. Other kinds of ODEs are called **boundary value problems**, where upper and lower boundary conditions are specified (i.e, when $x = x_0$ and $x = x_1$). This gives us more information this means that we have to make the solution meet at both points. The two we will cover are Fick's and Fourier's laws, which are laws of mass diffusion and heat conduction.

Example 9.4: Fourier's Law. Fourier stated that the flux of heat through a substance is proportional to the negative gradient of temperature in the substance (i.e. heat flows from hot to cold). We can express this, known as Fouriers law, as a differential equation:

$$q = -k\frac{dT}{dx}$$

where T is the temperature and x the distance variable.

Example 9.5: Fick's Law. Fick also found that the law that governs transport of mass by *Brownian motion* (i.e. random thermal motion) moves from high concentrations to low concentrations. We can express this, known as Fick's law, as a differential equation:

$$J = -D\frac{dC}{dx}$$

where J is the mass flux, C is the concentration of mass (e.g. mass of pollutant / gas or particulate) and D is a diffusion coefficient which depends on the properties of the substance that is diffusing.

For both of these equations, we assume steady state conditions. Remember from Chapter 5, a system is in steady-state when the key variables that describe the system's behavior do not change over time (i.e. $f(x) = c$ where c is constant). See Figure 10.1, consider the situation where water is poured into a container which has an outlet part on the side of the container. At first when the water level is below the outlet, the water level in the container rises with time. However, when the water level reaches a certain height, the amount of water entering the tank and the amount of discharged water are balanced and the water level remains constant. The system has reached a steady state. If the system is in steady state, we can differentiate again to get a derivative equal to zero (i.e. $\frac{df}{dx} = 0$. In Fick's and Fourier's Law, we specify that the rate of change remains constant, either over space or time, we can rewrite them as:

$$\frac{d^2T}{dx^2} = 0 \quad \frac{d^2C}{dt^2} = 0$$

These are **second-order ODEs**. Similar to Exercise 3, we will need to integrate them twice to arrive at expressions for concentrations and temperatures as a function of distance, however as you will see in the exercises, we will specify boundary conditions rather than intial conditions.

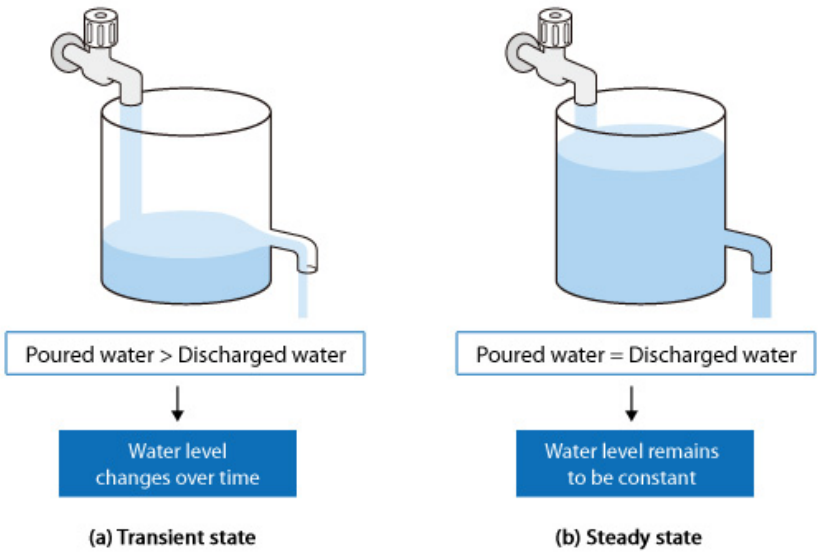


Figure 10.1: Diagram showing a system in (a) transient state and (b) steady state.

Exercises

Algebraic problems

For this section we are concerned with **homogeneous, linear, second-order ODEs** which resemble the following expression:

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

To find a general solution for y we first write out the **auxiliary** equation

$$a\lambda^2 + b\lambda + c = 0$$

and solve it to find the roots. If the roots are real and not equal ($\lambda_1 \neq \lambda_2$ then we can write a general solution as:

$$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$$

If our two roots are real and equal (i.e. $\lambda_1 = \lambda_2$) then we use the formula:

$$y = (A + Bx)e^{\lambda_1 x}$$

We don't consider what happens when the two roots are not real (i.e. $b^2 - 4ac < 0$) in this course.

Question 9.1: For the following second-order ordinary differential equations, write out the auxiliary questions, find the roots and express the general solution:

(a) $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} - 16y = 0$
(b) $8\frac{d^2y}{dt^2} - 2\frac{dy}{dt} - y = 0$
(c) $4\frac{d^2y}{dt^2} - 24\frac{dy}{dt} + 27y = 0$
(e) $\frac{d^2y}{dt^2} - 4y = 0$

Question 9.2: Return to the general solutions found in Question 9.1 and use the following initial or boundary conditions to find the constants A and B and then state the full solution.

(a) $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} - 16y = 0$ using $y = 2$ and $\frac{dy}{dx} = -6$ when $x = 0$.
(b) $8\frac{d^2y}{dt^2} - 2\frac{dy}{dt} - y = 0$ using $y = 10$ and $\frac{dy}{dx} = 2$ when $x = 0$.
(c) $4\frac{d^2y}{dt^2} - 24\frac{dy}{dt} + 27y = 0$ using $y = 0$ and $\frac{dy}{dx} = 12$ when $x = 0$.
(d) $\frac{d^2y}{dt^2} - 4y = 0$ using $y = -4$ and $\frac{dy}{dx} = 0$ when $x = 0$.

Question 9.3: Solve the following differential equation given the associated initial conditions associated with both the function value for y and gradient $\frac{dy}{dx}$.

$$3\frac{d^2y}{dt^2} + 2\frac{dy}{dt} - 8y = 0$$

where

$$y = -6 \text{ and } \frac{dy}{dx} = -18 \text{ when } x = 0$$

Question 9.4: For the following second-order ordinary differential equations, write out the auxiliary questions, find the roots and express the general solution:

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$$

Real-world problems

Question 9.5: Newtons laws of motion (as 2nd Order ODEs): We can write Newtons second law of motion as follows:

$$\begin{aligned} F &= m\frac{dv}{dt} \\ F &= m\frac{d^2x}{dt^2} \end{aligned}$$

(a) Use Newton's second law of motion to derive the equation of motion (distance versus time) of an object moving down a ramp inclined at 30°, starting with velocity $v = 0$. Do not consider air resistance or friction.

(b) Use Newton's second law of motion to derive the equation of motion (distance versus time) of an object moving down a ramp inclined at 30°, starting with velocity, $v = 0$. Do not consider air resistance but consider the frictional force to be equal to $\mu \times N$, where N is the magnitude of the contact force and $\mu = 0.1$. Solve this for initial velocity $= 0$ to get the position x vs time.

Question 9.6: Fick's Law of diffusion (as 2nd Order ODEs): An industrial incinerator produces a steady output of particulate matter having a concentration of $C_0 = 1.0$ mg m^{-3} of air at the chimney. 1 km away from the incinerator plant the concentration is $C_{1km} = 0.1$ mg m^{-3}. Some houses are a distance of 500 m away from the plant. Using Fick's law of 1-d diffusion, $\frac{d^2C}{dx^2} = 0$, what is the concentration of pollutant where the houses are? Note that you have been told the limit for particulate matter is 0.5 mg m^{-3}. Is the concentration at the location of the houses above 0.5 mg m^{-3}?

Question 9.7: Fouriers law of heat transfer (as 2nd Order ODEs): 3 km below the surface there is a hot granite bed, with a steady-state temperature of 250 °C. The temperature just below the surface the temperature stays relatively constant at 10°C, use the steady-state Fouriers law in 1-d, $\frac{d^2T}{dx^2} = 0$ to derive an equation that describes the temperature profile between the surface and the hot granite.

Question 9.8: Newtons laws of motion (as 2nd Order ODEs): Derive an equation for the displacement with time if an object moving at an initial velocity is experiencing a constant acceleration. In other words, we know that we can write Newtons second law as a second order ODE):

$$\frac{d}{dt}\left(\frac{dx}{dt}\right) = a$$

where a is now a constant. Proceed with integrating this expression by assuming that the starting position is $x = 0$ and initial velocity is v_0 m/s.

Solutions

Algebraic problems

Question 9.1:

(b) Write out the auxiliary equations:

$$8\lambda^2 - 2\lambda - 1 = 0$$

Find the roots of the equation:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \times (-1) \times 8}}{2 \times 8} = \frac{2 \pm \sqrt{36}}{16} = \frac{2 \pm 6}{16}$$

so the roots are $\lambda_1 = \frac{1}{2}$ and $\lambda_2 = -\frac{1}{4}$. So as the roots are real and distinct the general solution is:

$$y(t) = Ae^{\frac{t}{2}} + Be^{-\frac{t}{4}}$$

(c) Write out the auxiliary equations:

$$4\lambda^2 - 24\lambda + 27 = 0$$

Find the roots of the equation:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-24) \pm \sqrt{(-24)^2 - 4 \times 4 \times 27}}{2 \times 4} = \frac{24 \pm \sqrt{144}}{8} = \frac{24 \pm 12}{8}$$

so the roots are $\lambda_1 = \frac{9}{2}$ and $\lambda_2 = \frac{3}{2}$. So as the roots are real and distinct the general solution is:

$$y(t) = Ae^{\frac{9t}{2}} + Be^{-\frac{3t}{4}}$$

(d) Write out the auxiliary equations:

$$\lambda^2 - 4 = 0$$

Find the roots of the equation:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{0 \pm \sqrt{0^2 - 4 \times 1 \times (-4)}}{2 \times 1} = \frac{0 \pm \sqrt{16}}{2}$$

so the roots are $\lambda_1 = -2$ and $\lambda_2 = 2$. So as the roots are real and distinct the general solution is:

$$y(t) = Ae^{2t} + Be^{-2x}$$

Question 9.2:

(a) From Question 9.1(a), the general solution is:

$$y(t) = Ae^{2t} + Be^{-8x}$$

We know that $y = 2$ and $\frac{dy}{dx} = -6$ when $x = 0$. Write out y and $\frac{dy}{dx}$:

$$\begin{aligned} y &= Ae^{2t} + Be^{-8x} \\ \frac{dy}{dx} &= 2Ae^{2t} - 8Be^{-8x} \end{aligned}$$

Plug in the initial conditions into y and $\frac{dy}{dx}$:

$$\begin{aligned} A + B &= 2 \\ 2A - 8B &= -6 \end{aligned}$$

This leaves a system of simultaneous equations which we can solve:

$$\begin{aligned} 2(A+B) - (A-8B) = 4+6 \implies 6B = 6 \quad &\implies \quad B = 1 \\ A + B = 2 \implies A + 1 = 2 \quad &\implies \quad A = 1 \end{aligned}$$

Then the full general solution is given by:

$$y(t) = e^{2t} + e^{-8x}$$

(b) From Question 9.1(a), the general solution is:

$$y(t) = Ae^{\frac{t}{2}} + Be^{-\frac{t}{4}}$$

We know that $y = 10$ and $\frac{dy}{dx} = 2$ when $x = 0$. Write out y and $\frac{dy}{dx}$:

$$\begin{aligned} y &= Ae^{\frac{t}{2}} + Be^{-\frac{t}{4}} \\ \frac{dy}{dx} &= \frac{A}{2}e^{\frac{t}{2}} - \frac{B}{4}e^{-\frac{t}{4}} \end{aligned}$$

Plug in the initial conditions into y and $\frac{dy}{dx}$:

$$\begin{aligned} A + B &= 10 \\ \frac{A}{2} - \frac{B}{4} &= 2 \end{aligned}$$

This leaves a system of simultaneous equations which we can solve:

$$\begin{aligned} A + B + 4(\frac{A}{2} - \frac{B}{4}) = 10 + 8 \implies 3A = 18 \quad &\implies \quad A = 6 \\ A + B = 10 \implies 6 + B = 10 \quad &\implies \quad B = 4 \end{aligned}$$

Then the full general solution is given by:

$$y(t) = 6e^{\frac{t}{2}} + 4e^{-\frac{t}{4}}$$

(c) From Question 9.1(a), the general solution is:

$$y(t) = Ae^{\frac{9t}{2}} + Be^{-\frac{3t}{4}}$$

We know that $y = 0$ and $\frac{dy}{dx} = 12$ when $x = 0$. Write out y and $\frac{dy}{dx}$:

$$\begin{aligned} y &= Ae^{\frac{9t}{2}} + Be^{-\frac{3t}{4}} \\ \frac{dy}{dx} &= \frac{9A}{2}e^{\frac{9t}{2}} - \frac{3B}{4}e^{-\frac{3t}{4}} \end{aligned}$$

Plug in the initial conditions into y and $\frac{dy}{dx}$:

$$\begin{aligned} A + B &= 0 \\ \frac{9A}{2} - \frac{3B}{2} &= 12 \end{aligned}$$

This leaves a system of simultaneous equations which we can solve:

$$\begin{aligned} 3(A+B) + 2(\frac{9A}{2} - \frac{3B}{2}) = 0 - 24 \implies 12A = 24 \quad &\implies \quad A = 2 \\ A + B = 0 \implies 2 + B = 0 \quad &\implies \quad B = -2 \end{aligned}$$

Then the full general solution is given by:

$$y(t) = 2e^{\frac{9t}{2}} - 2e^{-\frac{3t}{4}}$$

(d) From Question 9.1(a), the general solution is:

$$y(t) = Ae^{2t} + Be^{-2x}$$

We know that $y = -4$ and $\frac{dy}{dx} = 0$ when $x = 0$. Write out y and $\frac{dy}{dx}$:

$$\begin{aligned} y &= Ae^{2t} + Be^{-2x} \\ \frac{dy}{dx} &= Ae^{2t} - 2Be^{-2x} \end{aligned}$$

Plug in the initial conditions into y and $\frac{dy}{dx}$:

$$\begin{aligned} A + B &= -4 \\ 2A - 2B &= 0 \end{aligned}$$

This leaves a system of simultaneous equations which we can solve:

$$\begin{aligned} 2(A+B) + 2A - 2B = -8 + 0 \implies 4A = -8 \quad &\implies \quad A = -2 \\ A + B = 0 \implies -2 + B = -4 \quad &\implies \quad B = -2 \end{aligned}$$

Then the full general solution is given by:

$$y(t) = -2\left(e^{2t} + e^{-2x}\right)$$

Question 9.3: Write out the auxiliary equations:

$$3\lambda^2 + 2\lambda - 8 = 0$$

Find the roots of the equation:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{2^2 - 4 \times 3 \times (-8)}}{2 \times 3} = \frac{-2 \pm \sqrt{100}}{6} = \frac{-2 \pm 10}{6}$$

so the roots are $\lambda_1 = \frac{4}{3}$ and $\lambda_2 = -2$. So as the roots are real and equal the general solution is:

$$y(x) = Ae^{\frac{4t}{3}} + Be^{-2t}$$

We know that $y = -6$ and $\frac{dy}{dx} = -18$ when $x = 0$. Write out y and $\frac{dy}{dx}$:

$$\begin{aligned} y &= Ae^{\frac{4t}{3}} + Be^{-2t} \\ \frac{dy}{dx} &= \frac{4A}{3}e^{\frac{4t}{3}} - 2Be^{-2t} \end{aligned}$$

Plug in the initial conditions into y and $\frac{dy}{dx}$:

$$\begin{aligned} A + B &= -6 \\ \frac{4A}{3} - 2B &= -18 \end{aligned}$$

Substitute to solve:

$$\begin{aligned} 2(A+B) + \frac{4A}{3} - 2B = -12 - 18 \implies \frac{10A}{3} = -30 \quad &\implies \quad A = -9 \\ A + B = -6 \implies -9 + B = -9 \quad &\implies \quad B = 3 \end{aligned}$$

Then the full general solution is given by:

$$y = 3e^{-2t} - 9Ae^{\frac{4t}{3}}$$

Question 9.4: Write out the auxiliary equations:

$$\lambda^2 - 4\lambda + 4 = 0$$

Find the roots of the equation:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-4) \pm \sqrt{(-4)^2 - 4 \times 1 \times 4}}{2 \times 1} = \frac{4 \pm \sqrt{0}}{2}$$

so the roots are $\lambda_1 = \lambda_2 = 2$. So as the roots are real and equal the general solution is:

$$y(x) = (A + Bx)e^{2x}$$

Real-world problems

Question 9.5: Newtons laws of motion (as 2nd Order ODEs):

(a) The total forces on the object are $F = mg\sin(30^\circ)$. We integrate to get the velocity (v) of this object (v from 0 to v and t from 0 to t):

$$\cancel{m}\frac{dv}{dt} = \cancel{m}g\sin(30^\circ) \implies \int_0^v dv = \frac{g}{2}\int_0^t dt \implies [v]_0^v = \frac{g}{2}[t]_0^t \implies v(t) = \frac{g}{2}t$$

We integrate again to get the distance travelled (x) of this object (x from 0 to x and t from 0 to t):

$$v = \frac{dx}{dt} = \frac{g}{2}t \int_0^x dx = \int_0^t \frac{g}{2}tdt \implies [x]_0^x = \frac{g}{2}\left[\frac{t^2}{2}\right]_0^t \implies x(t) = \frac{g}{4}t^2$$

So the velocity increases linearly and the distance quadratically in this problem.

(b) The total forces on the object are $F = mg\sin(30^\circ) - mg\cos(30^\circ)$. We integrate to get the velocity (v) of this object (v from 0 to v and t from 0 to t):

$$\begin{aligned}
&\cancel{m}\frac{dv}{dt} = \cancel{m}g\sin 30^\circ - \cancel{m}g\cos 30^\circ \\
&\implies \int_0^v dv = g(\sin(30^\circ) - \cos(30^\circ))\int_0^t dt \\
&\implies [v]_0^v = g(\sin(30^\circ) - \cos(30^\circ))[t]_0^t \\
&\implies v(t) = g(\sin(30^\circ) - \cos(30^\circ))t
\end{aligned}$$

We integrate again to get the distance travelled (x) of this object (x from 0 to x and t from 0 to t):

$$\begin{aligned}
&v = \frac{dx}{dt} = \frac{g}{2}g(\sin(30^\circ) - \cos(30^\circ))t \\
&\implies \int_0^x dx = g(\sin(30^\circ) - \cos(30^\circ))\int_0^t tdt \\
&\implies [x]_0^x = \frac{g}{2}(\sin(30^\circ) - \cos(30^\circ))\left[\frac{t^2}{2}\right]_0^t \\
&\implies x(t) = \frac{g}{2}(\sin(30^\circ) - \cos(30^\circ))t^2
\end{aligned}$$

Again, the velocity increases linearly and the distance quadratically in this problem.

Question 9.6: Fick's Law of diffusion (as 2nd Order ODEs): We need to solve:

$$\frac{d^2C}{dx^2} = 0$$

Differentiating once gives us:

$$\frac{dC}{dx} = A$$

Differentiating twice gives us:

$$C(x) = Ax + B$$

In general, the boundary conditions that $C(0) = C_0$ and $C(1000) = C_{1km}$ means that

$$\begin{aligned} C(0) &= B = C_0 \\ C(1000) &= 1000A + C_0 = C_{1km} \implies A = \frac{C_{1km} - C_0}{1000} \end{aligned}$$

Therefore the solution to Fick's law is:

$$C(x) = C_0 + \frac{C_{1\,km} - C_0}{1000} x$$

Using $C_0 = 1$ and $C_{1km} = 0.1$, houses 500m away have concentration

$$\begin{aligned} C(x) &= C_0 + \frac{C_{1\,km} - C_0}{1000} x \\ &= 1 + \frac{0.1 - 1}{1000} \times 500 \\ &= 0.55 \end{aligned}$$

So the concentration is 0.55mg/m^{-3}, which is above the 0.5 limit.

Question 9.7: Fouriers law of heat transfer (as 2nd Order ODEs): This is similar to Question 9.6. We need to solve:

$$\frac{d^2T}{dx^2} = 0$$

Differentiating once gives us:

$$\frac{dT}{dx} = A$$

Differentiating twice gives us:

$$T(x) = Ax + B$$

In general, the boundary conditions that $T(0) = T_0$ and $T(3000) = T_{3km}$ means that

$$\begin{aligned} T(0) &= B = T_0 \\ T(3000) &= 3000A + T_0 = T_{3km} \implies A = \frac{T_{3km} - T_0}{3000} \end{aligned}$$

Therefore the solution to Fick's law is:

$$T(x) = T_0 + \frac{T_{3km} - T_0}{3000} x$$

Plugging in $T_0 = 10$ and $T_{3km} = 250$ gives

$$T(x) = 10 + \frac{250 - 10}{3000} x = 10 + \frac{2}{25} x$$

Question 9.8: Newtons laws of motion (as 2nd Order ODEs): Integrate once

$$\begin{aligned}
\frac{d}{dt}\left(\frac{dx}{dt}\right) &= a \\
\int_0^{\frac{dx}{dt}} d\left(\frac{dx}{dt}\right) &= \int_0^t a dt \\
\frac{dx}{dt} &= at + B
\end{aligned}$$

When $t = 0$ we know that $v = v_0$ so then we see that $B = v_0$ then

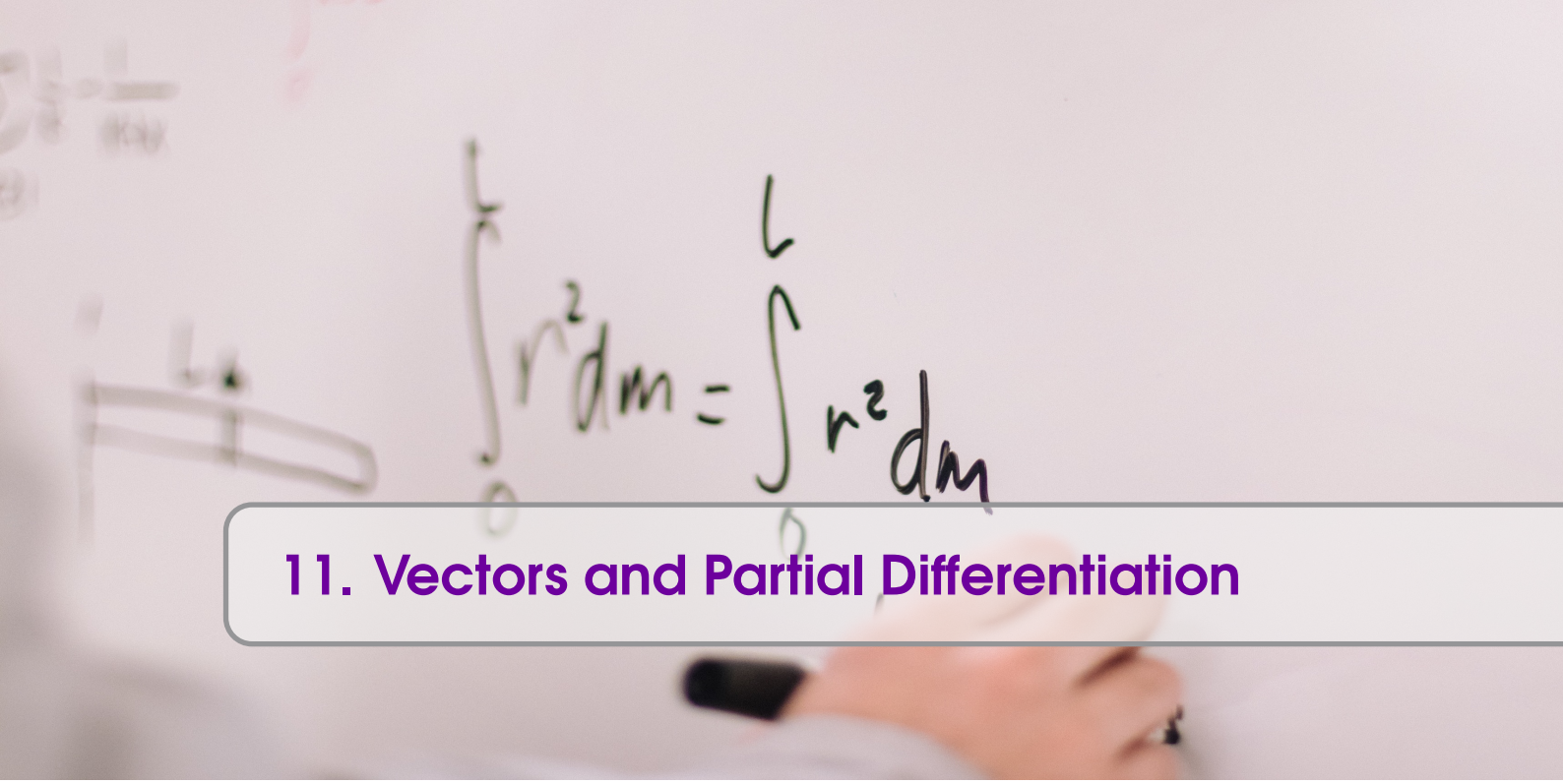
$$\frac{dx}{dt} = at + v_0$$

Integrate a second time:

$$\begin{aligned}
\frac{dx}{dt} &= at + v_0 \\
\int_0^x dx &= \int_0^t at + v_0 dt \\
x(t) &= \frac{a}{2}t^2 + v_0 t + C
\end{aligned}$$

When $t = 0$ we know that $x = 0$ so then we see that $C = 0$ then

$$x(t) = \frac{a}{2}t^2 + v_0 t$$



11. Vectors and Partial Differentiation

11.1 Overview

Now that we have concluded the topics Differentiation (Chapters 1–5) and Integration (Chapters 6–10) , we should note that these topics have been centred around a function in one variable or dimension (i.e. $y = f(x)$). We differentiated or integrated in terms of that one variable:

$$\frac{dy}{dx} \text{ or } \int f(x)\,dx$$

In many cases, we may need to work in more than one dimension (e.g. in space) such as $f(x,y) = x^2 + y^2$, therefore, in this Chapter we look at **scalar fields**, **vectors** and **partial differentiation**.

11.2 Scalar fields

In mathematics and physics, a **scalar field** is a function of two or more variables (i.e. $f(x,y)$) that assigns a single scalar value (a number) to every point in a space. Scalar fields represent a single value at different positions. They have magnitude but no direction and can be represented with a colour plot (e.g. temperature over a city, or for example wind speed – but not direction). In essence, they are functions in higher dimensions than we have seen thus far (i.e. $f(x)$). Take a look at the 2-D scalar fields displayed in Figure 11.1.

These figures show three scalar fields: $f(x,y) = x^2$, $f(x,y) = xy$, and $f(x,y) = \frac{1}{2}\left(x^2 + y^2\right)$. Which ones are which? Try to explain why. In the plots white represents high values and black low values. Where are the maximum and minimum values for each scalar field?

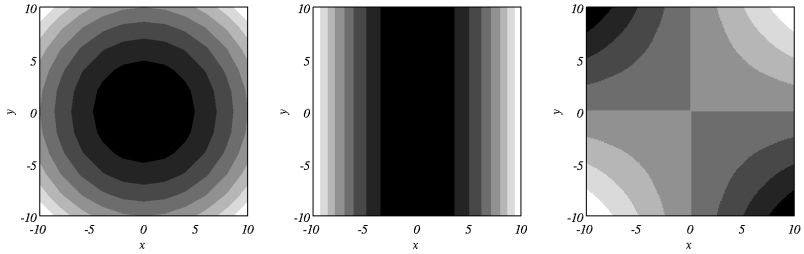


Figure 11.1: Three scalar fields: $f(x,y) = x^2$, $f(x,y) = xy$, and $f(x,y) = \frac{1}{2}\left(x^2 + y^2\right)$

11.3 Vectors

Now take a look at the plots in Figure 11.2 which represent **vector fields**. Can you see the similarity? Imagine these are wind fields, as you might see on a weather forecast. In the vector fields we can determine there is a direction, or flow. The magnitude of that direction can either be taken from the length of the arrow, or by referring to the scalar fields from the plots above. Which vector fields can be aligned to the scalar fields in Figure 11.1?

The left and middle vector field could both relate to the left hand scalar field. So you see here, these vector fields given opposite flow but have the same magnitude as a function of distance. SO how do we represent a vector field algebraically?

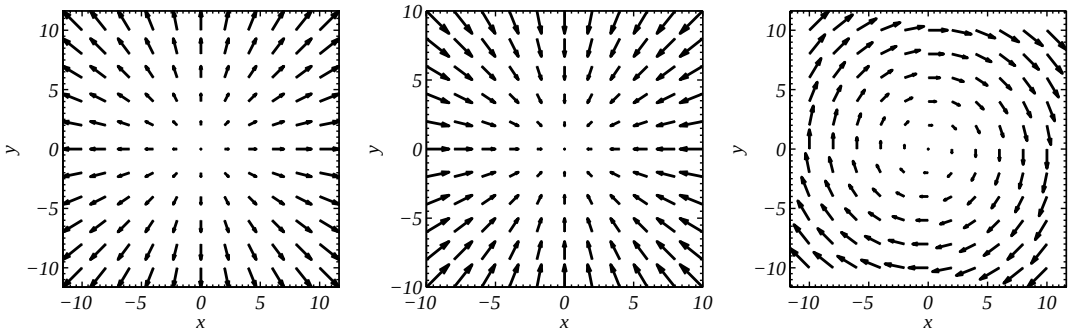


Figure 11.2: Vector fields, from which we can determine there is a direction, or flow. The magnitude of that direction can either be taken from the length of the arrow, or by referring to the scalar fields from the plots in Figure 11.1.

Vector notation

Vectors have **magnitude** and **direction**. We can use a simply graph to illustrate the concept. Consider Figure 11.3. Each line in this figure represents the same vector, but a different starting position. Wherever the starting position is, each vector moves 2 units to the left and 5 units up. If we write this procedure in vector form we write:

$$\vec{v} = \langle -2, 5 \rangle$$

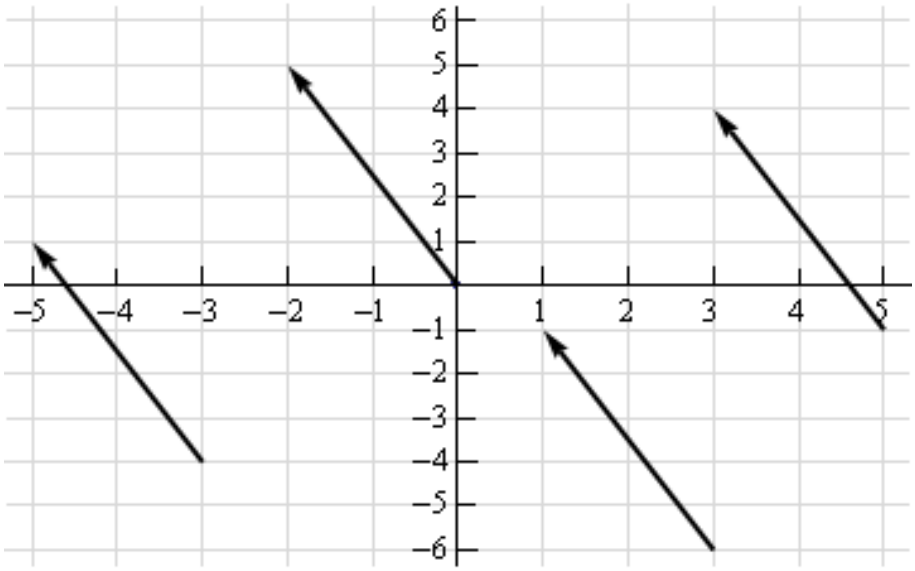


Figure 11.3: Vector fields. In this figure we use the same vector to draw a line but vary the starting position in each case.

Notice the distinction in style between the variable and the brackets. Specifically, the notation $\langle -2,5\rangle$ is fundamentally different from writing $(-2,5)$. The former conveys information about the magnitude and direction of a vector, while the latter represents a point in space defined by coordinates. A vector can be written more generally as:

$$\vec{v} = \langle a_1, a_2\rangle$$

where the constants a_1 and a_2 describe shifts along the horizontal and vertical axes, respectively, based on the assumed coordinate system.

Now, imagine this vector represents movement from point A to B. In this case, we write the vector as $\vec{AB}$. In two-dimensional space, if point A has coordinates $A = (x,y)$ and point B has coordinates $B = (x+a_1, y+a_2)$, the vector $\vec{AB}$ describes the displacement. Note the difference in notation between vectors and coordinates. The same principle extends to three-dimensional systems. If the vector becomes:

$$\vec{v} = \langle a_1, a_2, a_3\rangle$$

then the coordinates of points A and B can be written as $A = (x,y,z)$ and $B = (x+a_1, y+a_2, z+a_3)$, respectively, representing the vector $\vec{AB}$.

More generally, the direction indicated by the arrow over the vector, as in $\vec{AB}$, specifies movement from point A to B along a straight line. If the coordinates of A and B are given as $A = (a_1, a_2, a_3)$ and $B = (b_1, b_2, b_3)$, the vector $\vec{v}$ describing $\vec{AB}$ is:

$$\vec{v} = \langle b_1 - a_1, b_2 - a_2, b_3 - a_3\rangle$$

Conversely, if our vector now starts at B and moves to A, thus $\vec{BA}$, then we can write:

$$\vec{v} = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

Using a **start minus end** approach. You can practice these operations in the exercises.

Unit vectors and vector magnitude

In our set of problems we consider vectors in Cartesian form, that is broken down into components along x, y and z of a Cartesian domain. The component along x is component $\hat{i}$, along y is component $\hat{j}$ and along z is component $\hat{k}$. $\hat{i}$, $\hat{j}$ and $\hat{k}$ are all called unit vectors and have length equal to 1.For example, take a look at the following vector:

$$4\hat{i} + 3\hat{j}$$

This tells us that, whatever process is described using this vector, for every movement 4 units along the x axis there is 3 units of movement along the y axis. In our example, $\hat{i}$ and $\hat{j}$ represent what we call a unit vector so has a magnitude of 1. Note that this is equivalent to writing

$$\vec{v} = \langle 4, 3 \rangle$$

from above.

The size of a vector (or its magnitude) can be calculated by taking the square root of all of the square its components (i.e. Pythagoras' theorem). So if $\bar{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, then the magnitude, $|\bar{a}|$:

$$|\bar{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

A vector that has length that points in the same direction as a general vector $\bar{a}$ is called the unit vector of $\bar{a}$. It is calculated by dividing all components of $\bar{a}$ by the magnitude of $\bar{a}$.

Vector arithmetic

Vectors can be added or subtracted (add or subtract by component) or multiplied by a scalar (constant). For example, Figure 11.4 demonstrates adding two vectors a and b to create a new vector. When adding vectors, you simply add the components for $\hat{i}$ and $\hat{j}$.

Given the vectors $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ the addition of the two vectors is given by the following formula:

$$\vec{a} + \vec{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

Subtraction is similar. Given the vectors $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ the subtraction of the two vectors is given by the following formula:

$$\vec{a} - \vec{b} = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

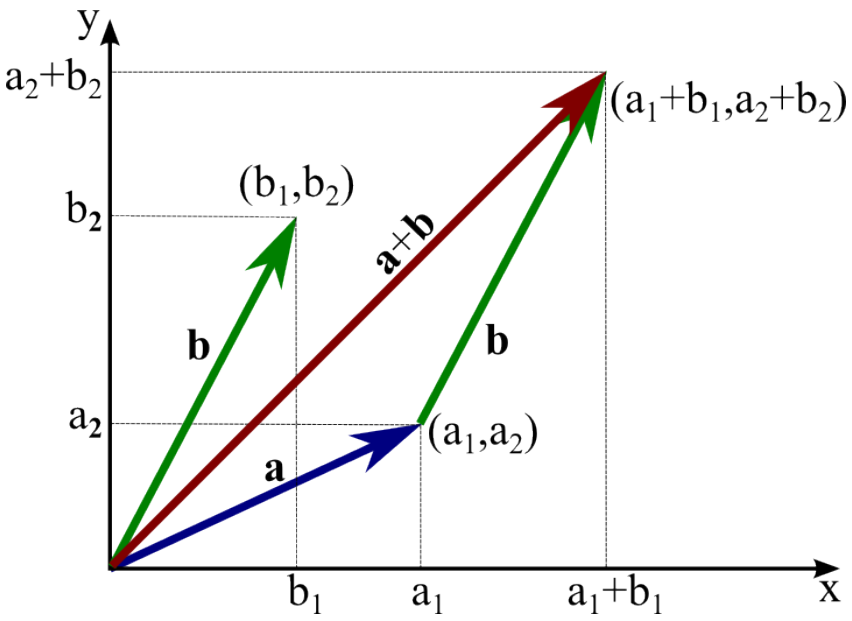


Figure 11.4

This can be harder to visualise. In Figure 11.5 you can see that if we consider we are adding the negative version of vector $\vec{b}$ to $\vec{a}$ (thus $\vec{a}+(-\vec{b})$ since we are performing an extraction), then you can visualise the net direction of travel:

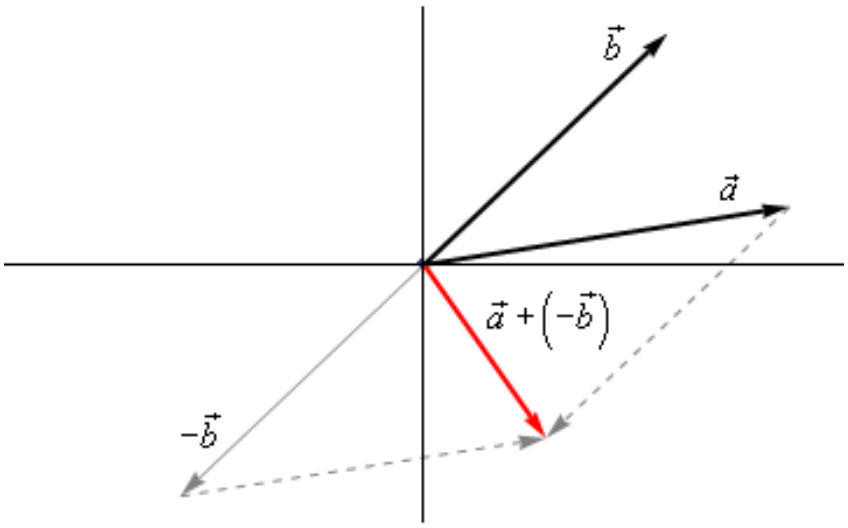


Figure 11.5: Visualisation of vector addition. Credit to https://tutorial.math.lamar.edu/Classes/CalcII/VectorArithmetic.aspx

We can finish on the ability to multiple vectors by a constant. Imagine a situation where a unit vector described the direction of travel every second across a two dimensional grid. If we knew the object was travelling at 15 metes per second, how might we find the absolute position over 1 second? We would multiple the unit vector by the speed. Combined, this new vector represents the velocity of the object. Take a look at Figure 11.6. We have 3

vectors that have been derived from the vector $\vec{a} = \langle 2, 4 \rangle$

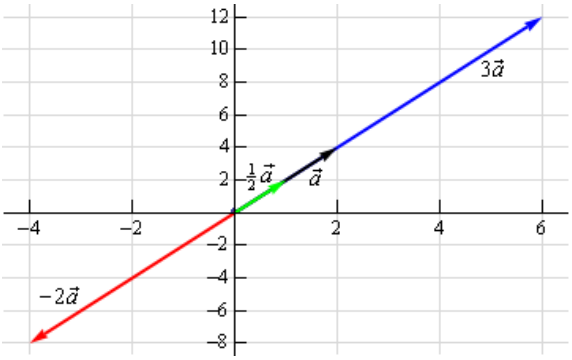


Figure 11.6: Visualisation of vector and scalar multiplication. Credit to https://tutorial.math.lamar.edu/Classes/CalcII/VectorArithmetic.aspx

These lines can be written as the following scalar multiplications:

$$3\vec{a} = \langle 6, 12 \rangle \quad \tfrac{1}{2}\vec{a} = \langle 1, 2 \rangle \quad -2\vec{a} = \langle -4, -8 \rangle$$

Noting that when we multiply our original vector by a negative number, we simply reverse the direction of travel.

11.4 Partial Differentiation

For the vast majority of this course, we have dealt with differentiation of functions in one dimension, or of one variable (i.e. $y = f(x)$). We now think about differentiation of functions of two (or more) variables (i.e. scalar field), $z = f(x, y)$, which now depend on x and y.

For a one dimensional curve the gradient $\frac{dy}{dx}$ is fairly straight forward. When there are more than one dimension we have to use something else to denote the gradient. The standard that is used is called 'grad', which is defined for a scalar field and can be written $grad(f)$ or also written ∇f. We note that ∇f (where $f = f(x, y, z)$ is a scalar field) is actually a vector. In Cartesian coordinates it is:

$$\nabla f \equiv \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k} \equiv \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

So the gradient has magnitude and direction. To describe the movement of mass, for example, this is important information. Notice the change in the derivative symbol. Specifically, let us highlight the gradient along the x axis:

$$\frac{\partial f}{\partial x}$$

Up to this point in our course, we have been using the expression:

$$\frac{df}{dx}$$

The change is important. The new expression is called a *partial differential equation*. When you have a function expressed using more than one dimension, this means that when calculating a derivative in say x, you assume other variables such as y or z remain constant. We are only interested in information along the axis given by the variable in the denominator. We can get a feel for this through the exercises and solutions provided in the later stages of today's material.

Even when we do not use vector notation, the partial derivatives of a function can be derived. Take the following example:

$$f(x,y) = 2x^2y^3 + y$$

where we have defined a function that varies with respect to both x and y. We have been asked to find the rate of change of the function at a set of coordinates given by (a,b). Conceptually we will need to hold y constant and vary x to determine the gradient along the x axis a vice versa. Indeed, if we were using the method of finite difference, we could use the following assumption:

$$\frac{\partial f}{\partial x}_{(x,y)=(a,b)} = \lim_{h \mapsto 0} \frac{f(a,b+h) - f(a,b)}{h}$$

Of course, we don't need to use finite difference in this instance since we can apply one of our rules of differentiation.

Example 10.1: Let's find the partial derviatives of the following function:

$$f(x,y) = 2x^2y^3 + y$$

If we take y as constant then we can derive the following expression for the partial derivative of f with respect to x:

$$\frac{\partial f}{\partial x} = 4xy^3$$

Note that the second term disappears here as it has no x within it. For the partial derivative of f with respect to y, we arrive at the following:

$$\frac{\partial f}{\partial y} = 6x^2y^2 + 1$$

Putting this into vector form gives us:

$$\begin{aligned} \nabla f &= \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} \\ &= 4xy^3\hat{i} + 6x^2y^2 + 1\hat{j} \end{aligned}$$

or in the form:

$$\begin{aligned}\nabla f &= \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle \\ &= \left\langle 4xy^3, 6x^2y^2 + 1 \right\rangle\end{aligned}$$

We can use this to calculate the magnitude of the gradient at any given point.

We could go even further and take second partial derivatives of f to obtain get $\nabla^2 f$ which would mean partial differentiating with respect to x and y:

$$\frac{\partial^2 f}{\partial x \partial y}$$

as we are differentiating twice with respect to all combinations of the variables, $\nabla^2 f$ becomes a matrix which has amazing properties in mathematics and physics, but this is outside the remit of this course!

Furthermore, integrating partial differential equations analytically can only be done for simple cases and requires knowledge of Fourier methods, which are outside of the remit of this course. Nearly all modern weather forecast models use approximate methods to find solutions of **partial differential equations**.

Exercises

Question 10.1: Find the magnitude of the following vectors:

(a) $\bar{a} = 4\hat{i} + 3\hat{j}$

(b) $\bar{b} = 4\hat{i} - 3\hat{j}$

(c) $\bar{c} = 3\hat{i} + 4\hat{j} + 5\hat{k}$

Question 10.2: What is the unit vector that points in the same direction as the vector $3\hat{i} + 4\hat{j} + 5\hat{k}$? A unit vector has a magnitude [length] of 1. Therefore you need to calculate the magnitude of this vector and then divide the vector by this value.

Question 10.3: Provide the vector of each of the following:

(a) The vector that represents a line from $\vec{a} = (2, -7, 0)$ to $\vec{b} = (1, -3, -5)$

(b) The vector that represents a line from $\vec{a} = (1, -3, -5)$ to $\vec{b} = (2, -7, 0)$

Question 10.4: Determine the magnitude of the following vectors:

(a) $\vec{a} = \langle 3, -5, 10 \rangle$

(b) $\vec{p} = \left\langle \frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right\rangle$

Question 10.5: At time $t = 0$ two cars are at the origin of a set of x-y coordinates (i.e. $0 = (0,0)$). Starting at $t = 0$, car A travels along the x-axis at 30 mph. At the same time car B travels in the direction $3\hat{i} + 4\hat{j}$ with a speed of 60 mph. After 3 hours, what is the distance between the two cars in miles? This requires some thought which may be helped by drawing a diagram.

Question 10.6: Calculate the gradient, *grad* of the following scalar fields:

(a) $f(x,y) = x^2 \sin(5y)$

(b) $f(x,y,z) = z \exp(-xy)$

(c) $f(x,y) = 1 \times 10^{-3} (x - 400)^2 + 1 \times 10^{-3} y^2$

Question 10.7: For the surface / scalar field: $f(x,y) = 1 \times 10^{-3} (x - 400)^2 + 1 \times 10^{-3} y^2$, calculate the magnitude of the gradient vector for all x and y. What is the magnitude of the gradient vector at $(x,y) = (800, 400)$? For this problem you will need to calculate the partial derivatives with respect to x and y and then create a vector, before arriving at an expression for the magnitude of that vector.

Question 10.8: Find all of the first order partial derivatives for the following functions. This means we are asking you to to find all of the partial derivatives with respect to each

variable.

(a) $f(x,y) = x^4 + 6\sqrt{y} - 10$

(b) $f(x,y) = cos\left(\frac{4}{x}\right) e^{x^2 y - 5y^3}$

(c) $z = \frac{9u}{u^2+5v}$

Question 10.9: An equation describing a bowl shaped hill is $f(x,y) = 100\exp\left(-1\times 10^{-5}\left[x^2+y^2\right]\right)$. You decide to walk along the line $y = 0$. What is the gradient $\frac{\partial f(x,y)}{\partial x}$?

Solutions

Question 10.1:

(a) $\bar{a} = 4\hat{i} + 3\hat{j} \implies ||\bar{a}|| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$

(b) $\bar{b} = 4\hat{i} - 3\hat{j} \implies ||\bar{b}|| = \sqrt{4^2 + (-3)^2} = \sqrt{25} = 5$

(c) $\bar{c} = 3\hat{i} + 4\hat{j} + 5\hat{k} \implies ||\bar{c}|| = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{25} = 7.07$

Question 10.2: For a vector to be a unit vector, we need $||\bar{v}|| = 1$ so we need to normalise the $\bar{v} = \sqrt{3^2 + 4^2 + 5^2}$ which is $||\bar{v}|| = \sqrt{50}$. Therefore the unit vector is

$$\frac{3}{\sqrt{50}}\hat{i} + \frac{4}{\sqrt{50}}\hat{j} + \frac{5}{\sqrt{50}}\hat{k}$$

Question 10.3: Provide the vector of each of the following:

(a) To construct this vector we subtract coordinates of the starting point from the ending point.

$$\vec{AB} = \langle 1-2, -3-(-7), -5-0 \rangle = \langle -1, 4, -5 \rangle$$

(b) To construct this vector we subtract coordinates of the starting point from the ending point.

$$\vec{AB} = \langle 2-1, -7-(-3), 0--(-5) \rangle = \langle 1, -4, 5 \rangle$$

Question 10.4:

1. $||\vec{a}|| = \sqrt{9 + 24 + 100} = \sqrt{134}$
2. $||\vec{p}|| = \sqrt{\frac{1}{5} + \frac{4}{5}} = 1$

Question 10.5:

- Car A will have moved $30 \times 3 = 90$ miles along the x-axis.
- Car B will have moved $60 \times 3 = 180$ miles along the direction of $3\hat{i} + 4\hat{j}$) which has magnitude 5.
- So it will have moved to $180 \times \frac{3}{5}\hat{i} + 180 \times \frac{4}{5}\hat{j}$.
- To find the distance between the two subtract one vector from the other, i.e. $90\hat{i} - 108\hat{i} - 144\hat{j}$.
- The magnitude of this vector is $\sqrt{18^2 + 144^2} = 145.1$ miles.

Question 10.6:

(a) Take partial derivatives of $f(x,y) = x^2 \sin(5y)$ (remember to treat the other variables as constant):

$$\frac{\partial f}{\partial x} = 18x \sin(5y) \qquad \frac{\partial f}{\partial y} = 45x^2 \cos(5y)$$

Then the gradient vector (∇f) can be constructed as follows:

$$\begin{aligned}\nabla f &= 18x\sin(5y)\,\hat{i} + 45x^2\cos(5y)\,\hat{j} \\ &= \left\langle 18x\sin(5y), 45x^2\cos(5y)\right\rangle\end{aligned}$$

(b) Take partial derivatives of $f(x,y,z) = z\exp(-xy)$ (remember to treat the other variables as constant):

$$\frac{\partial f}{\partial x} = -yz\exp(-xy) \quad \frac{\partial f}{\partial y} = -xz\exp(-xy) \quad \frac{\partial f}{\partial z} = \exp(-xy)$$

Then the gradient vector (∇f) can be constructed as follows:

$$\begin{aligned}\nabla f &= -yz\exp(-xy)\,\hat{i} - xz\exp(-xy)\,\hat{j} + \exp(-xy)\,\hat{k} \\ &= \langle -yz\exp(-xy), -xz\exp(-xy), \exp(-xy)\rangle\end{aligned}$$

(c) Take partial derivatives of $f(x,y) = 1\times10^{-3}(x-400)^2 + 1\times10^{-3}y^2$ (remember to treat the other variables as constant):

$$\frac{\partial f}{\partial x} = 2\times10^{-3}(x-400) \quad \frac{\partial f}{\partial y} = 2\times10^{-3}y$$

Then the gradient vector (∇f) can be constructed as follows:

$$\begin{aligned}\nabla f &= 2\times10^{-3}(x-400)\,\hat{i} + 2\times10^{-3}y\,\hat{j} \\ &= \left\langle 2\times10^{-3}(x-400), 2\times10^{-3}y\right\rangle\end{aligned}$$

Question 10.7: Take partial derivatives of $f(x,y) = 1\times10^{-3}(x-400)^2 + 1\times10^{-3}y^2$ (remember to treat the other variables as constant):

$$\frac{\partial f}{\partial x} = 2\times10^{-3}(x-400) \quad \frac{\partial f}{\partial y} = 2\times10^{-3}y$$

Then the gradient vector (∇f) can be constructed as follows:

$$\begin{aligned}\nabla f &= 2\times10^{-3}(x-400)\,\hat{i} + 2\times10^{-3}y\,\hat{j} \\ &= \left\langle 2\times10^{-3}(x-400), 2\times10^{-3}y\right\rangle\end{aligned}$$

At $(x,y) = (800,400)$ the gradient vector is:

$$\begin{aligned}\nabla f(x,y) &= \left\langle 2\times10^{-3}(800-400), 2\times10^{-3}\times400\right\rangle \\ &= \langle 0.8, 0.8\rangle\end{aligned}$$

Calculate the magnitude of the vector field:

$$\begin{aligned} ||\nabla f|| &= \sqrt{0.8^2 + 0.8^2} \\ &= 1.13 \end{aligned}$$

Question 10.8:

(a) Take partial derivatives of $f(x,y) = x^4 + 6\sqrt{y} - 10$. By taking the derivative with respect to x we hold other variables constant. In this case we hold y as a constant. The partial derivative with respect to x is then:

$$\frac{\partial f}{\partial x} = 4x^3$$

The second and third term in the expression always differentiate to 0 for x. To arrive at the partial derivative with respect to y we treat x as constant and thus:

$$\frac{\partial f}{\partial y} = \frac{3}{\sqrt{y}}$$

(b) Take partial derivatives of $f(x,y) = cos\left(\frac{4}{x}\right) e^{x^2y-5y^3}$. By taking the derivative with respect to x we hold other variables constant. Well lets start with our partial derivative with respect to x. We need to use the product rule as we have two expressions multiplied together here that contain x: a cosine function and an exponential. Split the function and differentiate:

$$\begin{aligned} u(x,y) = \cos\left(\frac{4}{x}\right), \quad &\Longrightarrow \quad \frac{\partial u}{\partial x} = -\frac{4}{x}\sin\left(\frac{4}{x}\right), \\ v(x,y) = \exp(x^2y - 5y^3) \quad &\Longrightarrow \quad \frac{\partial v}{\partial x} = 2xy\exp(x^2y - 5y^3) \end{aligned}$$

Apply the product rule:

$$\begin{aligned} \frac{\partial f}{\partial x} &= u \times \frac{\partial v}{\partial x} + v \times \frac{\partial u}{\partial x} \\ &= \frac{4}{x} sin\left(\frac{4}{x}\right)\exp(x^2y - 5y^3) + 2xy\cos\left(\frac{4}{x}\right)\exp(x^2y - 5y^3) \end{aligned}$$

For the partial derivative with respect to y, we do not need the product rule since variable y only appears in the exponential. We thus arrive at the following:

$$\frac{\partial f}{\partial y} = \left(x^2 - 15y^2\right)\cos\left(\frac{4}{x}\right)\exp(x^2y - 5y^3)$$

(c) Take partial derivatives of $z(u,v) = \frac{9u}{u^2+5v}$. By taking the derivative with respect to x we hold other variables constant. Well lets start with our partial derivative with

respect to u. We need to use the quotient rule as u appears both in the numerator and denominator. Split the function and differentiate:

$$\begin{aligned} f(u,v) = 9u, \quad &\implies \quad \frac{\partial f}{\partial u} = 9, \\ g(u,v) = u^2 + 5v \quad &\implies \quad \frac{\partial g}{\partial u} = 2u \end{aligned}$$

Apply the quotient rule:

$$\begin{aligned} \frac{\partial y}{\partial x} &= \frac{g \times \frac{\partial f}{\partial u} - f \times \frac{\partial g}{\partial u}}{g^2} \\ &= \frac{9(u^2+5v) - 9u(2u)}{(u^2+5v)^2} \\ &= \frac{-9u^2 + 45v}{(u^2+5v)^2} \end{aligned}$$

For the partial derivative with respect to v, we do not need the quotient rule since variable v only appears in the exponential. We thus arrive at the following:

$$\frac{\partial f}{\partial y} = \frac{-45u}{(u^2+5v)^2}$$

Question 10.9: Take the partial derivative of

$$f(x,y) = 100 \exp\left(-1 \times 10^{-5}\left[x^2 + y^2\right]\right)$$

with respect to x:

$$\frac{\partial f(x,y)}{\partial x} = -1 \times 10^{-5} \times 100 \times 2x \exp\left(-1 \times 10^{-5}\left[x^2 + y^2\right]\right)$$

Remember that taking the derivative with respect to x we hold other variables constant.



12. Crib Sheet

12.1 Overview

This Chapter contains a Crib Sheet, is a concise summary of key information, concepts, or formulas, that we have encountered during this course. It is created to serve as a quick reference guide and designed to aid in studying or solving problems efficiently. Please note that you will not be allowed this in the exam. Please see Chapter 14 for the table of equations that will be presented in the exam. If an equation does not appear in this table, then you are expected to learn it for the exam.

12.2 Differentiation

Here is a summary of Chapters 1-5 and 11 which covered the topic of Differentiation.

Finding gradients

We can use **differentiation** to find a function that calculates a gradient at any given point, which we might call $\frac{dy}{dx}$. For example, when given a function $f(x) = 2x^3 - 20x^2 + x + 4000$ we can use the rules of differentiation to arrive at the expression $\frac{dy}{dx} = 6x^2 - 40x + 1$. In this example we have arrived at an **analytical** expression for the gradient. In some cases we cannot arrive at an analytical expression but have to use a **numerical** approximation. The first approach is to take the current y value, subtracted from the previous y value and divided by the increment in x. Thus:

$$\frac{dy}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x}$$

OR, with regards to our current and previous y value:

$$\frac{dy}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x} \equiv \frac{y_{current} - y_{previous}}{x_{current} - x_{previous}}$$

We call this the **Euler backward step**. Rather than taking the value from a previous point, we can also take the value at a future point. The approximate formula then becomes:

$$\frac{dy}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x} \equiv \frac{y_{future} - y_{current}}{x_{future} - x_{current}}$$

We call this the **Euler forward step**.

Both of these approximations are attempting to calculate the same gradient but using different parts of the function curve, or function values. The **central difference** approximation is more accurate for smooth functions, though this accuracy can vary depending on the shape of our curves. In this approach we take function values either side of our current value and use the difference in a similar way. If we have two equally spaced function values, we can write the central approximation approach as:

$$\frac{dy}{dx} = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x} \equiv \frac{y_{x+\Delta x} - y_{x-\Delta x}}{2\Delta x}$$

As you can find in this crib sheet, you may also come across yet another slightly different notation when writing differential equations. Sometimes we write $\frac{dy}{dx}$ as y'. All of the following expressions are equally valid:

$$\frac{dy}{dx} = y' = \lim_{\Delta x \mapsto 0} \frac{\Delta y}{\Delta x} \equiv \frac{y_{x+\Delta x} - y_{x-\Delta x}}{2\Delta x}$$
$$\frac{dd}{dx} = f' = \lim_{\Delta x \mapsto 0} \frac{\Delta f}{\Delta x} \equiv \frac{f_x - f_{x-\Delta x}}{\Delta x}$$
$$\frac{dp}{dx} = p' = \lim_{\Delta x \mapsto 0} \frac{\Delta p}{\Delta t} \equiv \frac{p_{future} - p_{current}}{t_{future} - t_{current}}$$

Common differentials

Here is a table where we list all of the common differentials you will need for this course:

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Chain rule

The **chain rule** states that to obtain the derivative of a function $f(x)$, which can be written as a function of another function, (i.e. $f(g(x))$), we differentiate the function, f, with respect to the function g and multiply the result by the derivative of g with respect to x:

$$\frac{df}{dx} = \frac{df}{dg} \times \frac{dg}{dx}$$

Product rule

The **product rule** states that to obtain the derivative of a function $f(x)$, which can be written as the product of two functions $u(x)$ and $v(x)$ (i.e. $u(x)v(x)$), we calculate $\frac{du}{dx}$ and $\frac{dv}{dx}$ and use:

$$\frac{df}{dx} = u \times \frac{dv}{dg} + v \times \frac{du}{dx}$$

Quotient rule

The **quotient rule:** states that to obtain the derivative of a function $f(x)$, which can be written as function $u(x)$ divided by function $v(x)$ (i.e. $\frac{u(x)}{v(x)}$), we calculate $\frac{du}{dx}$ and $\frac{dv}{dx}$ and use:

$$\frac{df}{dx} = \frac{v \times \frac{du}{dx} - u \times \frac{dv}{dg}}{v^2}$$

Properties of a derivative and Stationary values

Remember, the derivative of a function at a given point represents the gradient of a function at that point, indicating the rate of change. Therefore, we can use the derivative to know where a function is increasing, decreasing or neither. We can see that if the rate of change is negative,

$$\frac{dy}{dx} < 0$$

then the function is decreasing and that if the rate of change is positive,

$$\frac{dy}{dx} > 0$$

then the function is increasing. If the derivative is zero,

$$\frac{dy}{dx} = 0$$

then it is likely the function will be changing direction either side. This means that the point on the function where the derivative is either a **maximum value**, **minimum value**

or a **saddle point**. This turns out to be a very useful property and more generally we call these **stationary values** or **stationary points**.

Sometimes it will not always be possible to solve an equation equal to zero by rearranging or factorising, due to the complexity of the function. We then must use an algorithm to find the stationary values (See Newton's method below). To find a stationary value we:

1. Differentiate the function, to find the gradient function, $f'(x)$ or $\frac{dy}{dx}$
2. Set the gradient function to zero and solve for x.
3. Substitute x into $f(x)$ to get what the value of the stationary points are.

We verify that we have found a minimum and maximum point using the second-order derviatives (see below)

Second-order derivatives

The **properties of derivatives** and how to use them to identify the **stationary values** of functions are listed above. These properties are invaluable for optimising functions or processes. However, to determine whether a stationary value represents a maximum or a minimum, a formal method for verifying this involves something called the **second-order derivative** which involves differentiating a function $f(x)$ twice:

$$f''(x) = \frac{d}{dx}\left(\frac{df}{dx}\right) = \frac{d^2f}{dx^2}$$

The second-order derivative has some very interesting and useful properties. Previously we saw that the first-order derivative, $f'(x)$ tells us about the rate of change of a function, $f(x)$. The second-order derivative, $f''(x)$, goes a step further by telling us how the rate of change, $f'(x)$, itself is changing.

The properties of second-order derivative make them very useful for understanding the nature of stationary points, which are the points where $f'(x) = 0$. Here's how $f''(x)$ helps us classify these points:

- If $f''(x) < 0$, then $f(x)$ has a local maximum at the stationary value x
- If $f''(x) > 0$, then $f(x)$ has a local minimum at the stationary value x
- If $f''(x) = 0$, then it is not possible to conclude anything about the point x. It could be saddle point (neither a maximum nor a minimum) so we need to either visually inspect the stationary value or or further evaluation of $f(x)$ is needed.

Steady state

When a system is in steady-state, the key variables that describe the system's behavior do not change over time. In a chemical system, this means that even though processes

are happening that could cause change, the system's variables stay the same. For this to happen, there must be a continuous flow of materials through the system. This rate of change can be written as:

$$\frac{dw}{dt} = f(t, x_1, x_2, ...)$$

This equation means that the rate of change of w is a function of time and multiple sources and sinks of water which I have expressed as x_1,x_2. Using the above definition of steady state, here the rate of change does not change over time, we can take the derivative of the equation above:

$$\frac{d}{dt}\left(\frac{dw}{dt}\right) = \frac{d^2w}{dt^2} = w'' = 0$$

This equation tells us that the second derivative of w with respect to time is zero. In other words, the rate of change of the volume of water is not changing—it remains constant. This implies that there is a steady flow, and the total amount of water in the system remains constant.

Finding roots of equations: Newton-Raphson method

We know how derivatives help us understand change. We need to **find the roots** of a function, which is where a **function equals zero**. You have encountered these when finding stationary values. We might be able to solve the equation directly to find the root (see above), however, in many cases, we will not. In these situations, we must use Newton's method which generally is written as:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

where $f'(x)$ is the derivative of $f(x)$. We iterate through until we reach a number of desired iterations or we have converged based on some criteria (i.e. three decimal places do not change).

Ordinary Differential Equations

The theoretical base of many of the processes we are interested in are written in the form of Ordinary Differential Equations (ODEs). These are the derivative functions we have been working with. To predict values or properties in the past or future, we can integrate ODEs either without limits, or between specific limits. Some examples of first-order ODEs have been:

- The **radioactive decay equation**: The number of decays per unit time is proportional to the number of unstable nuclei:

  $$\frac{dN}{dt} = -\lambda N$$

  where λ is a constant for a specific isotope and N is the number of unstable nuclei.

- **The hydrostatic equation**: The density of air depends on the pressure, P, and temperature of the air, T, (through the ideal gas law): $\rho = \frac{P}{R_a T}$, where $R_a = 287$ J kg^{-1} K^{-1}:
$$\frac{dP}{dz} = -\frac{P}{R_a T} g$$
- **Newton's Laws of Motion**: They state that:
$$F = ma \quad F = m\frac{dv}{dt} \quad F = mv\frac{dv}{dx}$$
where a is accelaration, v is velocity, x is distance, m is mass and F is the total forces acting on a object. Each expression is equivalent to the other.
- **Fourier's Law**: Fourier stated that the flux of heat through a substance is proportional to the negative gradient of temperature in the substance (i.e. heat flows from hot to cold). We can express this, known as Fouriers law, as a differential equation:
$$q = -k\frac{dT}{dx}$$
where T is the temperature and x the distance variable.
- **Fick's Law**: Fick also found that the law that governs transport of mass by *Brownian motion* (i.e. random thermal motion) moves from high concentrations to low concentrations. We can express this, known as Fick's law, as a differential equation:
$$J = -D\frac{dC}{dx}$$
where J is the mass flux, C is the concentration of mass (e.g. mass of pollutant / gas or particulate) and D is a diffusion coefficient which depends on the properties of the substance that is diffusing.

Other examples of ODEs that we have seen are **homogeneous, linear, second-order ODEs** which resemble the following expression:
$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$
Some examples of second-order ODEs have been:

- **Newton's Laws of Motion**:
$$F = ma \qquad F = m\frac{d^2x}{dt^2}$$
where a is accelaration, x is distance, m is mass and F is the total forces acting on a object. Each expression is equivalent to the other.
- **Fourier's and Fick's Laws**: Assuming steady state, we specify that the rate of change remains constant, either over space or time, we can rewrite them as:
$$\frac{d^2T}{dx^2} = 0 \quad \frac{d^2C}{dt^2} = 0$$

More information can be seen in Chapter 13.

Initial and Boundary Value Problems

Some ODEs are called *initial value problems*, where the initial conditions are specified and you solve to find something in the future or a certain distance away. Other kinds of differential equations are called *boundary value problems*, where upper and lower boundary conditions are specified.

Partial differentiation

When there are more than one dimension we have to use something else to denote the gradient. The standard that is used is called 'grad', which is defined for a scalar field and can be written $grad(f)$ or also written ∇f. We usually called this *'grad-f'*.

The ∇f (where $f = f(x,y,z)$ is a scalar field) is actually a vector. In Cartesian coordinates and vector form it is:

$$\begin{aligned}\nabla f &\equiv \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k} \\ &\equiv \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle\end{aligned}$$

Vectors have **magnitude** and **direction**. In our set of problems we consider vectors in Cartesian form, that is broken down into components along x, y and z of a Cartesian domain. The component along x is component $\hat{i}$, along y is component $\hat{j}$ and along z is component $\hat{k}$. So the gradient has magnitude and direction. Notice the change in the derivative symbol. Specifically, let us highlight the gradient along the x axis:

$$\frac{\partial f}{\partial x}$$

Up to this point in our course, we have been using the expression.

$$\frac{df}{dx}$$

The change is important. The new expression is called a *partial differential equation*. When you have a function expressed using more than one dimension, this means that when calculating a derivative in say x, you assume other variables such as y or z remain constant. We are only interested in information along the axis given by the variable in the denominator.

We can use this to calculate the magnitude of the gradient at any given point. The **magnitude** of a vector $\bar{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ can be calculated by taking the square root of all of the square its components:

$$||\bar{a}|| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

12.3 Integration

Integration is the reverse of differentiation. Since the derivative function provides us with a **gradient**, integration therefore provides us with a function that can be used to **calculate absolute values**. For example, if a derivative function tells us how much water is changing over time, integrating that function tells us how much water is at a given time or how much total water has changed over a specific time period. To integrate a differential equation therefore requires us to reverse the steps taken to get to that point.

Because integration can be seen as a **summing up** of lots of smaller rectangles we write it formally with an S-shaped symbol:

$$\int_0^x \left(-0.1x^2 + 10x\right) dx = \frac{-0.1x^3}{3} + 5x^2$$

here the $\int$ symbol represents a **summing up**. The dx represents the length of the base of each rectangle (i.e. Δx) and the sub and super-scripts on the $\int$ are known as *limits of integration*.

Common integrals

Here is a table where we list all of the common integrals you will need for this course:

Differential equation $\frac{dy}{dx}$	Integral, y
constant	$constant \times x + d$
x^n	$\frac{x^{n+1}}{n+1} + d$
e^x	$e^x + d$
ke^{kx}	$e^{kx} + d$
$\frac{1}{x}$	$ln(x) + d$
$cos(x)$	$sin(x) + d$
$sin(x)$	$-cos(x) + d$

where d is a constant.

Indefinite integration

In **indefinite integration**, we do not prescribe limits on our integral, as per the table above. We add a constant in the integration here because integration is the reverse process of differentiation, and differentiating a constant yields zero. This means that the original function before differentiation could have any constant value, which cannot be determined from the integral without limits, or initial conditions alone. For example, if we want to **integrate** $y = x^2$, we write this out as follows:

$$\int x^2 dx = \frac{x^3}{3} + constant$$

Definite integration

In **definite integration**, we derive the integral then evaluate the this for a lower and upper limit, and then calculate the difference. For example, if we want to **integrate** $y = x^2$ between $x = 1$ to $x = 2$, we write this out as follows:

$$\int_1^2 x^2 dx = \left[\frac{x^3}{3}\right]_0^2 \tag{12.1}$$

$$= \left[\frac{2^3}{3}\right] - \left[\frac{1^3}{3}\right] \tag{12.2}$$

$$= \left[\frac{8}{3}\right] - \left[\frac{1}{3}\right] \tag{12.3}$$

$$= \frac{7}{3} \tag{12.4}$$

note that we calculate the integral and then evaluate the function by putting in both limits and calculating the difference.

Integration by substitution

There are occasions when making a **substitution** in an integration enables us to perform what seems to be an otherwise difficult integration. For example, when the integrand contains a factor $ax+b$, often substituting $u = ax+b$ will enable you to evaluate the integral. However, we also need to write down what dx is in terms of u. For example, lets integrate $(x+4)^2$:

1. Write out the integral in its simplest form:

$$y = \int (x+4)^2 \, dx$$

2. Find a new variable (u) as a function of the old variable (x) as a substitution:

$$u = x+4$$

3. Rearrange to get an expression for the old variable (x) in terms of the new variable (u):

$$x = u-4$$

4. Differentiate with respect to the new variable u:

$$\frac{dx}{du} = 1$$

5. Get an expression for dx:

$$dx = du$$

6. Make the substitution in the original integral

$$y = \int (x+4)^2 dx \implies y = \int u^2 du \implies y = \frac{1}{3}\int u^4 \, du$$

7. Integrate as usual:
$$y = \int u^2 \, du \implies \frac{1}{3}u^3 + d$$
8. Substitute the original expression back in to get the final expression:
$$y = \frac{1}{3}(x+4)^3 + d$$

So we see that we can integrate more complicated functions using substitution. We want to use a substitution so that the integration becomes something we can use our table of standard integrals for. However, this is integration by substitution for **indefinite integrals**. In many cases, we want to find the **area under a curve** so the integral has limits, therefore we have to perform **definite integration**. For example, lets integrate $(x+4)^2$ between $x = 0$ and $x = 1$:

1. Write out the integral in its simplest form:
$$y = \int_0^1 (x+4)^2 \, dx$$
2. Find a new variable (u) as a function of the old variable (x) as a substitution:
$$u = x + 4$$
3. Rearrange to get an expression for the old variable (x) in terms of the new variable (u):
$$x = u - 4$$
4. Differentiate with respect to the new variable u:
$$\frac{dx}{du} = 1$$
5. Get an expression for dx:
$$dx = du$$
6. Find the new limits of the integral:
$$\text{When } x = 0, \;\; u = 0 + 4 = 4$$
$$\text{When } x = 1, \;\; u = 1 + 4 = 5$$
7. Make the substitution in the original integral
$$y = \int_0^1 (x+4)^2 dx \implies y = \int_4^5 u^2 \frac{du}{3} \implies y = \frac{1}{3}\int_4^5 u^2 \, du$$
8. Integrate as usual:
$$\begin{aligned} y = \frac{1}{3}\int_4^5 u^2 \, du &= \frac{1}{3}\left[u^3\right]_4^5 \\ &= \frac{1}{3} \times (5^3 - 4^3) \\ &= \frac{61}{3} = 20.33 \end{aligned}$$

Integrating Second-order ODEs

Second order differential equations are differential equations that contain a *second order derivative*. To solve second order ODEs we often have to integrate the equation twice. Firstly to get the first gradient and secondly to get the solution. See above for examples.

In the cases, we are concerned with **homogeneous, linear, second-order ODEs** which resemble the following expression:

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

To find a general solution for y we first write out the **auxiliary** equation

$$a\lambda^2 + b\lambda + c = 0$$

and solve it to find the roots. If the roots are real and not equal ($\lambda_1 \neq \lambda_2$ then we can write a general solution as:

$$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$$

If our two roots are real and equal (i.e. $\lambda_1 = \lambda_2$) then we use the formula:

$$y = (A + Bx)e^{\lambda_1 x}$$

We don't consider what happens when the two roots are not real (i.e. $b^2 - 4ac < 0$) in this course. In both general solutions, we have added two new constants A and B as we are combining multiple solutions we need to know how much each will contribute to the final solution. To calculate their values, we need some initial conditions on y and y' such as:

$$y = y_0 \text{ and } y' = y'_0 \text{ when } x = 0$$

The second-order ODEs above are called **initial value problems**. Other kinds of ODEs are called **boundary value problems**, where upper and lower boundary conditions are specified on y and y' such as:

$$y = y_0 \text{ when } x = 0 \text{ and } y' = y'_1 \text{ when } x = 1$$

This gives us more information this means that we have to make the solution meet at both points. Examples include Newton's Laws of Motion, Fick's and Fourier's laws, which are described above.



13. Constants and Physical Laws

13.1 Overview

This Chapter contains a concise summary of key information on the constants and physical laws that are used throughout the course. You are not expected to memorise the contents of this Chapter, as it is designed to supplement the course.

13.2 Constants, units and common formulae

Table 13.1: Derived Units

Entity	**Abbreviation**	**Name**
Force	$\mathrm{N} = \mathrm{kg} \cdot \mathrm{m}/\mathrm{s}^2$	newton
Energy	$\mathrm{J} = \mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}^2$	joule
Power	$\mathrm{W} = \mathrm{J}/\mathrm{s}$	watt
Pressure	$\mathrm{Pa} = \mathrm{N}/\mathrm{m}^2$	pascal
Frequency	$\mathrm{Hz} = 1/\mathrm{s}$	hertz
Electronic potential	$\mathrm{V} = \mathrm{J}/\mathrm{C}$	volt
Capacitance	$\mathrm{F} = \mathrm{C}/\mathrm{V}$	farad
Charge	$\mathrm{C} = \mathrm{s} \cdot \mathrm{A}$	coulomb
Resistance	$\Omega = \mathrm{V}/\mathrm{A}$	ohm
Magnetic field	$\mathrm{T} = \mathrm{N}/(\mathrm{A} \cdot \mathrm{m})$	tesla
Nuclear decay rate	$\mathrm{Bq} = 1/\mathrm{s}$	becquerel

Table 13.2: Fundamental Units

Entity	Abbreviation	Name
Mass	*kg*	kilogram
Time	*s*	second
Current	*A*	ampere

Table 13.3: Physical constants

Constant	Definition	Value and unit
g	earths gravitational constant	$9.80\ \mathrm{m\ s^{-2}}$
c	speed of light in vacuum	$3.00 \times 10^{8}\ \mathrm{ms^{-1}}$
e	charge of electron	$1.60 \times 10^{-19}\ \mathrm{C}$
m_e	mass of electron	$9.11 \times 10^{-31}\ \mathrm{kg}$
m_n	mass of neutron or proton	$1.67 \times 10^{-27}\ \mathrm{kg}$
N_A	Avogadro's number	6.022×10^{23} molecules/mol
k_B	Boltzmann's constant	$1.38 \times 10^{-23}\mathrm{J/K}$
R	universal gas constant	$8.31 \frac{\mathrm{J}}{\mathrm{mol\ K}} = 0.0821 \frac{\mathrm{l\ atm}}{\mathrm{mol\ K}}$
c_{W}	specific heat of water	$1\ \mathrm{cal}/(\mathrm{g\ K})]$
1 cal	calorie	4.186 J
σ	Stefan-Boltzmann constant	$5.67 \times 10^{-8}\ \frac{\mathrm{W}}{\mathrm{m^2 K^4}}$

Table 13.4: Powers of Ten and Their Symbols

Metric Prefixes		
Prefix	**Symbol**	**Value**
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
hecto	h	10^{2}
deka	da	10^{1}
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}

13.3 Common formulae

Table 13.5: Useful Formulae

Formulae	
Definition	**Formula**
Circumference of a circle with radius r, r diameter d	$C = 2\pi r = \pi d$
Area of a circle with radius r, or diameter d	$A = \pi r^2 = \pi d^2/4$
Area of a sphere with radius, r	$A = 4\pi r^2$
Volume of a sphere with radius, r	$V = (4/3)(\pi r^3)$

13.4 Radioactive decay

Certain isotopes of atoms are unstable because they dont contain enough 'glue' or neutrons to hold the protons together in order to rectify this lack of stability the unstable isotopes spontaneously decay into daughter nuclei. There are three types of decay that are referred to as α decay, β decay and γ decay, which are the high energy particles or radiation that is given off during the decay.

Alpha decay

Alpha decay is a type of radioactive decay in which an atomic nucleus emits an alpha particle, or rather a helium nucleus, and thereby transforms (or 'decays') into an atom with a mass number 4 less and atomic number 2 less. For example:

$$^{238}U \rightarrow ^{234}Th + ^{4}_{2}He^{+2}$$

which can also be written

$$^{238}U \rightarrow ^{234}Th + \alpha$$

Note this is not a chemical reaction as it doesn't involve movement of electrons per-se.

Why should this happen? Well before the decay the ^{238}U atom had 92 protons and 146 neutrons the ratio of neutrons to protons was low for such a large nucleus: $\frac{146}{92} = 1.587\%$. Whereas after the ratio is $\frac{144}{90} = 1.60$. Although the difference doesn't seem a lot it is enough to make the daughter nucleus more stable and less likely to decay.

Beta decay

Beta decay is a type of radioactive decay in which a beta particle (an electron or a positron) is emitted. In the case of electron emission, it is referred to as beta minus (β^-), while in the case of a positron emission as beta plus (β^+). In electron emission, an electron antineutrino

is also emitted, while positron emission is accompanied by an electron neutrino. For β^- decay a neutron is converted to a proton, electron and an anti-neutrino

$$n \rightarrow p + e^- + \bar{\nu}$$

Again this is not a chemical reaction as it doesn't involve electrons taking part in bonds. Note though that when β^- decay occurs the daughter nucleus is often left in an excited state and this can then decay to give off electro-magnetic radiation or gamma rays.

For β^+ decay a proton is converted to a neutron, positron and a neutrino.

$$p \rightarrow n + e^+ + \nu$$

In the same way that too few neutrons cause the nucleus to become unstable, too many also cause the nucleus to become unstable as they try to pull everything together too tightly. In this case a neutron can change into a proton and emit a β particle. This can only occur when the binding energy of the daughter nucleus is higher than the mother.

Gamma decay

Gamma decay is a type of radioactive decay when a nucleus in an excited state decays to a lower state. Gamma rays are very penetrating high energy radiation.

Radioactive decay equations

For α and β decay we can apply the radioactive decay equations. The law states that the number of decays per unit time is proportional to the number of unstable nuclei:

$$\frac{dN}{dt} = -\lambda N$$

where λ is a constant for a specific isotope and N is the number of unstable nuclei. This kind of law is quite common in physics. It is to be expected since if you have more unstable nuclei you would expect to observe more decays per second and vice versa. Another statement of this law is that the number of unstable nuclei decays away with time:

$$N(t) = N_0 \exp(-\lambda t)$$

where N_0 is the number of unstable nuclei at time $t = 0$. The constant λ is used to characterise how quickly the unstable nucleus decays it can be related to half-life (which is the time it takes for the initial number of unstable nuclei to decay to half the initial number) as

$$\lambda = \frac{\ln(2)}{t_{1/2}}$$

where $t_{1/2}$ is the half-life.

Table 13.6: Typical unstable isotopes, with their decay mode and half-life

Parent atom	Radioactive particle	Energy	Half-life
$^{2}35$U	α	4.52 MeV	7.1×10^{8} years
$^{2}38$U	α	4.18 MeV	4.5×10^{9} years
$^{2}32$Th	α	3.98 MeV	1.4×10^{10} years
$^{2}22$Rn	α	5.49 MeV	3.825 days
$^{1}4$C	β	0.56 MeV	5670 years

13.5 Newton's laws

Before the 20th century, our understanding of the physical world was built entirely on classical mechanics. More specifically - **Newtonian Physics**. This framework emerged from the collective efforts of the "Scientific Revolution", spanning figures like Nicolaus Copernicus, Galileo Galilei, and Isaac Newton.

Newton's laws of motion, a cornerstone of classical mechanics, were first presented in `Philosophiae Naturalis Principia Mathematica (1687)`, commonly known as the `Principia`. In 1543, Nicolaus Copernicus proposed a heliocentric model of the universe, placing the Sun rather than Earth at its center. This radical idea laid the groundwork for subsequent contributions by Galileo, Johannes Kepler, and René Descartes, who collectively challenged and replaced the Aristotelian worldview.

In the `Principia`, Newton formulated **three laws** to explain planetary motion, in order to explain why the orbits of the planets are ellipses rather than circles. These laws, however, extend far beyond planetary motion, offering a universal framework for understanding force and motion. Newton's laws are as follows:

1. **Law of Inertia:** A body at rest or moving at a constant speed in a straight line will remain in that state unless acted upon by an external force.
2. **Law of Force and Momentum:** The rate of change of a body's momentum is directly proportional to the force applied, with the change occurring in the same direction as the force (i.e. if a force acts on a body, that body will experience a change in momentum). This is captured in equations you see throughout this course.
3. **Law of Action and Reaction:** When two bodies interact, they exert forces on each other that are equal in magnitude and opposite in direction. You will have likely come across this law before; also known as the law of action and reaction.

In a number of our problems, we solve Newtons second law to predict how fast or far a body will travel down an incline. Since gravity is pulling the body downwards, why doesn't the body move through the surface? Well, in very simplistic terms the surface

applies an equal and opposite force to the body resting on it. In reality, molecular forces that dictate the properties of both the object and surface also play a role but these laws are adequate to explain our observed motion.

Newton's second law

Newton's second law of motion is given by the following equation:

$$F = ma$$

where F is the net force acting on a body (in Newton's, or N) that has a mass m (in kg) and therefore acceleration a (in $m.s^{-2}$). We can re-write this equation as an ordinary differential equation (ODE) as follows:

$$F = m\frac{dv}{dt}$$
$$F = mv\frac{dv}{dx}$$

where these expressions are equal, and we use the fact that $a = v\frac{dv}{dx}$ (because $v = \frac{dx}{dt}$, so $v\frac{dv}{dx} = \frac{dv}{dt}$, which is equal to a).

Kinetic energy

As we talk about forces and movement we should also consider the transfer of energy. This is obvious when we discuss transfer of heat, but the same applies for movement of mass. It takes energy to move something. As we consider the movement of a body down an incline, we can consider the gravitational force imparting a kinetic energy to our body. We often talk about the work done by something. In Physics, work done can be calculated through the following equation:

$$W = Fs$$

where W is the work done (Joules, J or newton-metres, (Nm)), F the force [Newtons (N))] and s the distance travelled (in metres (m)). As an example, as we push a body against a frictional surface we transfer energy into heat. If a force of 20N acts along a path of distance 3m, then 60J of work has been transferred. This sounds rather abstract. We have an amount of energy that has been transferred but we know, through our real-world experience, that the same amount of energy would surely not lead to a 20kg ball and a 20g ball of mass moving at the same speed, would it? We expect the smaller mass to end up with a higher speed with the same transfer of energy. We can calculate this directly.

Newton's second law $F = ma = m\frac{d^2x}{dt^2}$ can be made to look like a 1st order ODE because $\frac{d^2x}{dt^2} = \frac{d}{dt}\left(\frac{dx}{dt}\right) = \frac{dv}{dt}$, which also equals $v\frac{dv}{dx}$, by chain rule.

If we then consider that the work done along a path dx is equal to $F \times dx$, the total work done is $\int F dx$. If we proceed using the 1st order ODE variant of Newton's second law and integrate this equation, we arrive at the following equation the describes the change in kinetic energy:

$$\Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

where v would be final velocity and u the initial velocity.

13.6 Heat transfer and Fourier's laws

We know from our own experience that heat migrates from warm to cold regions. There is a molecular and thermodynamic description that explains why this is the case. In 1822 a french Physicist and Mathematician, Joseph Fourier, concluded that **the heat flux resulting from thermal conduction is proportional to the magnitude of the temperature gradient and opposite to it in sign**. There are a number of terms here, but we can see the use of a gradient and reference to the sign of the gradient. In other terms, this is saying that heat flux moves in the opposite direction to the gradient in heat - it moves from hot to cold. What is thermal conduction? Thermal conduction is the transfer of heat from hotter to cooler parts of a body resulting in equalizing of temperature. In contrast to heat transfer by convection, thermal conduction has nothing to do with macroscopic displacements in the body, but is a result of a direct energy transfer between particles, such as molecules, atoms, and electrons, with higher energy and ones with lower energy. Contrary to heat transfer by radiation, there is no thermal conduction in vacuum. Materials have a measurable thermal conductivity; a property that quantifies the potential for dissipation of heat through it. The table below lists thermal conductivity of some common materials.

Table 13.7: Thermal conductivity of some common materials

Material	**Thermal Conductivity** ($W.m^{-1}K^{-1}$)
Air at $0 \circ C$	0.024
Aluminum	205.0
Brass	109.0
Concrete	0.8
Copper	385.0
Glass	0.8
Styrofoam	0.01
Wood	0.12-0.04

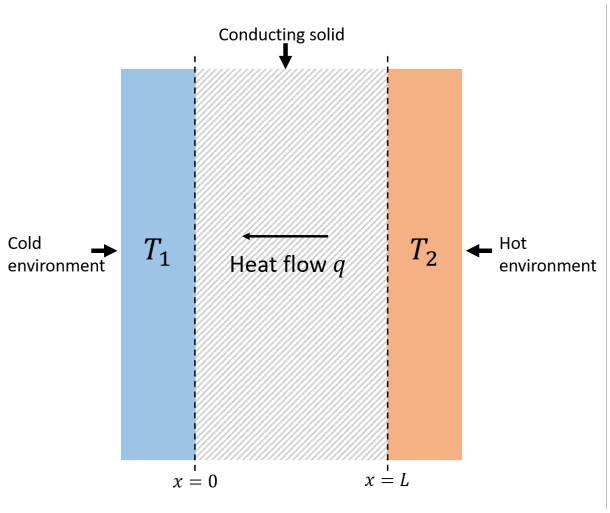


Figure 13.1: Schematic illustrating transfer of heat normal to the surface area of a substance. Source: https://en.wikipedia.org/w/index.php?curid=59712855

We note that Thermal conductivity (which we have denoted k) is defined as the quantity of heat (Q, measured in Watts or W) transmitted through a unit thickness (L - a value of 1) in a direction normal to a surface of unit area (A, in m^{-2}) due to a unit temperature gradient (ΔT, in K) under steady state conditions and when the heat transfer is dependent only on the temperature gradient. In equation form this becomes the following:

$$k = \frac{QL}{A\Delta T}$$

This is shown schematically in Figure 13.1

How do we relate this to a rate of change with distance? We only need to modify this equation slightly. Heat flux is a flow of energy per unit area per unit time (thus in $W.m^{-2}$, knowing that one Watt is one Joule per second!). If we re-arrange the above equation by moving the ΔT to the left hand side we arrive at this definition:

$$q = -k\Delta T = -k\frac{dT(x)}{dx}$$

Where are variable q is given in $W.m^{-2}$. This is in one dimension. Heat flux however has both strength and direction. If we re-write our equation as a vector quantity (please see Chapter 11), we arrive at the following:

$$\mathbf{q} = -k\nabla T$$

where $\mathbf{q}$ is still the flux of heat W m^{-2} and k is the thermal conductivity W m^{-1} K^{-1}, T is the temperature. The symbol ∇ represents the Grad operator covered in Chapter 11. The above equations represent **Fourier's first law**.

If we consider heat conduction in 1-D entering and leaving two sides of a box, volume 1 m^3, of thickness δx then the rate that heat enters the box, R (W m^{-3}), is equal to the difference in the heat entering through one side and exiting through the other side (i.e. a **gradient of a gradient**):

$$\begin{aligned} R &= \frac{\mathbf{q}_{right} - \mathbf{q}_{left}}{\delta x} \\ &= \frac{-k\frac{dT}{dx}_{right} - -k\frac{dT}{dx}_{left}}{\delta x} \end{aligned}$$

If we assume that the box's temperature is in steady state, $R = 0$, we can write this as:

$$\frac{d^2T}{dx^2} = 0$$

which is **Fourier's second law**. Using vector calculus notation we can write it as:

$$\begin{aligned} \nabla \cdot \nabla T &= 0 \text{ , which is the same as} \\ \nabla^2 T &= 0 \end{aligned}$$

13.7 Mass transfer and Fick's laws

Following Fourier, a physiologist called Adolf Fick also found that the law that governs transport of mass by **Brownian motion** (i.e. random thermal motion) is very similar (i.e. mass moves from high concentrations to low concentrations). Again from our real-world experience we can see that mass moves from high to low concentrations. Again, there are molecular and fluid properties that are important to understand. We can qualitatively understand some of those properties before we take a look at equations that describe Ficks laws.

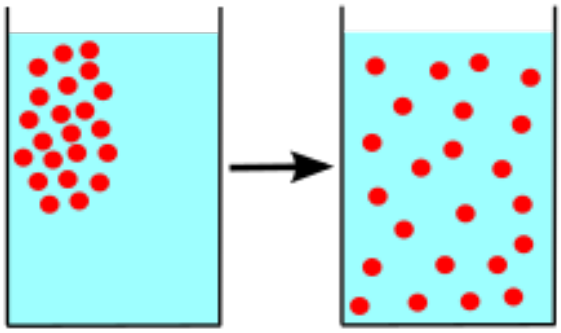


Figure 13.2: If the particles randomly move around ("diffuse") in the water, they eventually become distributed randomly and uniformly from an area of high concentration to an area of low, and organized (diffusion continues, but with no net flux). Source: https://creativecommons.org/licenses/by/3.0

In a gas we know that the molecules in that gas are moving according to a thermal energy they have at a set temperature. There are relationships that can be used to calculate that

thermal speed as a function of temperature and molecular weight. There are also some very interesting distributions of speeds of a given type of molecule but that isn't for this course. These molecules will collide with each other and change direction accordingly. If we keep that gas within a sealed box we know, through the ideal gas equation, that the gas exerts a pressure on the walls of this box through those collisions. If we were able to inject an initial cluster of molecules in the middle of that box, through this thermal motion they will diffuse out within the box. If we suddenly open a side of the box, the gas will eventually diffuse out of this opening.

The following equation presents **Ficks first law** in one dimension. Can you see the similarity with Fourier's law and the use of the negative gradient?:

$$J = -D\frac{d\Theta}{dx}$$

J is the diffusive flux (amount of substance per unit area per unit time -e.g $g.cm^{-2}.s^{-1}$. Variable Θ is the concentration (amount per unit volume) along a given axis, in this case it could be the x axis, and is given in units of e.g $molecules.cm^{-3}$. The derivative would give us units of $molecules.cm^{-4}$. Variable D is the diffusion coefficient, and is given in units of $cm^2.s^{-1}$. The diffusion coefficient D is proportional to the squared velocity of the diffusing particles, which depends on the temperature, viscosity of the fluid and the size of the colliding molecules. A table listing some common diffusion coefficients of gases in air can be found below.

The previous equation is in one dimension. As with heat transfer, if we re-write our equation as a vector quantity, we arrive at the following:

$$J = -D\nabla\Theta$$

The symbol ∇ represents the Grad operator covered in Chapter 11. Fick's **second law** predicts how diffusion causes the concentration to change with respect to time. Notice how Fick's first law provides us with a flux. We can relate that flux as another differential equation:

$$\frac{d\Theta}{dt} = D\frac{d^2\Theta}{dx^2}$$

This is a **partial differential equation**. We do not solve these types of equations in our course. Indeed, they are very difficult to solve. We can however use this to frame a steady-state scenario. If the system is in steady-state, Ficks second law this becomes:

$$\nabla^2\Theta = 0$$

Or alternatively we can write (in 1 dimension):

$$\frac{d^2\Theta}{dx^2} = 0$$

We canintegrate this equation to derive expressions that describe the change in concentration along a given dimension.

13.8 Atmospheric pressure and the hydrostatic equation

The pressure above the earth's surface can be found based on the fact that it arises due to the weight of the air above us (this is similar for the pressure in deep water, except it is the weight of water above). We know that the weight of any object is given by the following equation:

$$\text{weight} = m.g$$

where m is the objects mass (kg) and g is the Earths gravitational factor. Weight is equivalent to the definition of gravitational force we have been using in Newtons second laws.

The weight of a 1 m^3 cube or air that has density ρ is thus equal to $\rho \times 1 \times g$, since we simply calculate the mass as $V.\rho$ where $V = 1m^3$ and ρ is given in $kg.m^{-3}$. We know that the definition of pressure is force per unit area and, as we are working with $1m^3$ of air, we can divide by $1m^2$ and write an expression for pressure as follows:

$$P = \rho \times g$$

The pressure is force per unit area so that the change in pressure (within a column that has area 1 m^2) as we move, $\Delta z = 1$ m, up through the atmosphere is $\Delta P = -\rho \times g \times \Delta z$. We can therefore write that:

$$\frac{\Delta P}{\Delta z} = \frac{dP}{dz} = -\rho g$$

that is the gradient in pressure with height is equal to minus the density of the air multiplied by gravity, where gravity, $g = 9.8$ m s^{-2}.

The density of air depends on the pressure, P, and temperature of the air, T, (through the ideal gas law): $\rho = \frac{P}{R_a T}$, where $R_a = 287$ J kg^{-1} K^{-1}. So we can write:

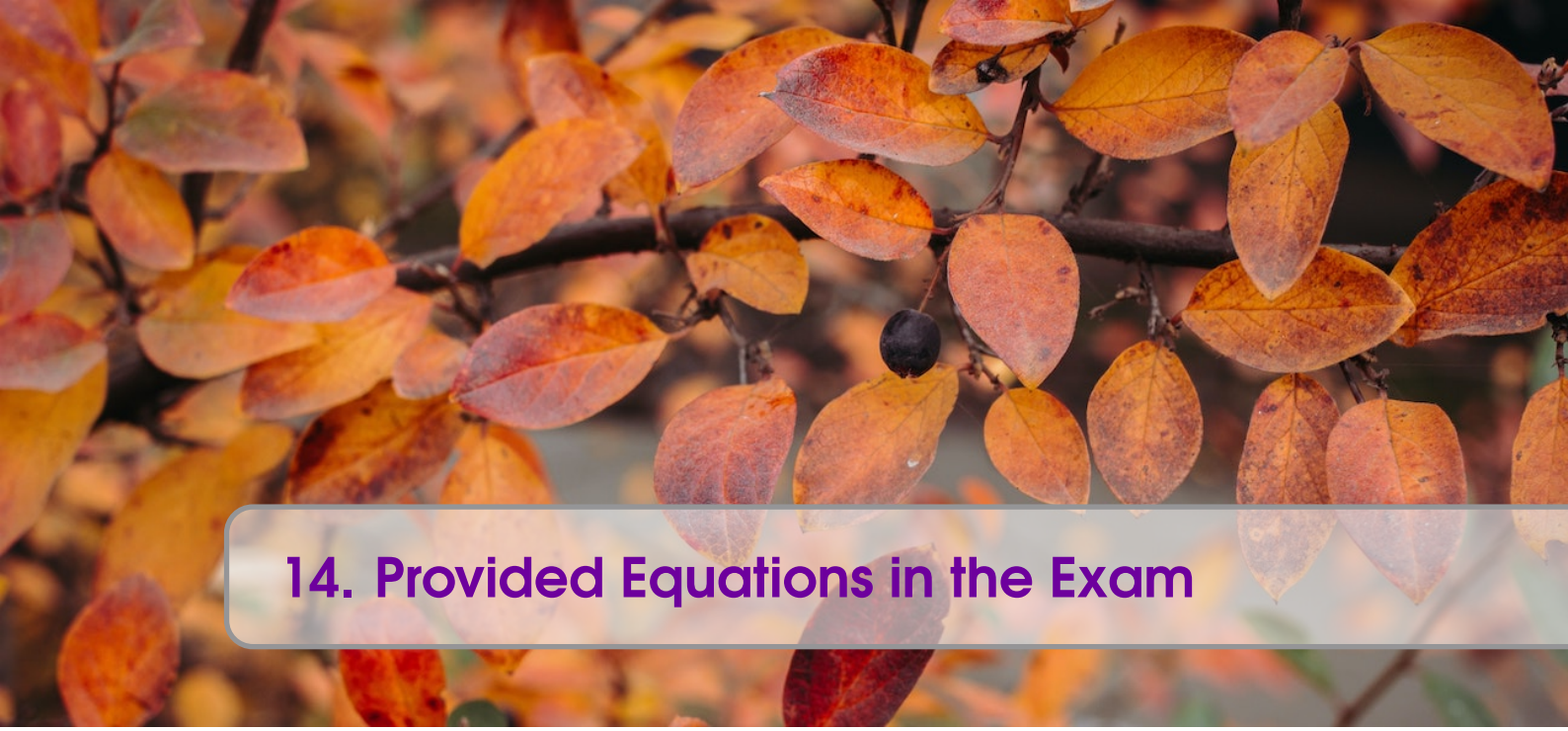
$$\frac{dP}{dz} = -\frac{P}{R_a T} g$$

In actual fact the assumption that the temperature is constant is not true as it usually decreases with height in the atmosphere at about 10°C km^{-1} (i.e. a drop of 10 °C for every km). When we take into account that temperature drops with altitude the hydrostatic equation becomes:

$$\frac{dP}{dz} = -\frac{P}{R_a (T_0 - 0.01z)} g$$

which has an analytical solution of:

$$P(z) = P_0 \exp\left(-\frac{100g}{R_a} \ln\left[\frac{T_0}{T_0 - 0.01z}\right]\right)$$



14. Provided Equations in the Exam

This Chapter contains equations that will be presented to you in the exam. If an equation does not appear in this Chapter, then you are expected to learn it for the exam.

Newton's method: Recall that we can write Newtons formula for root finding, generally, as:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

where $f'(x)$ is the derivative of $f(x)$.

Chain rule: To obtain the derivative of a function $f(x)$, which can be written as a function of another function, (i.e. $f(g(x))$), we differentiate the function, f, with respect to the function g and multiply the result by the derivative of g with respect to x:

$$\frac{df}{dx} = \frac{df}{dg} \times \frac{dg}{dx}$$

Product rule: To obtain the derivative of a function $f(x)$, which can be written as the product of two functions $u(x)$ and $v(x)$ (i.e. $u(x)v(x)$), we calculate $\frac{du}{dx}$ and $\frac{dv}{dx}$ and use:

$$\frac{df}{dx} = u \times \frac{dv}{dg} + v \times \frac{du}{dx}$$

Quotient rule: To obtain the derivative of a function $f(x)$, which can be written as function $u(x)$ divided by function $v(x)$ (i.e. $\frac{u(x)}{v(x)}$), we calculate $\frac{du}{dx}$ and $\frac{dv}{dx}$ and use:

$$\frac{df}{dx} = \frac{v \times \frac{du}{dx} - u \times \frac{dv}{dg}}{v^2}$$

Common differentials:

$$
\begin{aligned}
y = x^n \qquad & \frac{dy}{dx} = nx^{n-1} \\
y = e^{f(x)} \qquad & \frac{dy}{dx} = \frac{df}{dx} e^{f(x)} \\
y = \ln(x) \qquad & \frac{dy}{dx} = \frac{1}{x} \\
y = \sin(x) \qquad & \frac{dy}{dx} = \cos(x) \\
y = \cos(x) \qquad & \frac{dy}{dx} = -\sin(x)
\end{aligned}
$$

Common integrals:

$$
\begin{aligned}
\frac{dy}{dx} = x^n \qquad & y = \frac{x^{n+1}}{n+1} + \text{constant} \\
\frac{dy}{dx} = ke^{kx} \qquad & y = e^{kx} + \text{constant} \\
\frac{dy}{dx} = \frac{1}{x} \qquad & y = \ln(x) + \text{constant} \\
\frac{dy}{dx} = \cos(x) \qquad & y = \sin(x) + \text{constant} \\
\frac{dy}{dx} = \sin(x) \qquad & y = -\cos(x) + \text{constant}
\end{aligned}
$$



15. Acknowledgments

This course and the course textbook builds on the historical course developed and delivered by Prof David Topping and Prof Paul Connolly in the Department of Earth and Environmental Sciences. I would also like to make reference to a the online Calculus repository provided by Dr Paul Dawkins (Link)